

Verification and Testing Methodologies Real Time Embedded Systems

Agenda

- Introduction: Test and Verification
- Verification
 - Introduction and Basic Operations on Temporal Logic
 - Syntax and Semantics of CTL
 - CTL Model Checking Algorithm
 - RTCTL for Real Time ES
- Test
 - Functional vs. Structural Test
 - Combination and Sequential Circuit Test
 - Delay Fault Testing for Real Time ES

Verification vs. Test

- Verifies correctness of design i.e., check if the design meets the specifications.
- Simulation or formal methods.
- Performed once prior to manufacturing.
- Required for reliability of design.
- Checks correctness of manufactured hardware.
- Two-stage process:
 - 1. Test generation: CAD tools executed once during design for ATPG
 - 2. Test application: TPs tests applied to ALL hardware samples
- Test application performed on every manufactured device.
- Responsible for reliability of devices.

Design Cycle

Specification

Design

Verification

Implementation

Testing

Installation/marketing

Maintenance

Verification is used to capture the bugs at the early phase of design cycle.

Simulation

Simulation:

Exhaustive Simulation

Non exhaustive Simulation

Number of test cases are exponential to the number of state variables.

Non Exhaustive Simulation

Instead of using all possible combinations, simulation is done for some selected input combinations.

To find the appropriate subset is a complex problem.

- Test case generation

We may not cover all possible error cases.

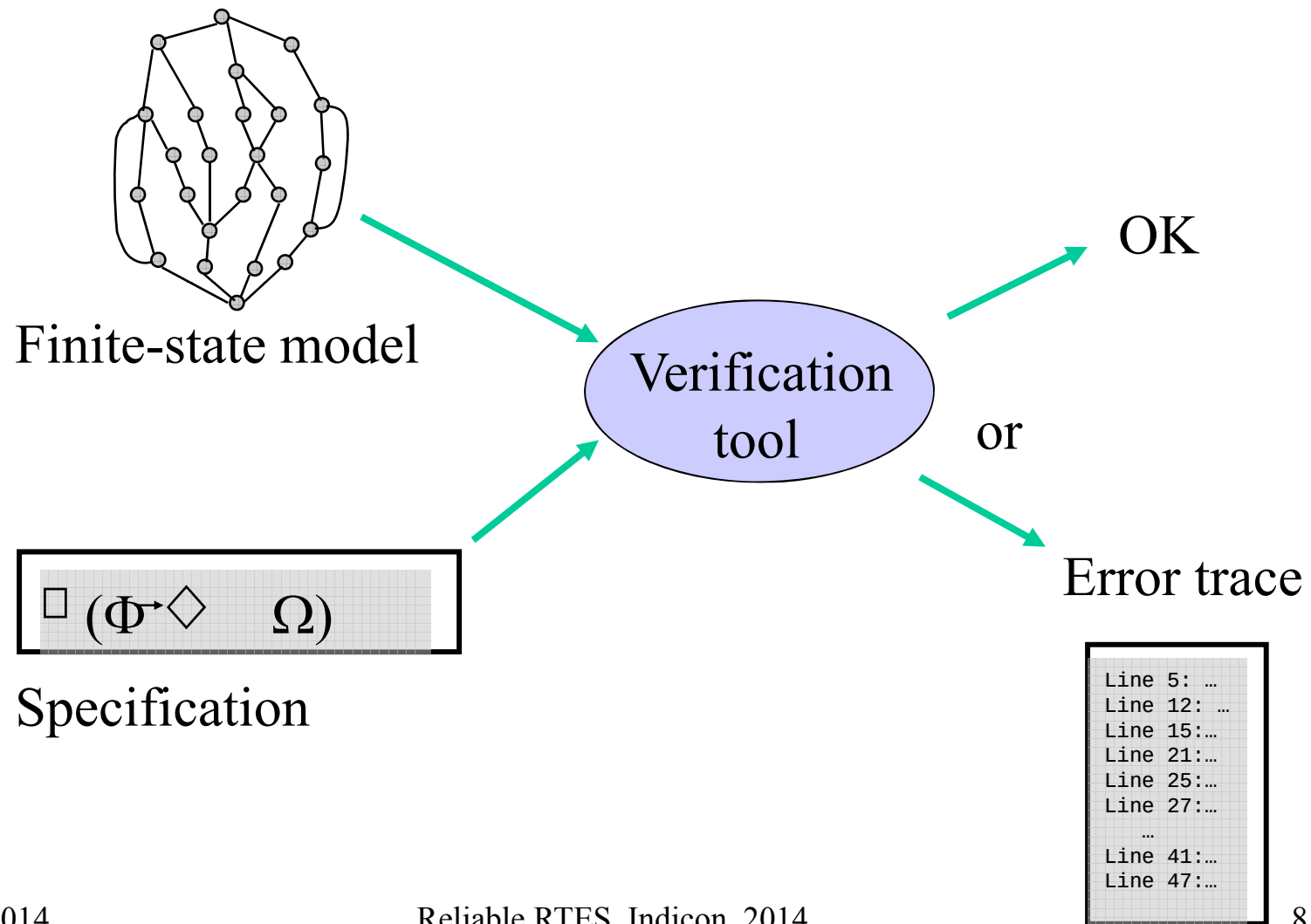
Non Exhaustive Simulation

Non Exhaustive Simulation: **Pentium Bug**

The intel pentium TM processor for IBM compatible PC was first introduced into the market in May of 1993.

A year later an estimated two million had been sold, it was discovered that there was a flaw in the hardware of floating point division.

Basic picture of Model Checking (Formal Verification)



Logic for Formal verification

- Propositional Logic

- p : ‘I qualified GATE exam in 2010’
- q : ‘I prepared well’
- r : ‘I got a non-scholarship seat in IIT Guwahati ’
- Dually, the formula denotes conjunction of p and r : ‘I qualified GATE exam in 2010 and I got a non-scholarship seat in IIT Guwahati.’ (may be his score was not high!!!)

- Predicate Logic

- $W(X)$: ‘ X gets fail grade.’
- $R(X)$: “ X is working hard”,
- In predicate logic, we also use two quantifiers, one is universal quantifier, for all ($\forall$), and the other is existential quantifier, there exists ($\exists$).
- We can write: $\forall x [R(x) \rightarrow U(x)]$ for all those who work hard will get good marks.

Temporal Logic

- The truth value of a temporal logic is defined with respect to a model.
- Temporal logic formula is not *statically* true or false in a model.
- The models of temporal logic contain several states and a formula can be true in some states and false in others.

Temporal Logic

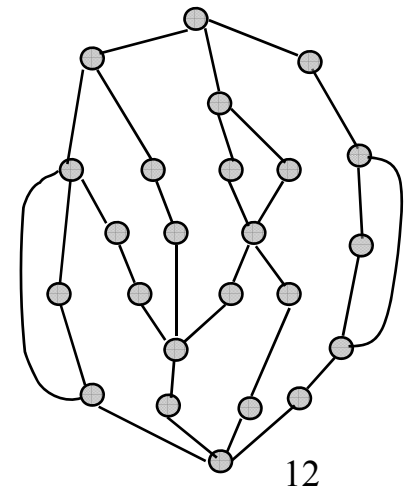
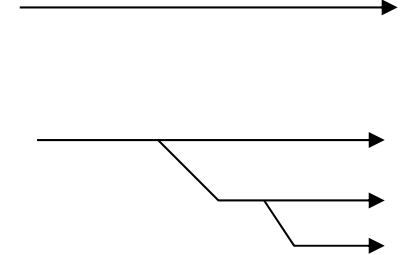
In temporal logic we can express statements like:

- "I am *always* happy",
- "I will *eventually* be happy",
- "I will be happy *until* I do something wrong"
- "I am happy."

Temporal logic

Branching vs. linear time:

- *Linear time* logic thinks of time as a set of paths, where a path is a sequence of time instances.
- *Branching-time* logic represents time as a tree, rooted at the present instance and branching out into the future.



Temporal Operator

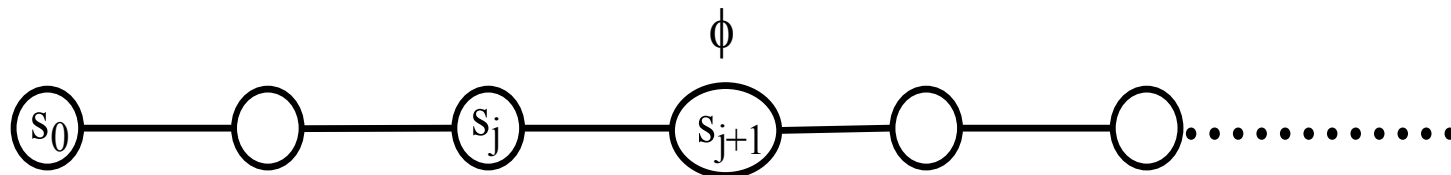
Operator	Textual Notation	Meaning
$\bigcirc$	$X \varphi$	Φ holds at next state
$\Diamond$	$F \varphi$	Φ eventually holds
$\Box$	$G \varphi$	Φ holds globally
U	$\varphi U \psi$	Φ holds until ψ holds

Temporal Operator

Next : $X \varphi$

$$(M, S_j) \models X \varphi \Leftrightarrow (M, S_{j+1}) \models \varphi$$

State S_j satisfies as its next state S_{j+1} satisfies φ .

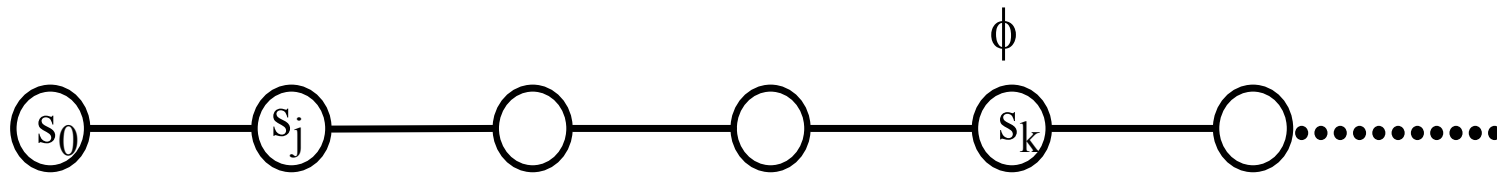


Temporal Operator

Future : $F \varphi$

$$(M, S_j) \models F \varphi \Leftrightarrow \exists k, k \geq j, (M, S_k) \models \varphi$$

state S_j satisfies as future state S_k satisfies φ .

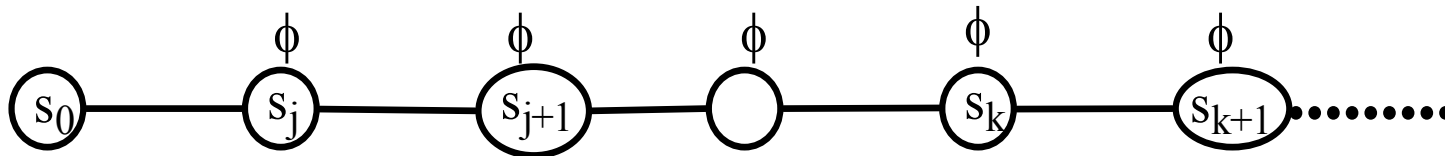


Temporal Operator

Globally : $G \varphi$

$$(M, S_j) \models G\varphi \Leftrightarrow \forall k, k \geq j, (M, S_k) \models \varphi$$

S_j satisfies $G \varphi$ as all states satisfies φ .



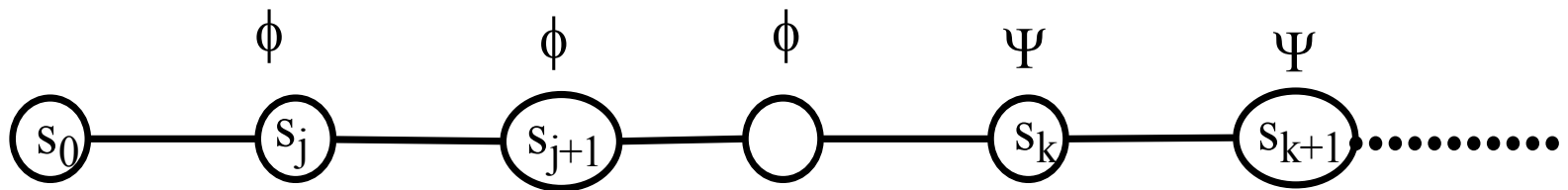
Temporal Operator

Until : $\Psi \text{ U } \varphi$

$$(M, S_j) \models \Psi \text{ U } \varphi \Leftrightarrow \exists k, (M, S_k) \models \varphi \text{ and}$$

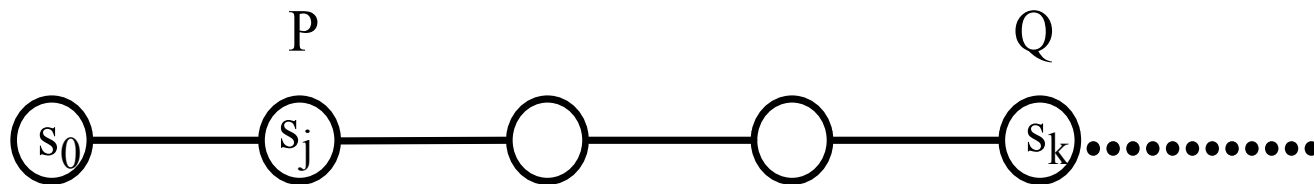
$$\forall k, j < k (M, S_j) \models \Psi$$

S_j satisfies $(\Psi \text{ U } \varphi)$ because Ψ is true for all states S_i $j \leq i < k$ and then φ is true for state S_k .

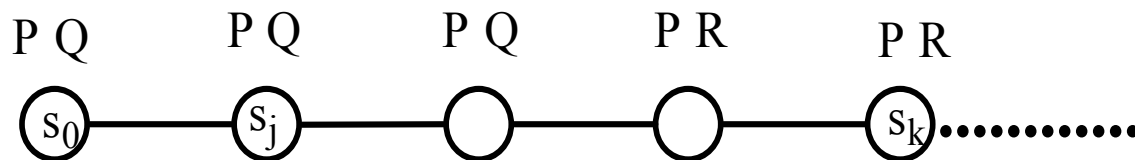


Example

$(P \rightarrow FQ)$: If P is true in a state then in a future state Q is true.



$(P \wedge (Q \cup R))$: P holds in a state and also $Q \cup R$ (Q until R) holds in the state



Syntax of CTL

Syntax of CTL

A CTL formula comprises

1. Atomic propositions such as $\{p, q, r, \dots\}$
2. Path Quantifiers $\{A, E\}$
 - a. A : all paths starting from a given state.
 - b. E : there exists at least one path from a given state.
3. Propositional logic operators such as AND ($\wedge$), OR ($\vee$), NOT ($\neg$)
4. Temporal operators $\{X, F, G, U\}$
 - a. **NEXT**: next states of current state.
 - b. **FUTURE**: any one of future states from the current state.
 - c. **GLOBAL**: all future states from the current state.
 - d. **UNTIL**: Some CTL formula holds until another CTL formula, from the current state.

CTL Formulas

We can define CTL formulas as:

$$\Phi ::= \perp \mid \top \mid P \mid (\neg\varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi) \mid AX\varphi$$
$$\mid EX\varphi \mid AF\varphi \mid EF\varphi \mid AG\varphi \mid EG\varphi \mid A[\varphi \ U \ \varphi] \mid E[\varphi \ U \ \varphi];$$

where

- The symbol $\top$ means truth value ‘true’ and symbol $\perp$ means truth value ‘false’.
- P ranges over a set of atomic propositions

CTL Formulas

AGf_1 : Always globally f_1 holds.

EGf_1 : There exists a path where f_1 holds globally

AFf_1 : f_1 holds in all path in future.

EFf_1 : There exists a path in which f_1 holds in future.

CTL Formulas

$AX f_1$ means: holds in state s° iff f_1 holds in all successor states of s°

$EX f_1$ means: There exists a successor such that f_1 holds

$A[f_1 U f_2]$ means: always until, in all paths such that f_1 holds until f_2 is satisfied.

$E[f_1 U f_2]$ means: There exists a path such that f_1 holds until f_2 is satisfied.

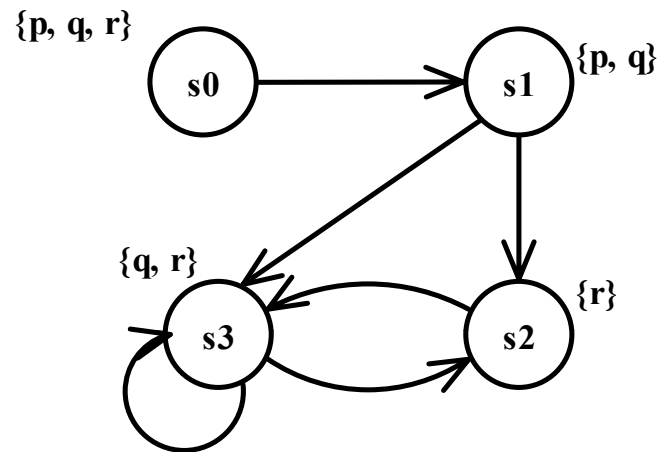
Temporal structure

- The semantics of CTL is defined over a model M , which is defined as 3-tuple $M = (S, \rightarrow, L)$
- **Definition:** A temporal structure $M := (S, \rightarrow, L)$ consists of
 1. A finite set of states S
 2. A transition relation $\rightarrow \subseteq S \times S$ with $\forall s \in S \exists s' \in S: (s, s') \in \rightarrow$
 3. A labeling function $L: S \rightarrow \wp(V)$, with V being the set of propositional variables (atomic formulas)
- This structure is often called Kripke structure.

Kripke structure

- A Kripke structure is similar to a state transition diagram, with
 - All states must have at least one outgoing edge.
 - Each state is labeled with one of the element of the power set of atomic propositions.

Kripke structure



- $S = \{s0, s1, s2, s3\}$
- $\rightarrow = \{\{s0, s1\}, \{s1, s2\}, \{s1, s3\}, \{s2, s3\}, \{s3, s2\}, \{s3, s3\}\}.$
- $L: L(s0) = \{p, q, r\}, L(s1) = \{p, q\}, L(s2) = \{r\}, L(s3) = \{q, r\}.$

Illustration of basic CTL temporal operators

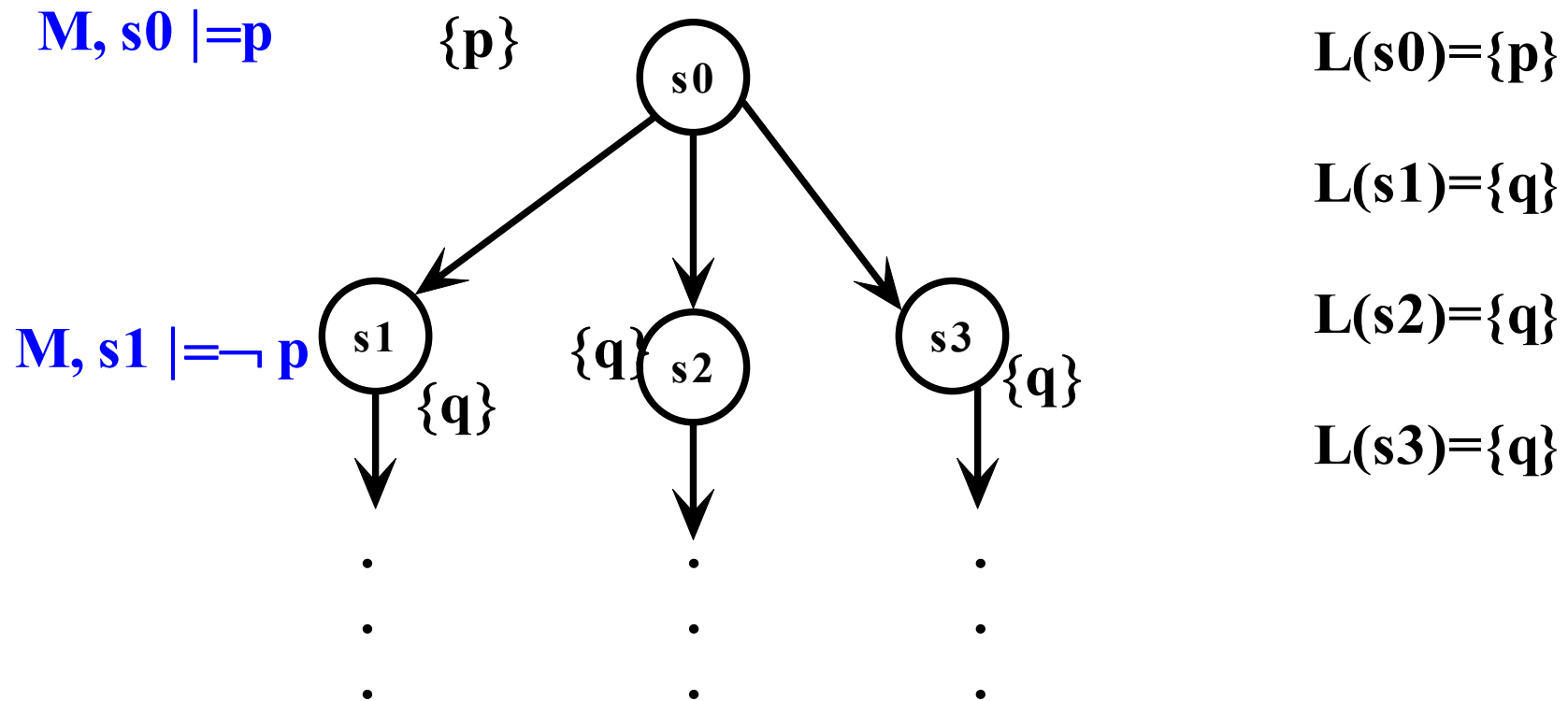
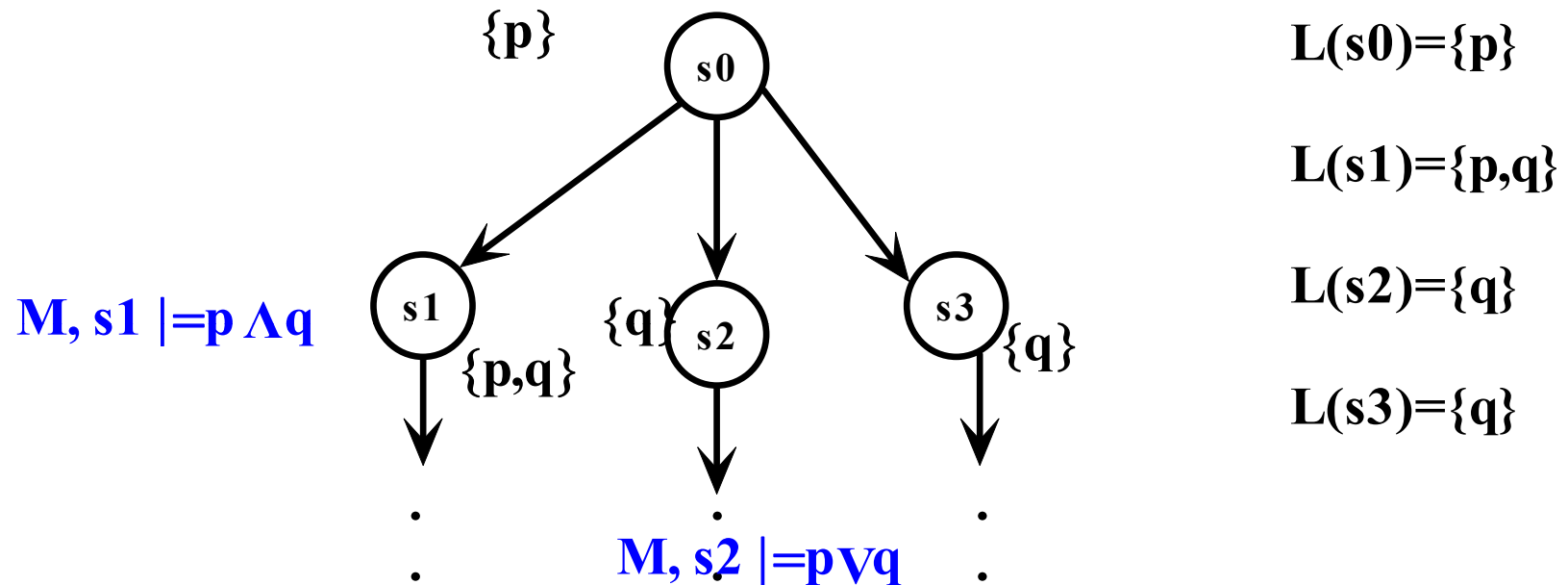


Illustration of basic CTL temporal operators



$M, s \models \varphi_1 \rightarrow \varphi_2$ iff $M, s \not\models \varphi_1$ or $M, s \models \varphi_2$;

$\varphi_1 \rightarrow \varphi_2$ is satisfied at s if in s either φ_1 is not satisfied or φ_2 is satisfied.

Illustration of basic CTL temporal operators

$M, s \models AX\varphi$ iff for all $s1$ such that $s \rightarrow s1$, we have $M, s1 \models \varphi$; $AX\varphi$ is satisfied at s if in all next states of s , φ is satisfied.

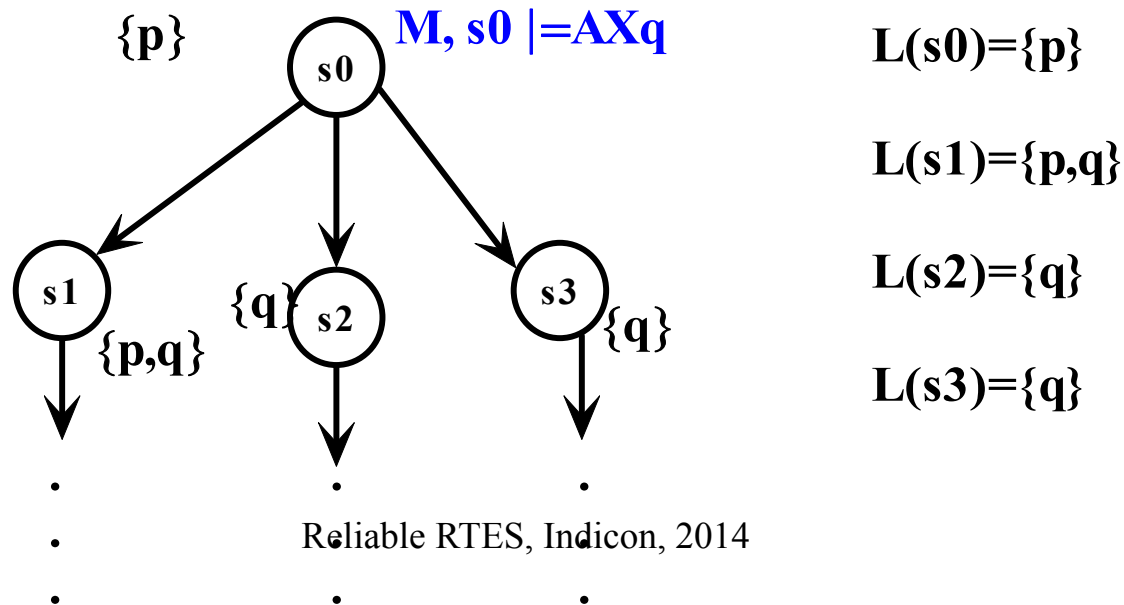


Illustration of basic CTL temporal operators

$M, s \models EX\varphi$ iff for one state $s1$ such that $s \rightarrow s1$ we have $M, s1 \models \varphi$; $EX\varphi$ is satisfied at s , if in some next state of s , φ is satisfied.

$s0 \models EXp$.

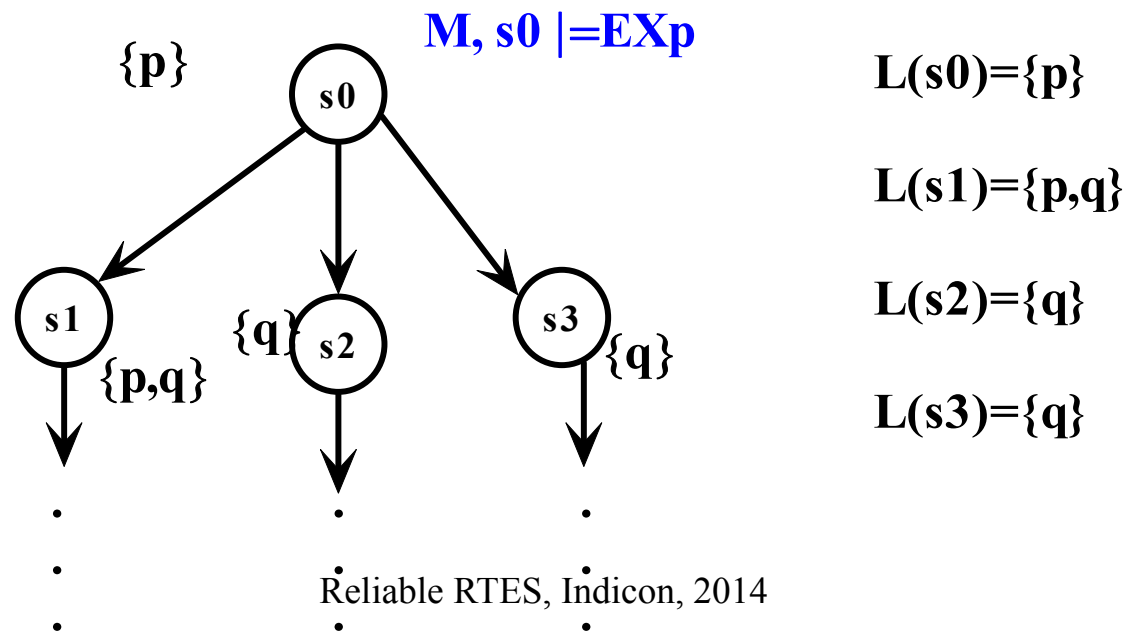
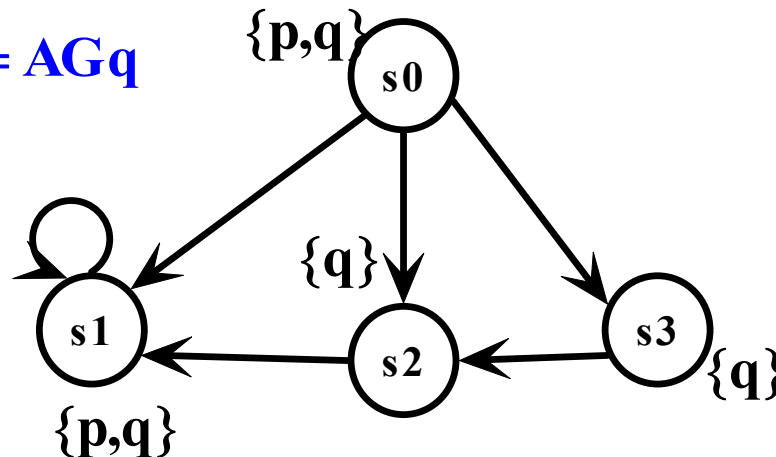


Illustration of basic CTL temporal operators

$M, s \models AG\varphi$ holds iff for all paths $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \dots$, where $s=s_1$, and all s_i along the path, $M, s_i \models \varphi$. $AG\varphi$ is satisfied at s if all states of all paths from s satisfies φ .

$M, s_0 \models AGq$



$L(s_0)=\{p,q\}$

$L(s_1)=\{p,q\}$

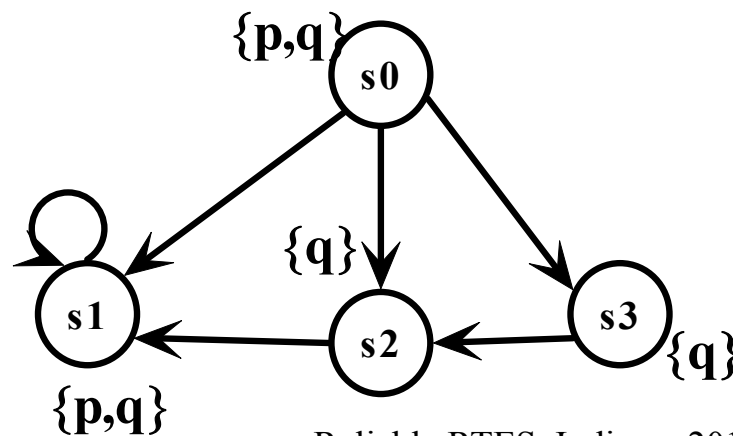
$L(s_2)=\{q\}$

$L(s_3)=\{q\}$

Illustration of basic CTL temporal operators

$M, s \models EG\varphi$ holds iff there is a path $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \dots$, where $s=s_1$, and all s_i along the path, $M, s_i \models \varphi$. $EG\varphi$ is satisfied at s if all states of at least one path from s satisfies φ .

state s_0 satisfies EGp (the path $s_0 \rightarrow s_1 \rightarrow s_1 \rightarrow \dots$).



$L(s_0)=\{p,q\}$

$L(s_1)=\{p,q\}$

$L(s_2)=\{q\}$

$L(s_3)=\{q\}$

Illustration of basic CTL temporal operators

$M, s \models AF\varphi$ holds iff for all paths $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \dots$, where $s=s_1$, and for at least one s_i along the path, $M, s_i \models \varphi$. $AF\varphi$ is satisfied at s if some “future” state of all paths from s satisfies φ .

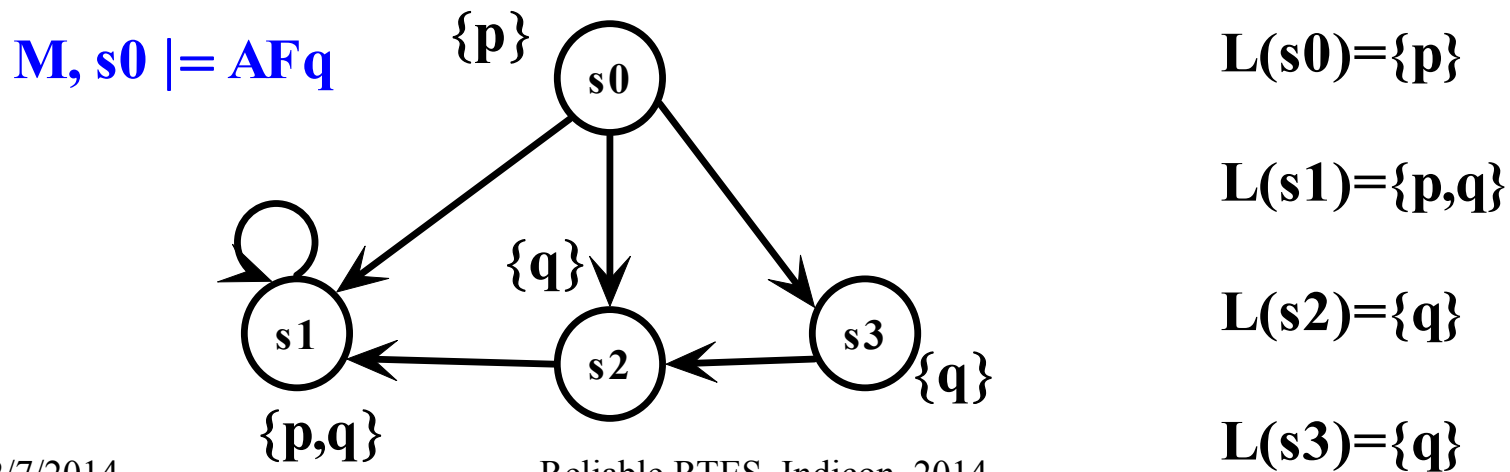


Illustration of basic CTL temporal operators

$M, s \models A[\varphi 1 \ U \ \varphi 2]$ holds iff for all paths $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \dots$, where $s = s_1$, $\varphi 1 \ U \ \varphi 2$ is satisfied, i.e., there is some s_i along the path, such that $M, s_i \models \varphi 2$, and for each $j < i$, we have $M, s_j \models \varphi 1$

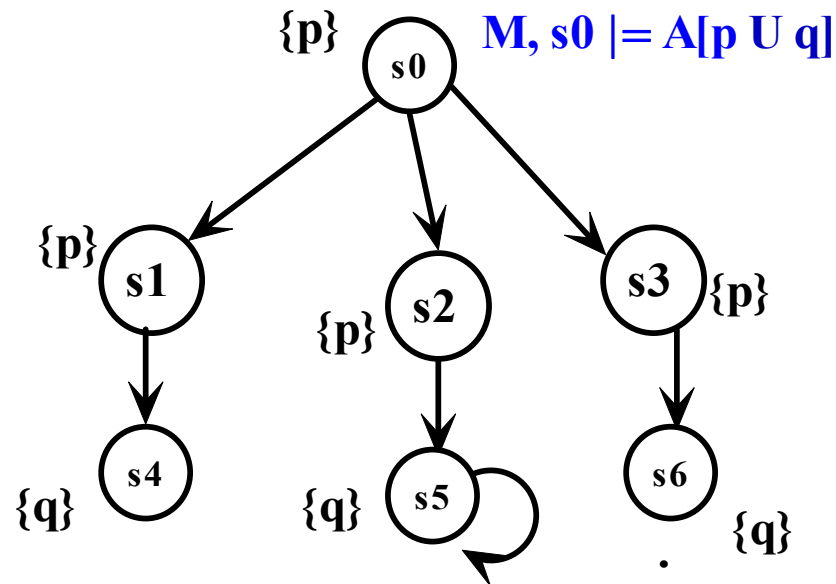
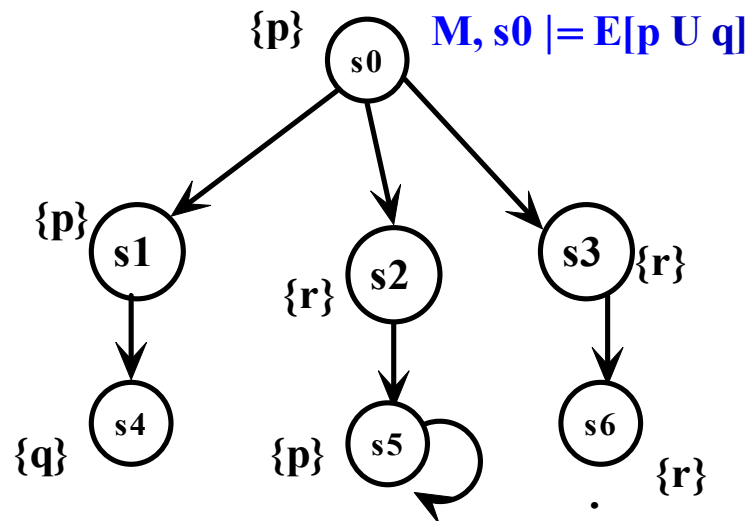


Illustration of basic CTL temporal operators

$M, s \models E[\varphi 1 \ U \ \varphi 2]$ holds iff for at least one path $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \dots$, where $s=s_1$, $\varphi 1 \ U \ \varphi 2$ is satisfied, i.e., there is some s_i along the path, such that $M, s_i \models \varphi 2$, and for each $j < i$, we have $M, s_j \models \varphi 1$



Verification Technique: Model checking (CTL)

- **Process of Model Checking:**
 - Modeling
 - Specification
 - Verification Method: Model Checking Algorithm

Labeling Algorithm

CTL model checking algorithm basically works by iteratively determining (i.e., labeling) states which satisfy a given CTL formula.

The basic input/output of labeling algorithm are as follows:

INPUT : A CTL model ' M ' = $(S, \rightarrow, L)$ where S is the set of states, $\rightarrow$ is the transition relation and L is the labeling function and a CTL formula Φ .

OUTPUT : The set of states of M which satisfy Φ .

Labeling Algorithm

- The adequate set of temporal operators for CTL is AF, EU and EX.
- First, we write the given formula Φ in terms of the connectives AF, EU and EX along with other logical connectives and truth value T.

Labeling Algorithm

- Atomic proposition
 - p : label state s with p if $p \in L(s)$
- Logical connectives
 - $p \wedge q$: label s with $p \wedge q$ if s is already labeled with p and q

Labeling Algorithm

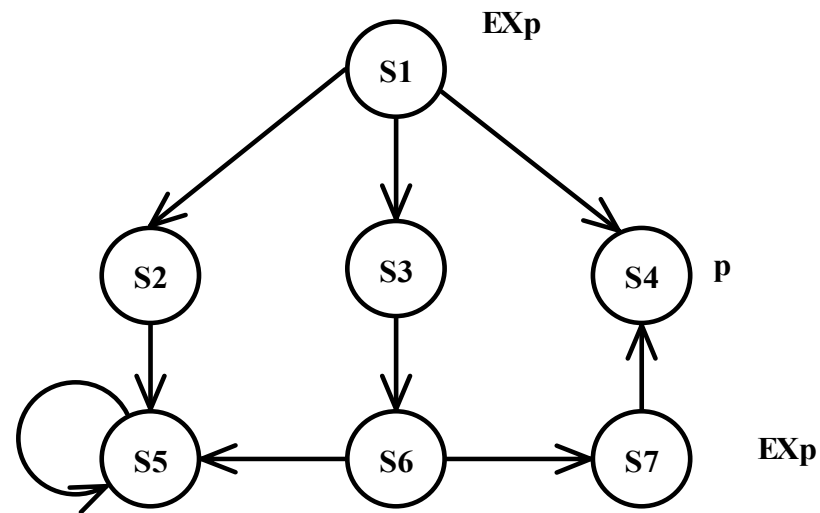
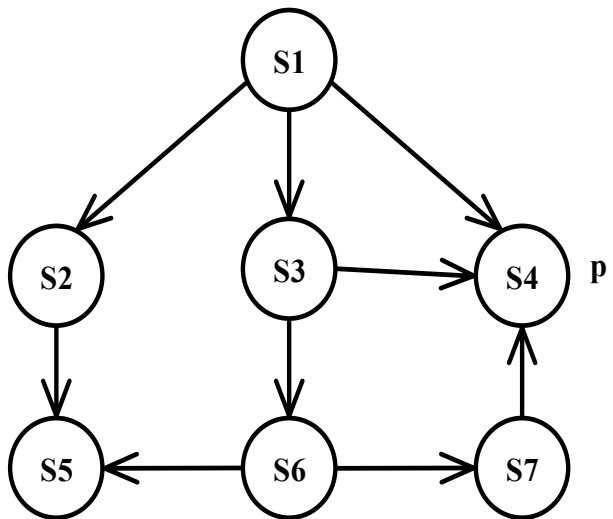
Temporal Operator:

$EX\ p$

Label any state with $EX\ p$ if one of its successor is labeled with p

Labeling Algorithm Example

EF p



Labeling Algorithm

Function $\text{SAT}_{\text{EX}}(p)$

/* determines the set of states satisfying $\text{EX}p$ */

local var X, Y

begin

$X := \text{SAT}(p)$

$Y := \{s_0 \in S \mid s_0 \rightarrow s_1 \text{ for some } s_1 \in X\}$

return Y

end

Labeling Algorithm

AF p

Temporal Operator:

AF p

- If any state s is labeled with p, label it with AF p
- Repeat: label any state with AF p if all successor states are labeled with AF p until there is no change.

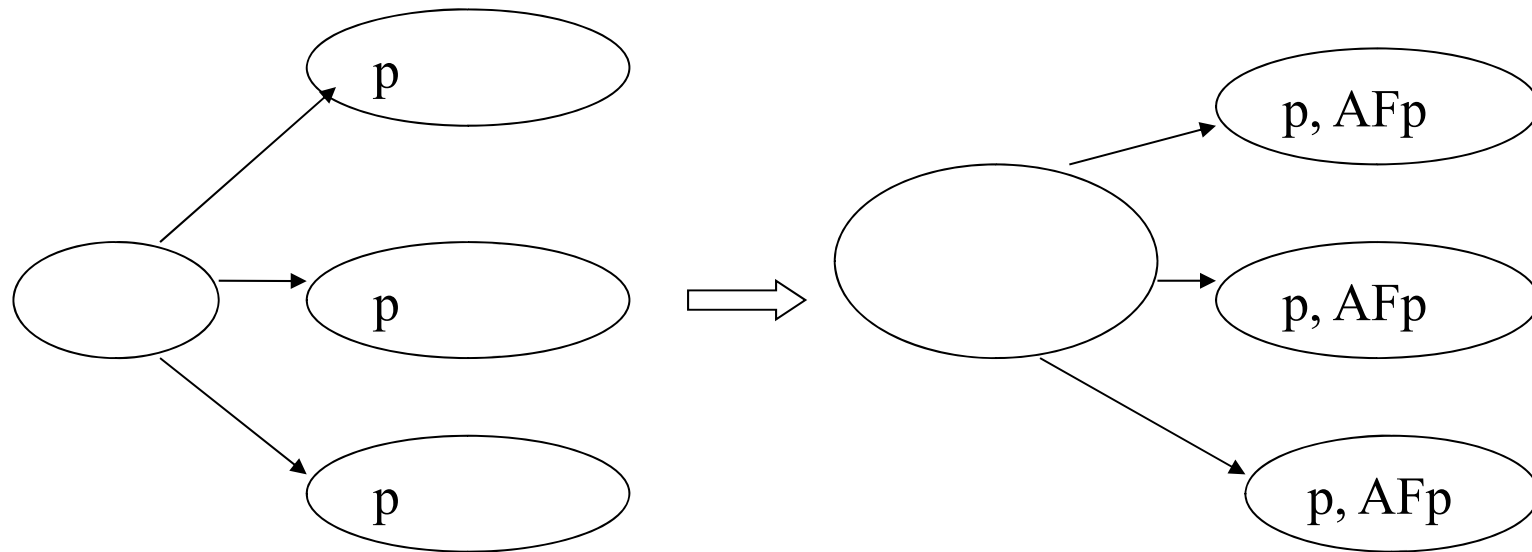
$$\text{AF } p \equiv p \vee \text{AX AF } p$$

Labeling Algorithm

AF p

Function $\text{SAT}_{\text{AF}}(p)$

/ determines the set of states satisfying AFp */*



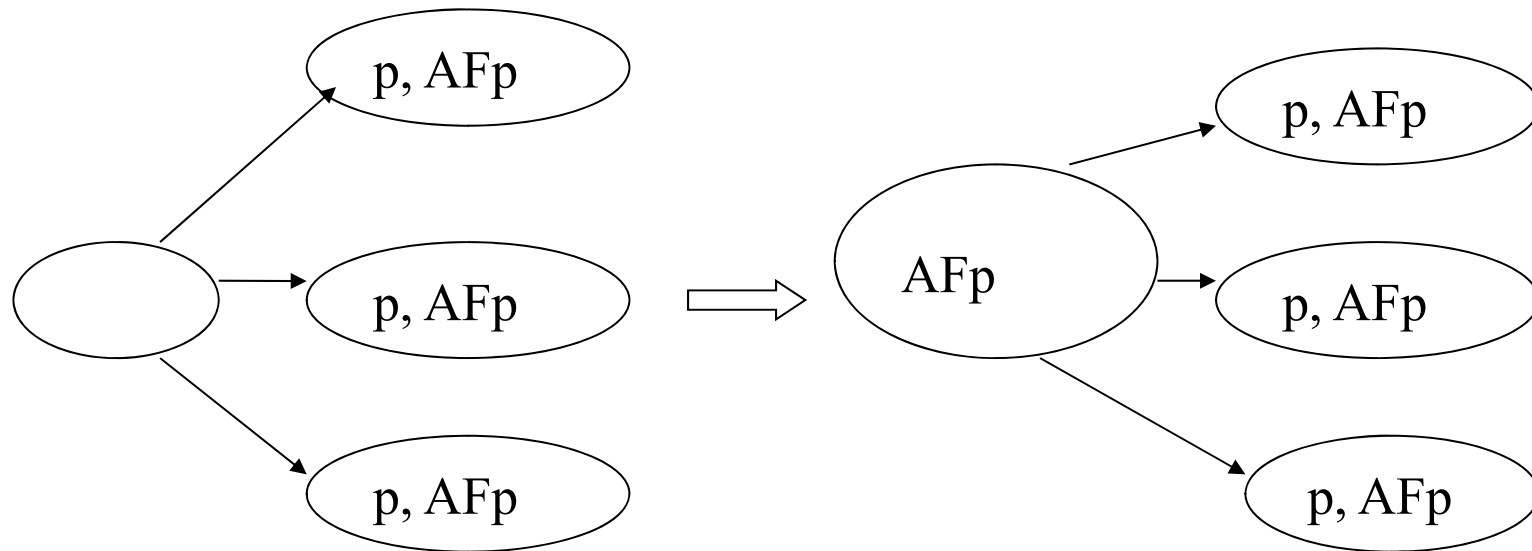
...Until no change

Labeling Algorithm

AF p

Function $\text{SAT}_{\text{AF}}(p)$

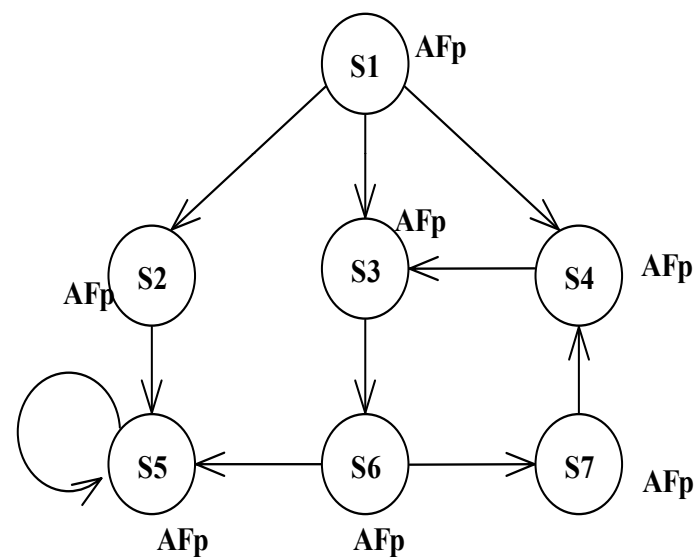
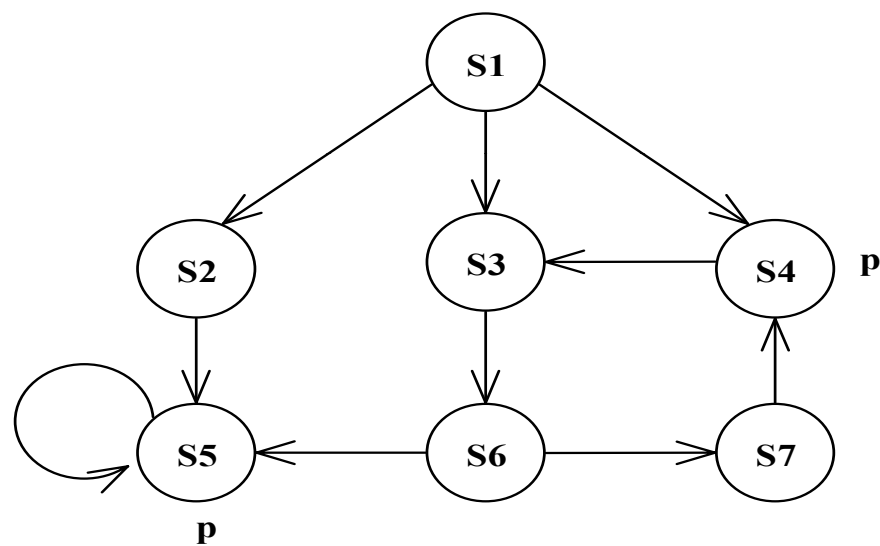
/* determines the set of states satisfying AFp */



...Until no change

Labeling Algorithm

AF p



Labeling Algorithm

$E(p \cup q)$

Temporal Operator:

$E(p \cup q)$

- If any state s is labeled with q , label it with $E(p \cup q)$
- Repeat: label any state with $E(p \cup q)$ if it is labeled with p and at least one of its successor is labeled with $E(p \cup q)$ until there is no change.

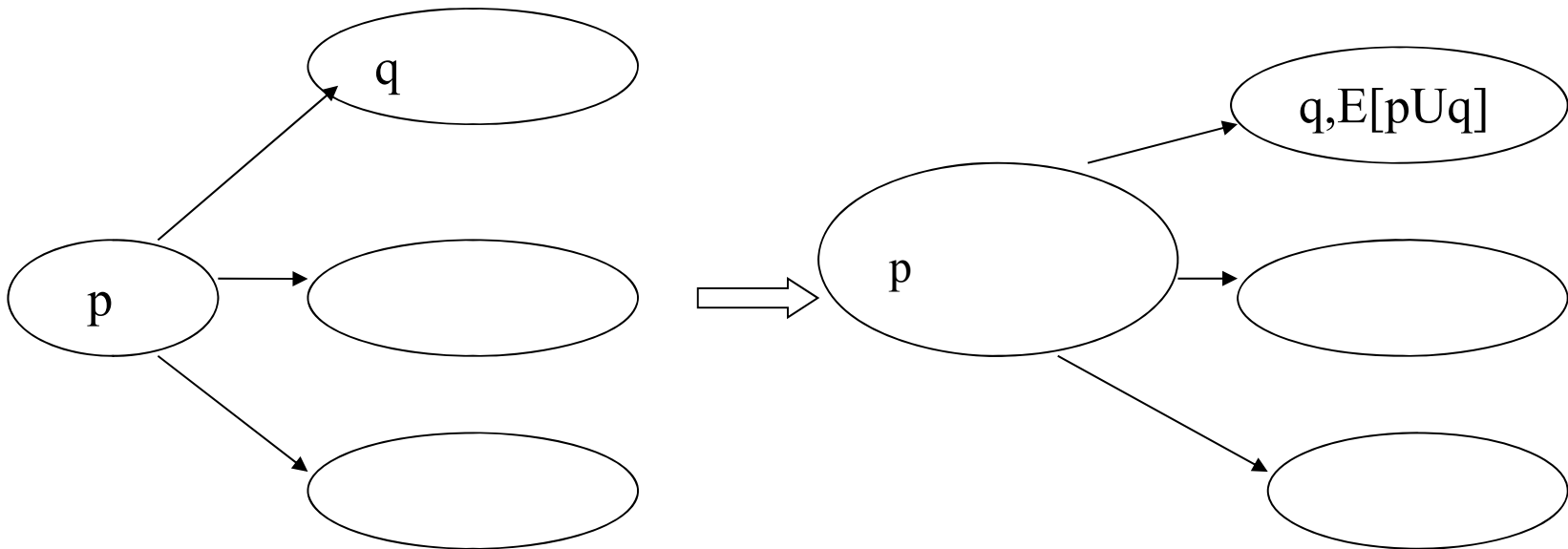
$$E[p \cup q] \equiv q \vee (p \wedge EX E[p \cup q])$$

Labeling Algorithm

$E(p \cup q)$

Function $SAT_{EU}(p, q)$

/ determines the set of states satisfying $E(p \cup q)$ */*



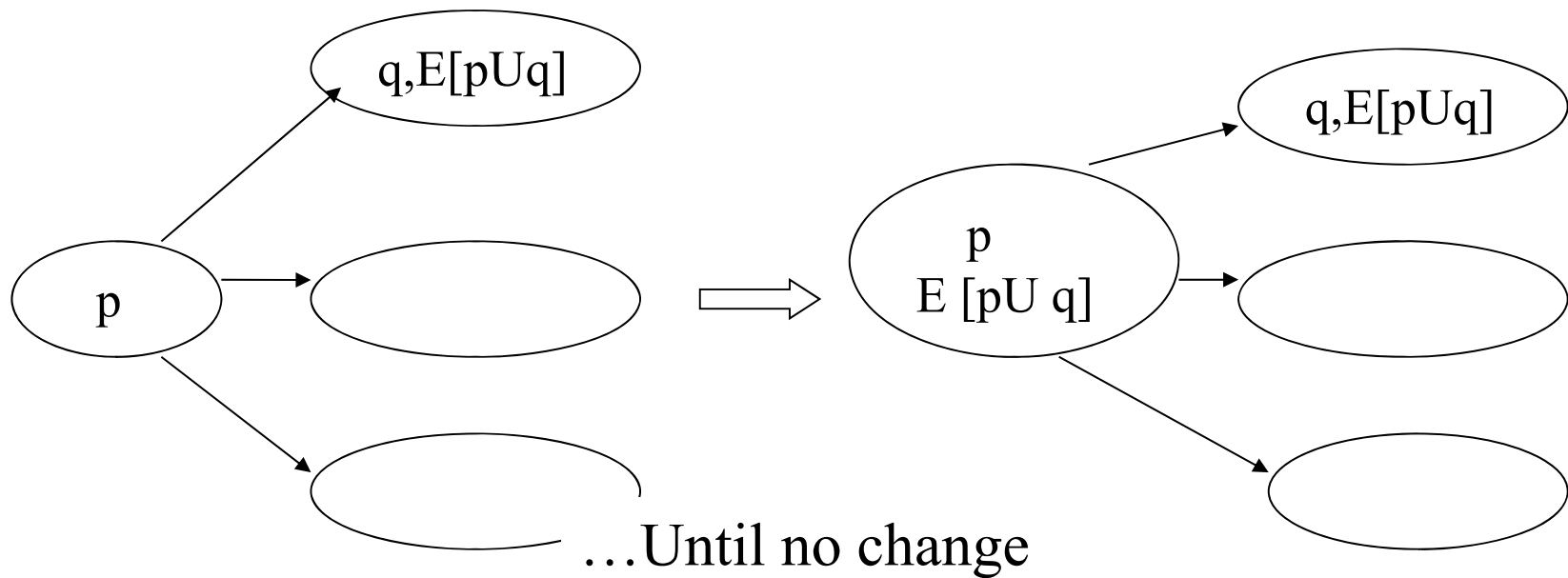
...Until no change

Labeling Algorithm

$E(p \cup q)$

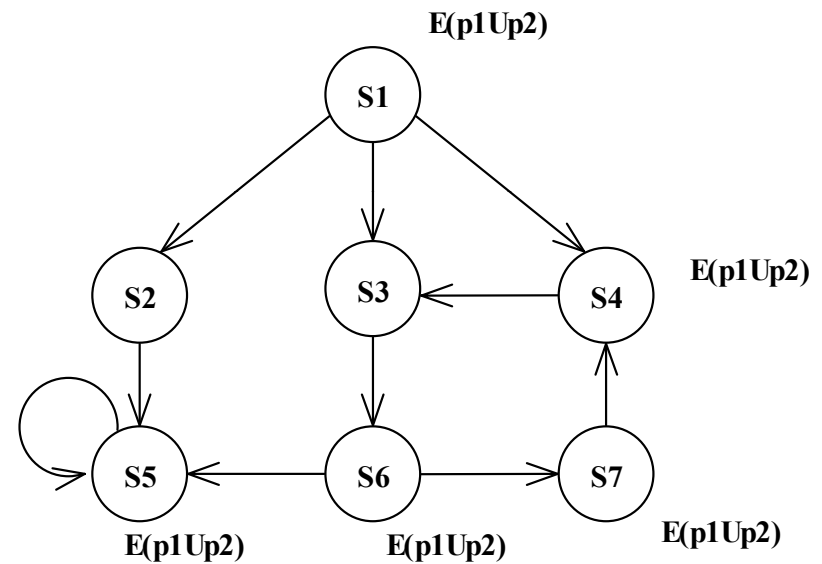
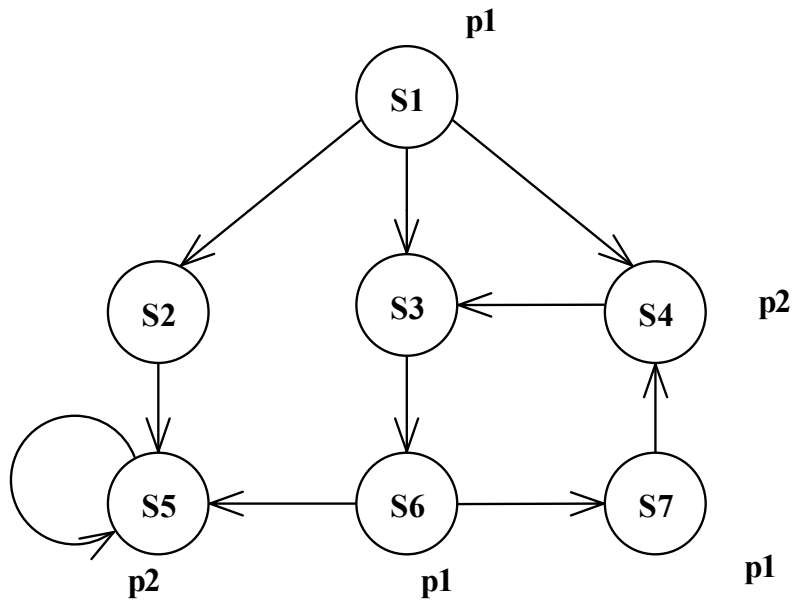
Function $SAT_{EU}(p, q)$

/* determines the set of states satisfying $E(p \cup q)$ */



Labeling Algorithm

$E(p1 \cup p2)$



Labeling Algorithm

$E(p \cup q)$

Function $SAT_{EU}(p,q)$

/ determines the set of states satisfying $E(p \cup q)$ */*

local var W, X, Y

begin

$W := SAT(p), X := S, Y := SAT(q)$

repeat until $X = Y$

begin

$X := Y$

$Y := Y \cup (W \cap \{s \mid \text{exists } s' \text{ such that } s \rightarrow s' \text{ and } s' \in Y\})$

end

return Y

end

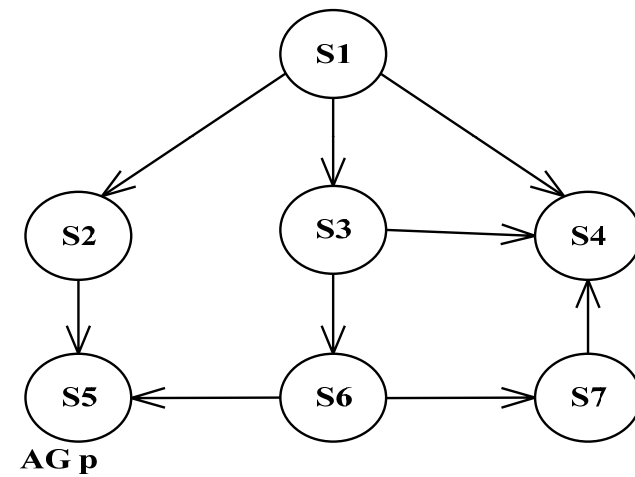
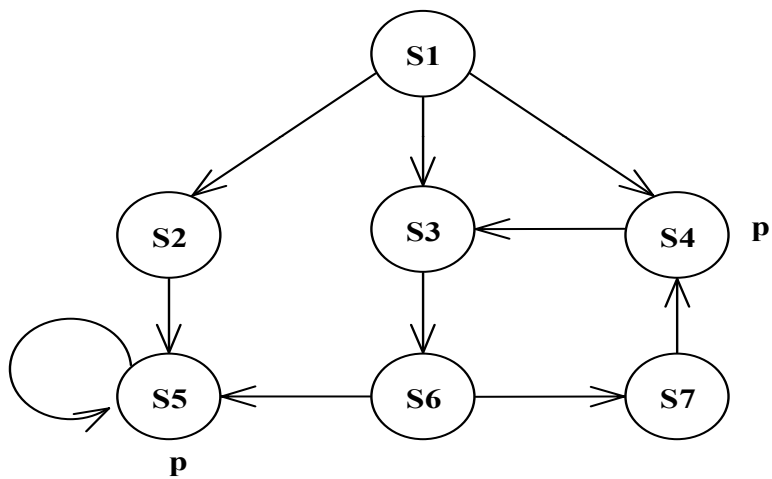
Labeling Algorithm AGp

Step1: Label all the states with AGp.

Step2: If any state s is not labeled with p , delete the label AGp.

Step3: Repeat: delete the label AGp from any state if all of its successors are not labeled with AGp until there is no change.

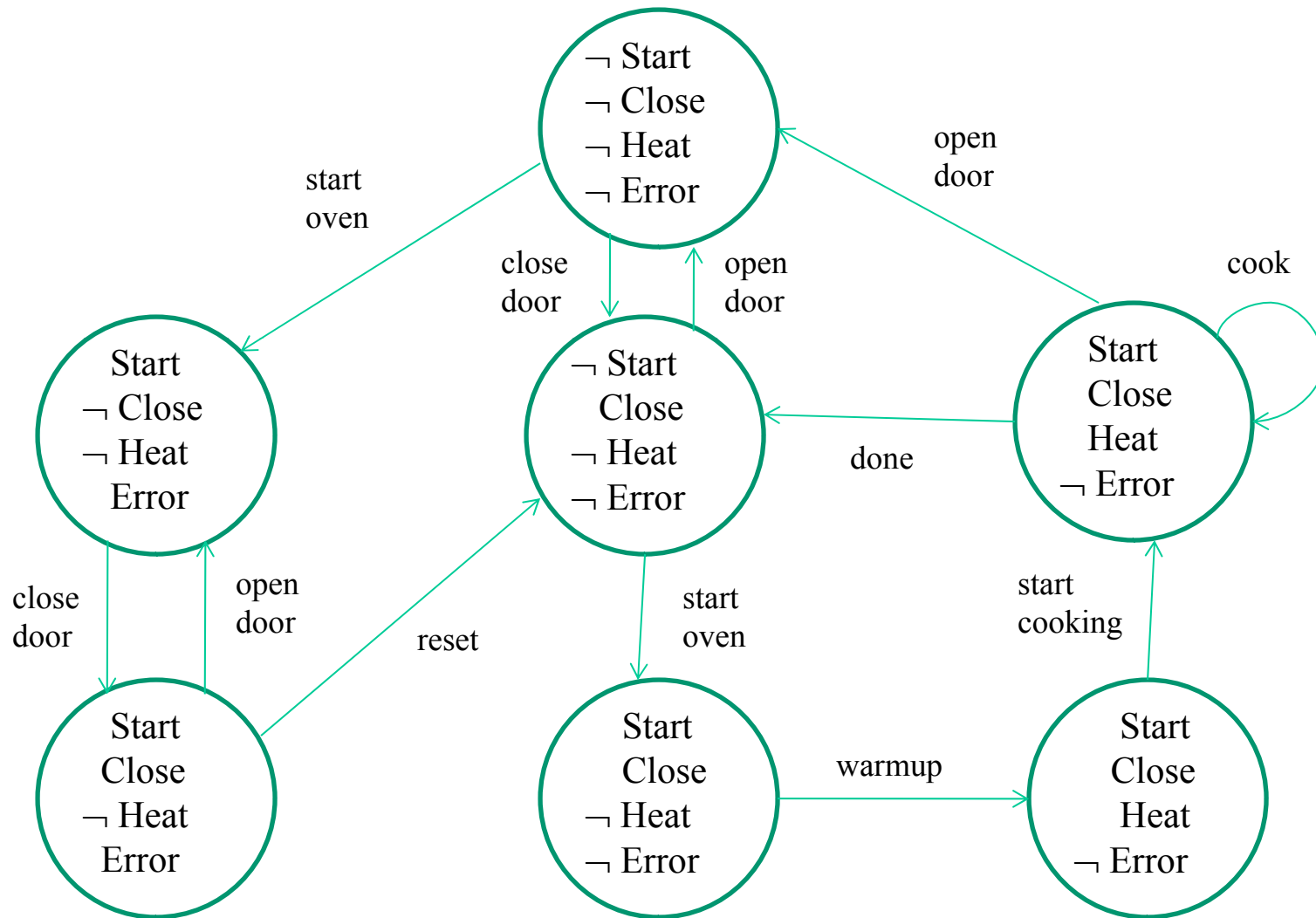
Labeling Algorithm AGp



Example

- Design a controller for a microwave oven.
 - Door of the oven: either open or close
 - Start of the oven
 - reset

Model



Conditions to be Checked

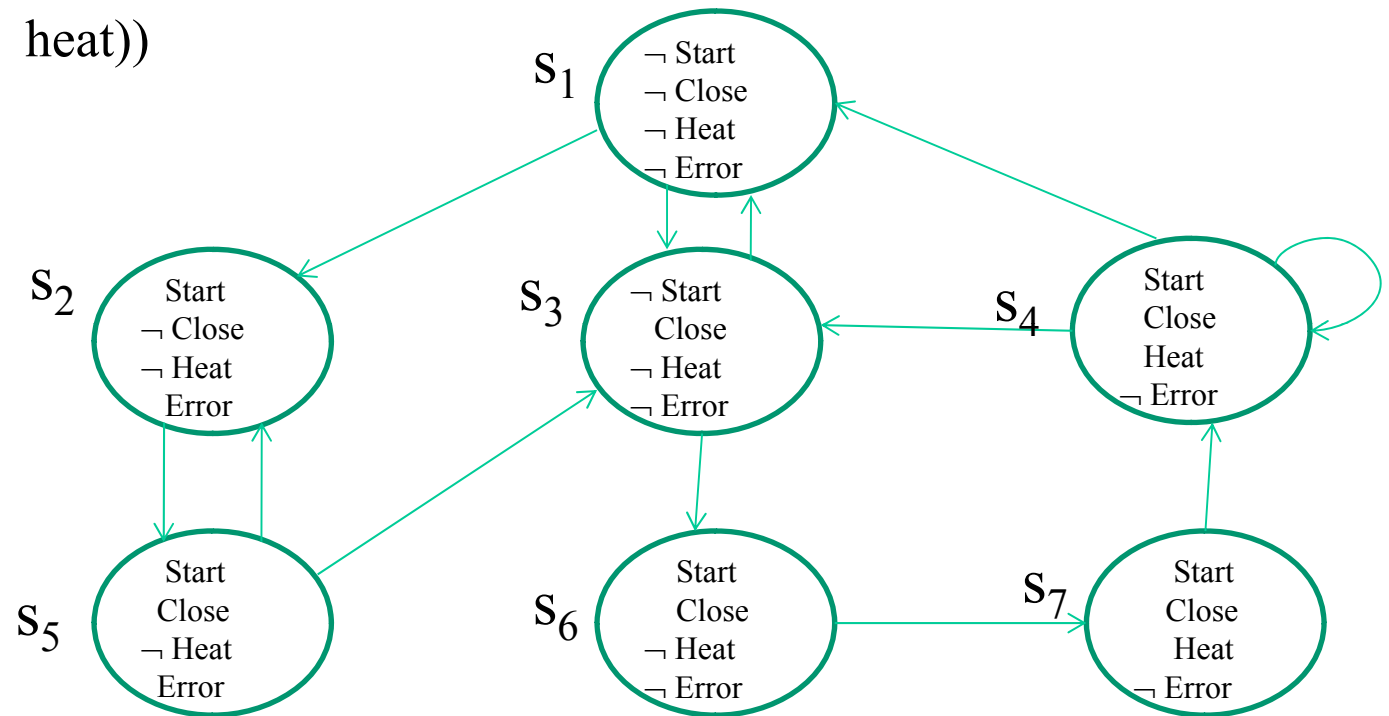
- Microwave oven should not heat up with its door open.
- Once we start the oven, eventually it must start heating.

CTL formulas for the conditions

- Microwave oven should not heat up with its door open.
 - $AG (\neg (\neg \text{close} \wedge \text{heat}))$
- Once we start the oven, eventually it must start heating.
 - $AG(\text{start} \rightarrow AF \text{ heat})$

Model Checking for the conditions

$AG (\neg (\neg \text{close} \wedge \text{heat}))$



$(\neg \text{close} \wedge \text{heat})$

$S1 = \{\}$

$\neg(\neg \text{close} \wedge \text{heat})$

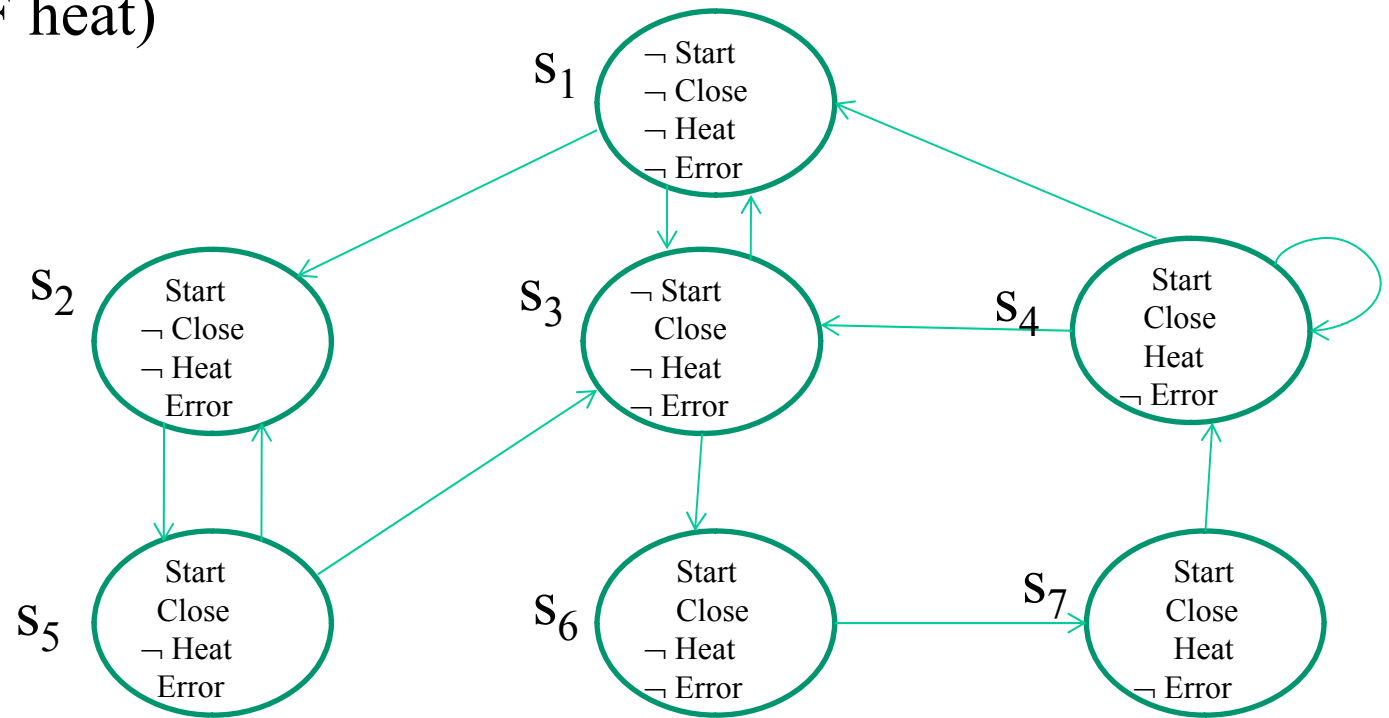
$S2 = S$

$AG (\neg (\neg \text{close} \wedge \text{heat}))$

$S2 = S$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$

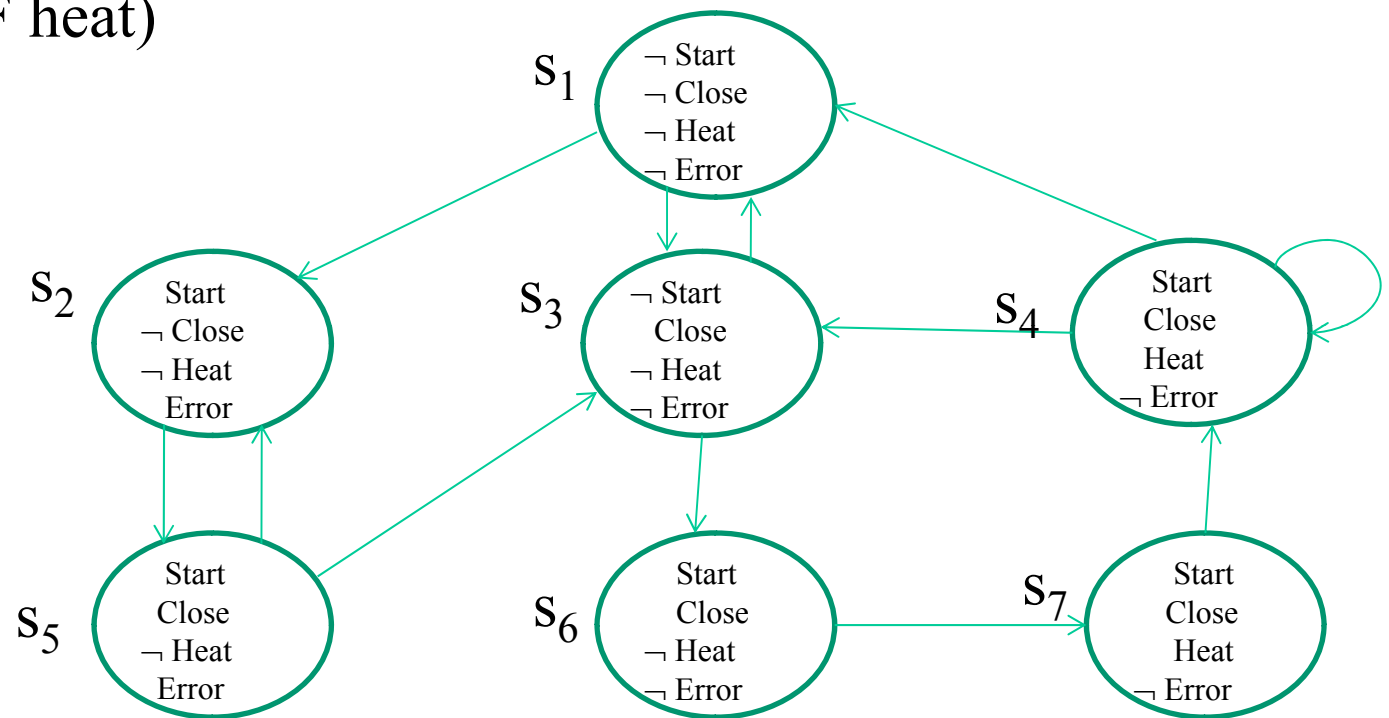


(heat)

$S1 = \{s_4, s_7\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



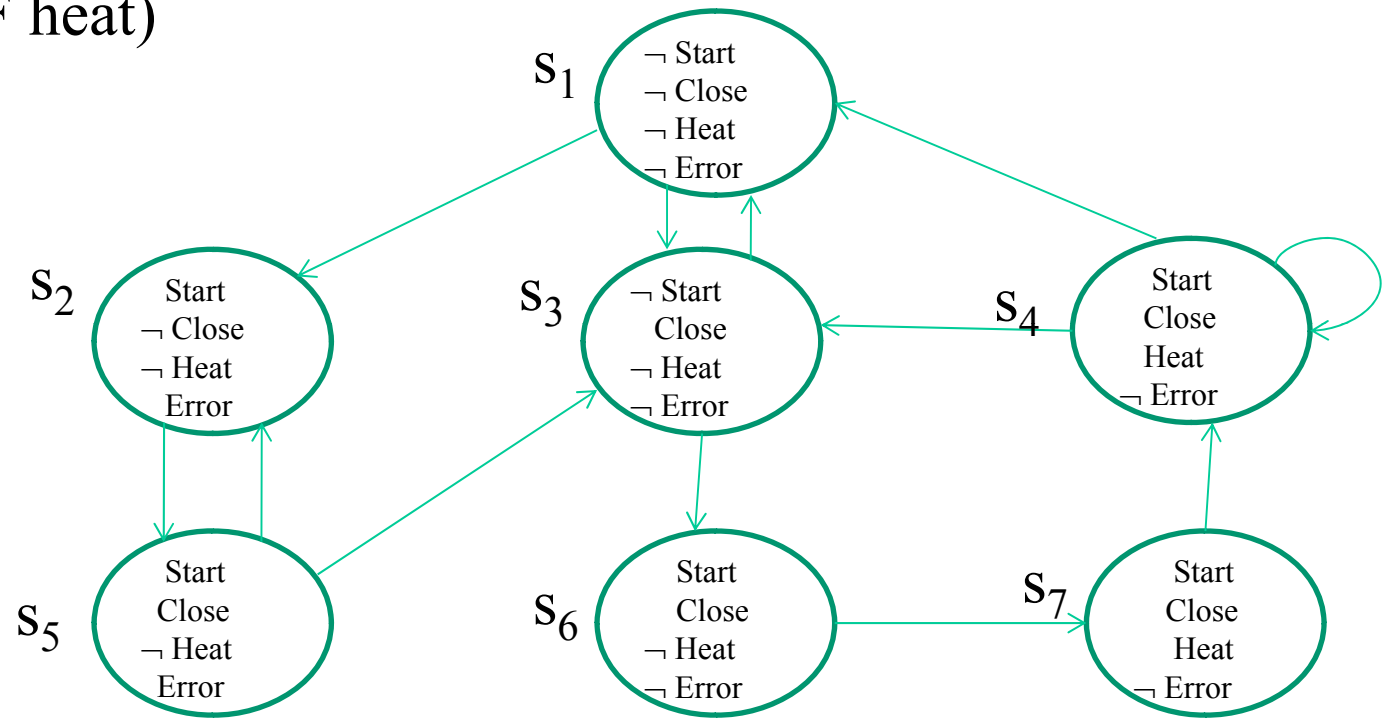
heat

$S1 = \{s_4, s_7\}$

AF heat

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



heat

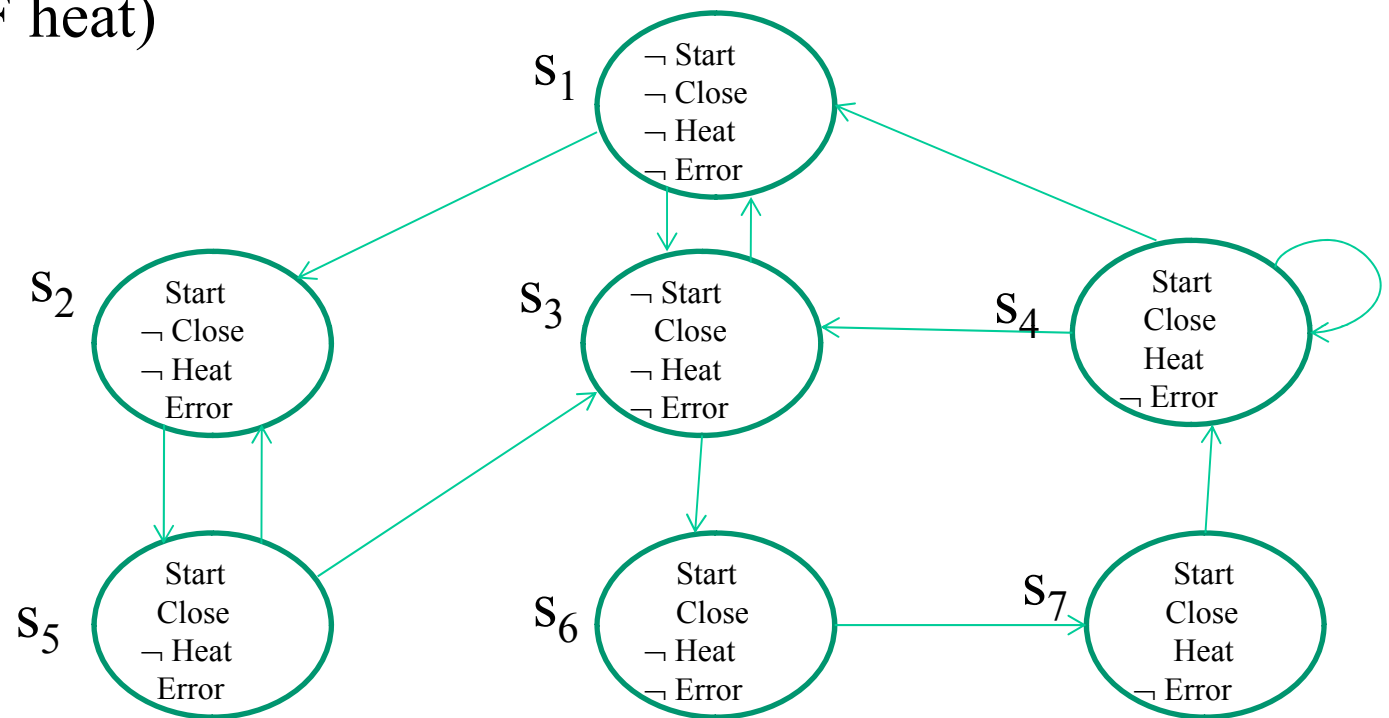
$S1 = \{s_4, s_7\}$

AF heat

$S1 = \{s_4, s_7\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



heat

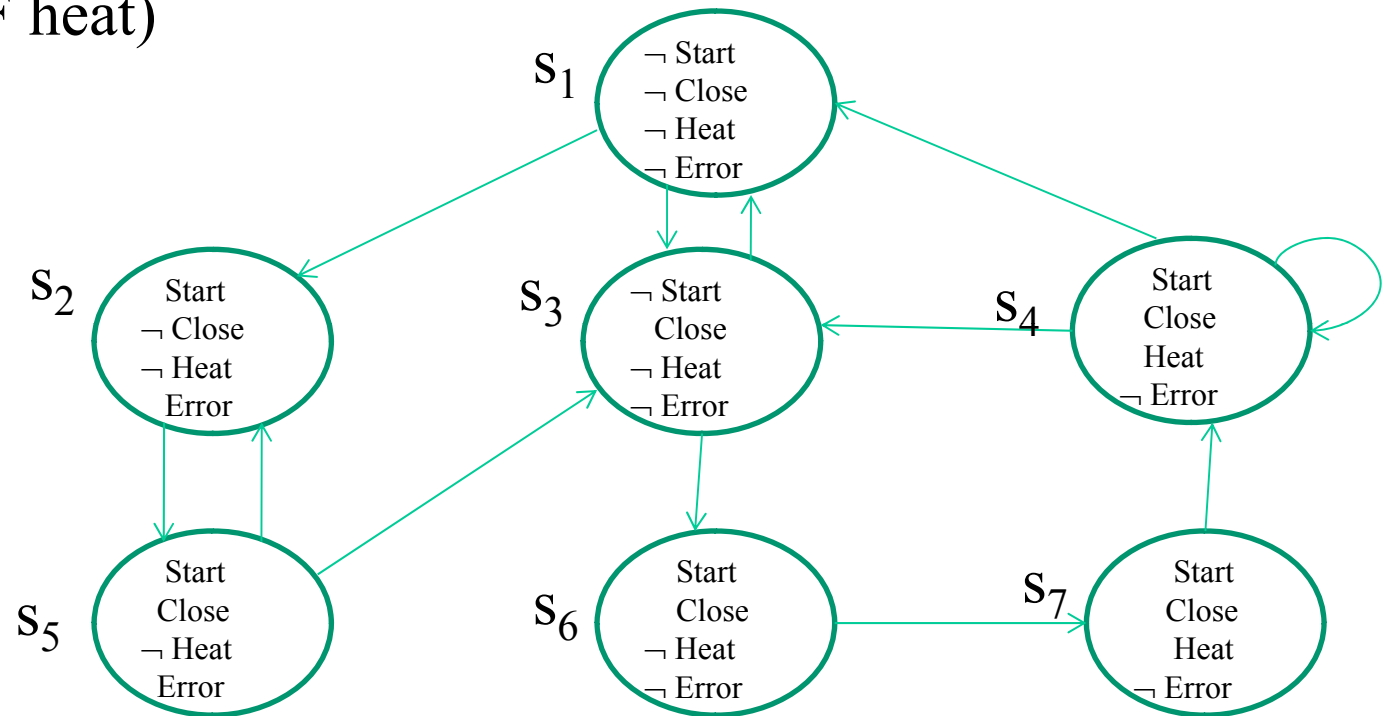
$$S1 = \{s_4, s_7\}$$

AF heat

$$S1 = \{s_4, s_7\} \quad S2 = \{s_4, s_7, s_6\}$$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



heat

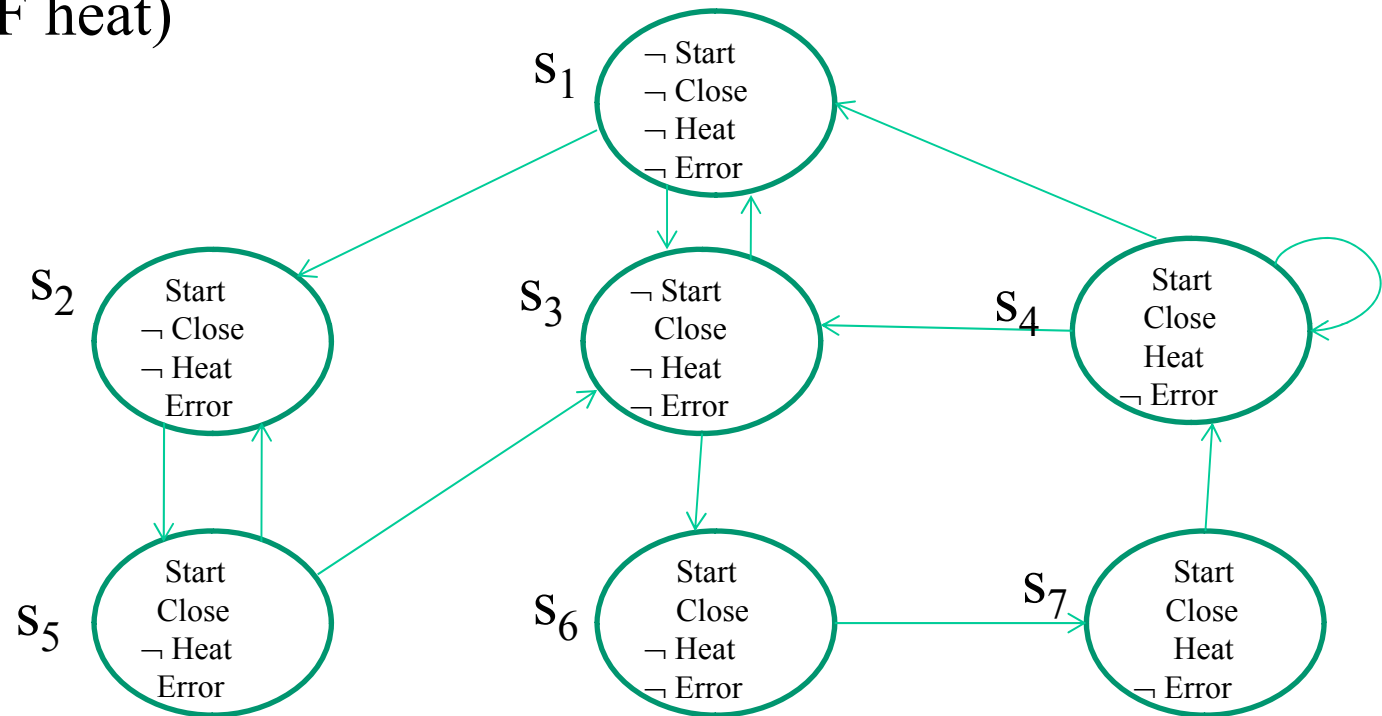
$$S1 = \{s_4, s_7\}$$

AF heat

$$S1 = \{s_4, s_7\} \quad S2 = \{s_4, s_7, s_6\} \quad S3 = \{s_4, s_7, s_6\}$$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



heat

$$S1 = \{s_4, s_7\}$$

AF heat

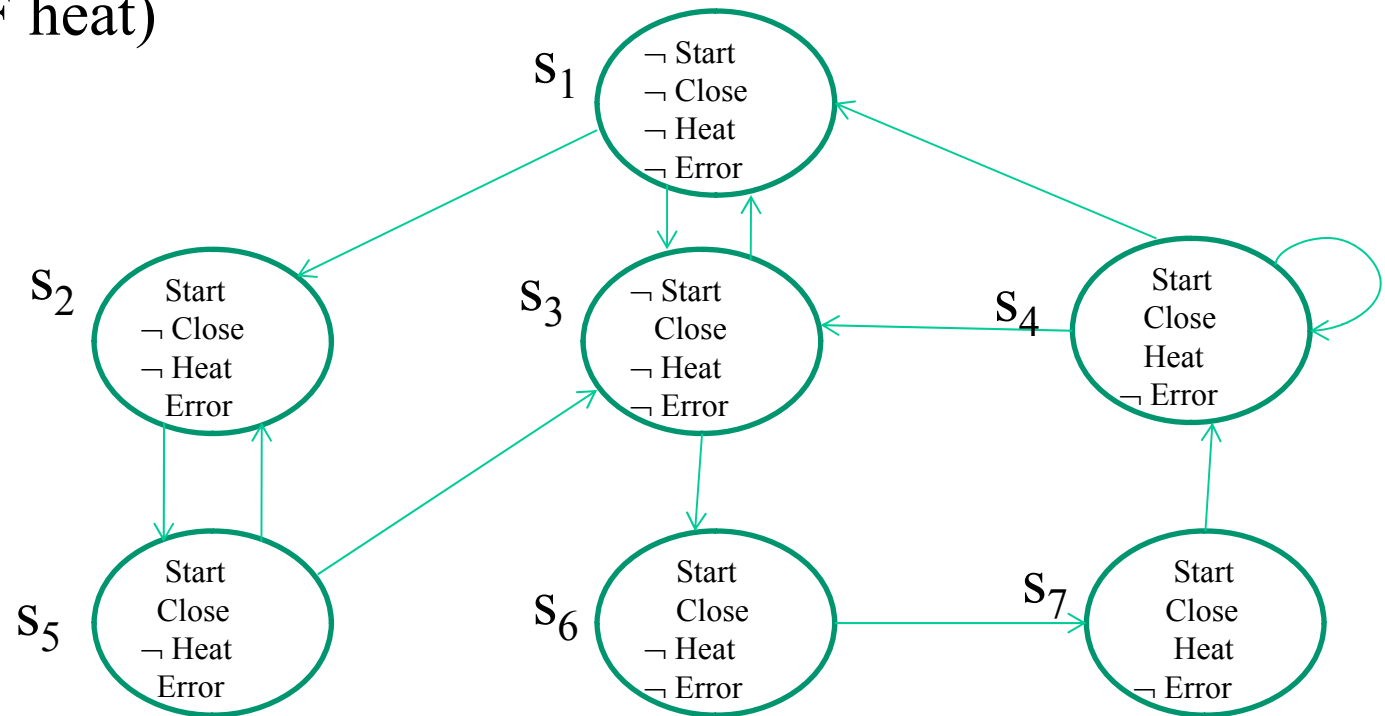
$$S2 = \{s_4, s_7, s_6\}$$

start

$$S3 = \{s_4, s_7, s_6, s_2, s_5\}$$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



heat

$S1 = \{s_4, s_7\}$

start

$S3 = \{s_4, s_7, s_6, s_2, s_5\}$

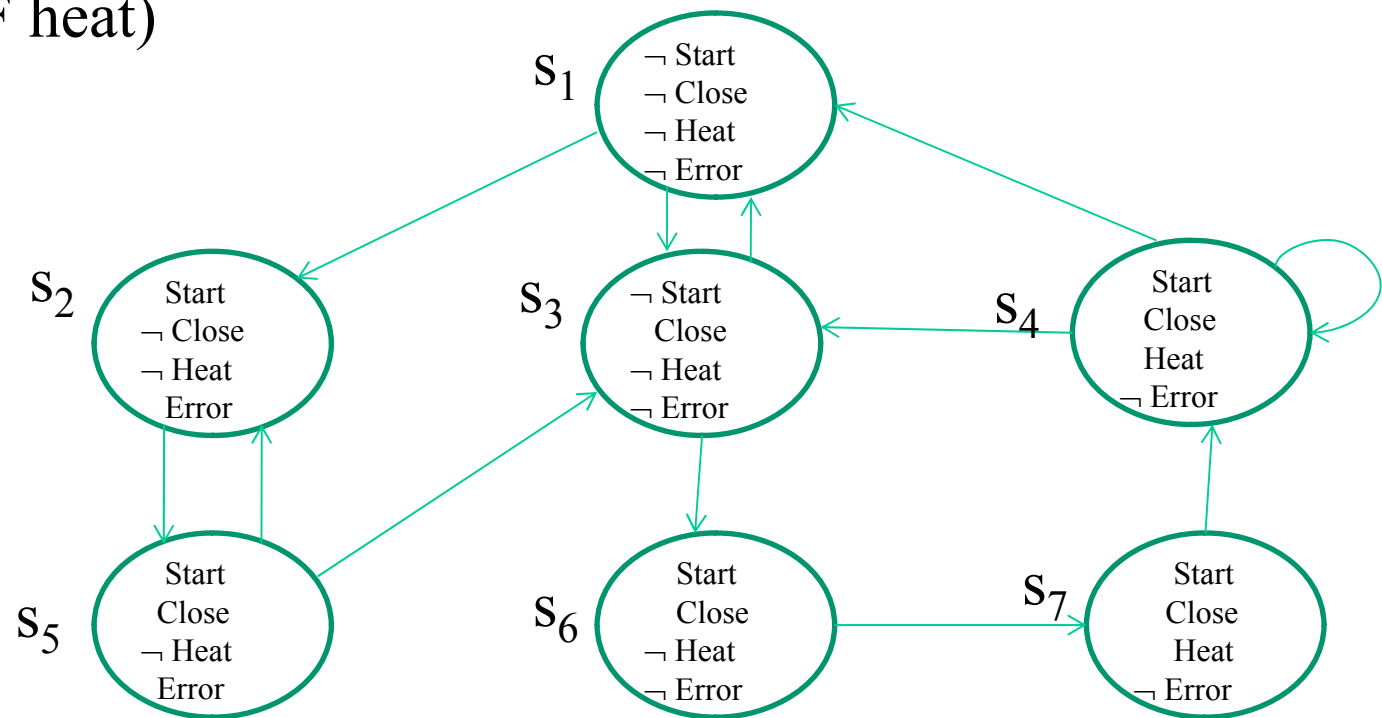
AF heat

$S2 = \{s_4, s_7, s_6\}$

$(start \rightarrow AF\ heat) \quad S4 = \{s_4, s_7, s_6, s_3, s_1\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$

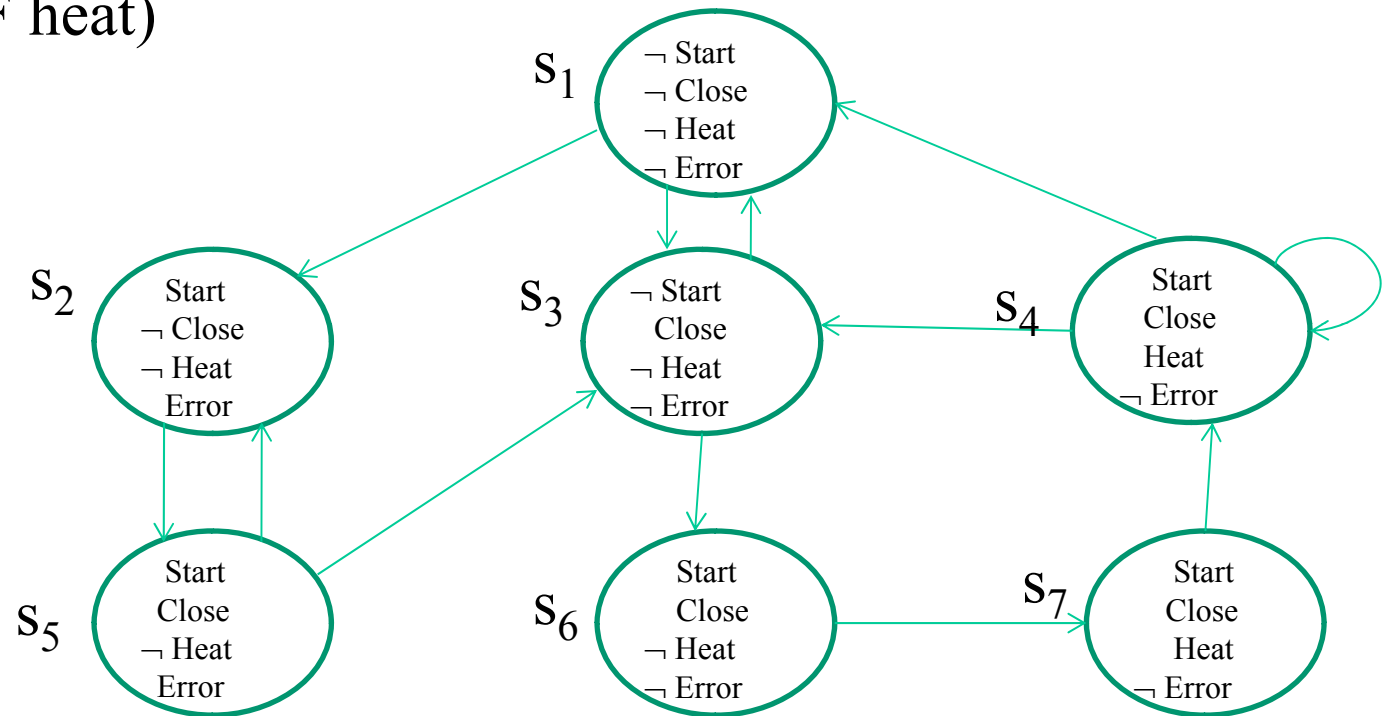


$AG(start \rightarrow AF\ heat)$ $S5 = \{s_4, s_7, s_6, s_3, s_1\}$

$S6 = \{s_4, s_7, s_6, s_3\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$

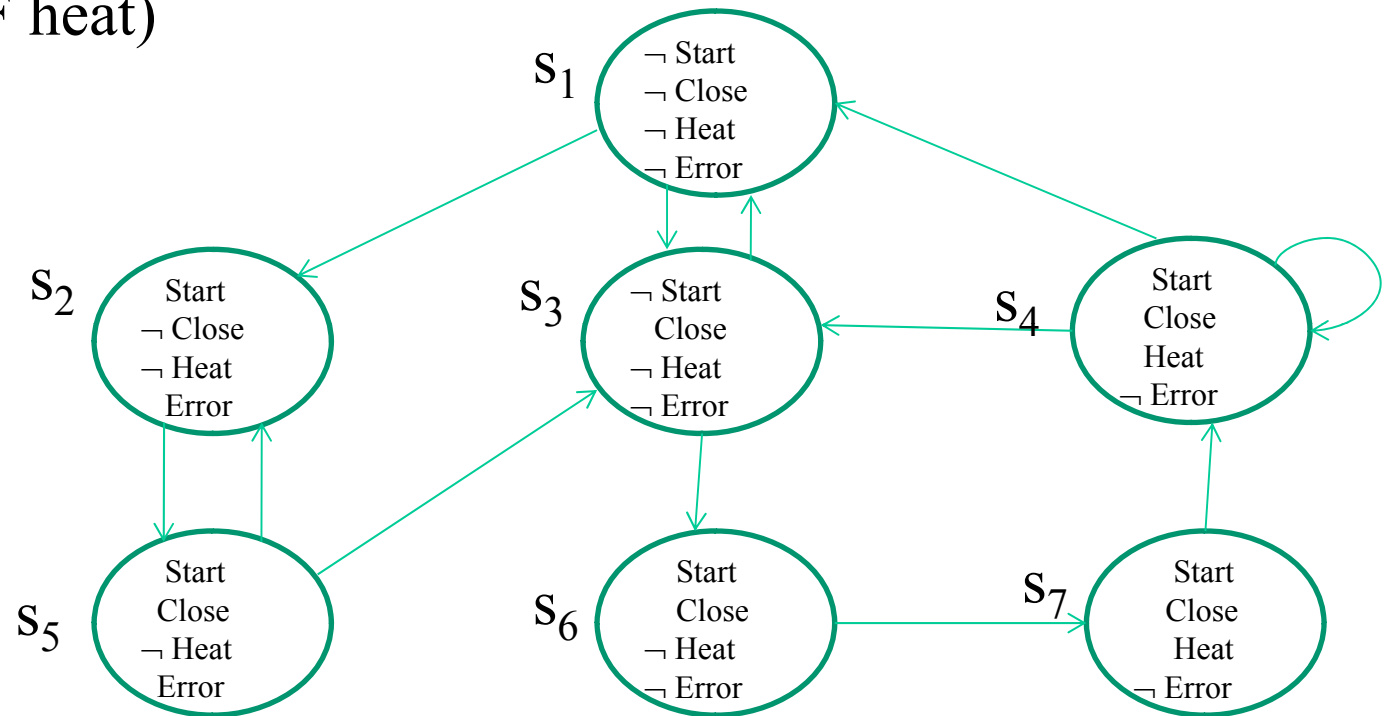


$AG(start \rightarrow AF\ heat)$ $S6 = \{s_4, s_7, s_6, s_3\}$

$S7 = \{s_4, s_7, s_6\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$

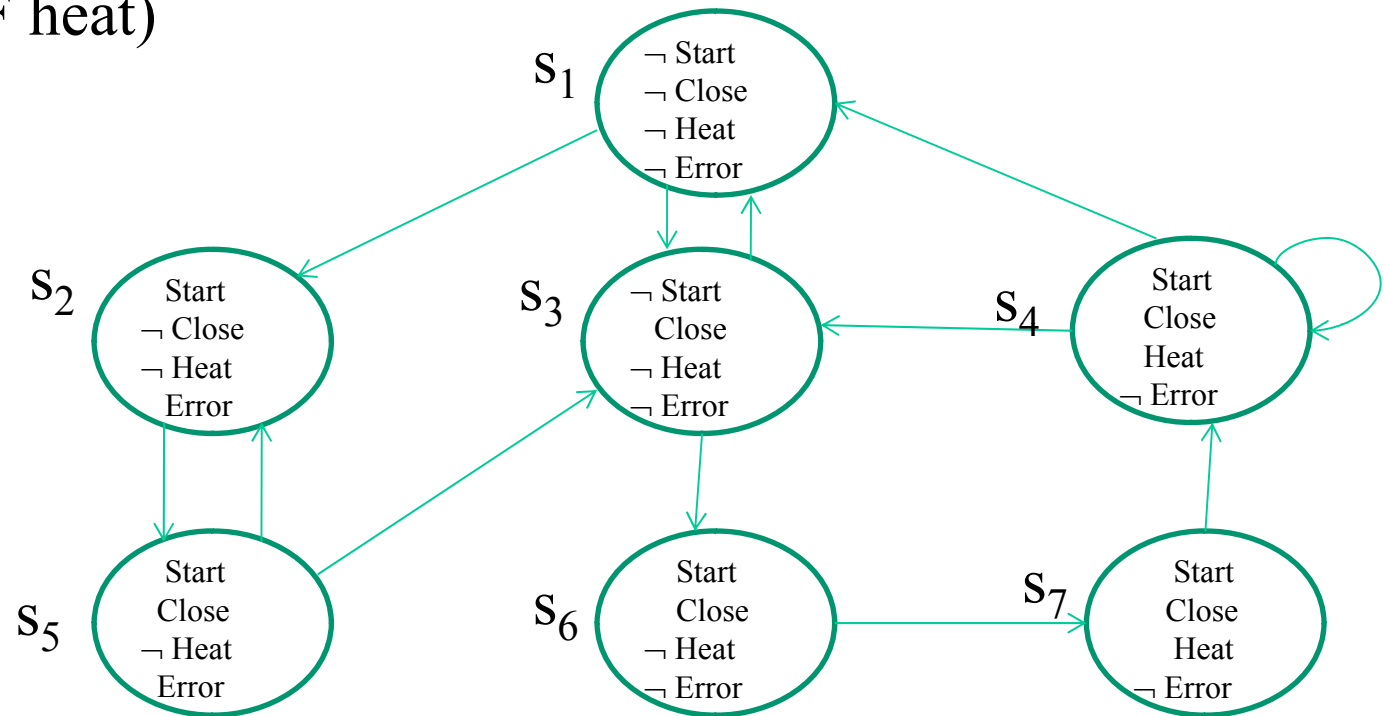


$AG(start \rightarrow AF\ heat) \quad S7 = \{s_4, s_7, s_6\}$

$S8 = \{s_7, s_6\}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$

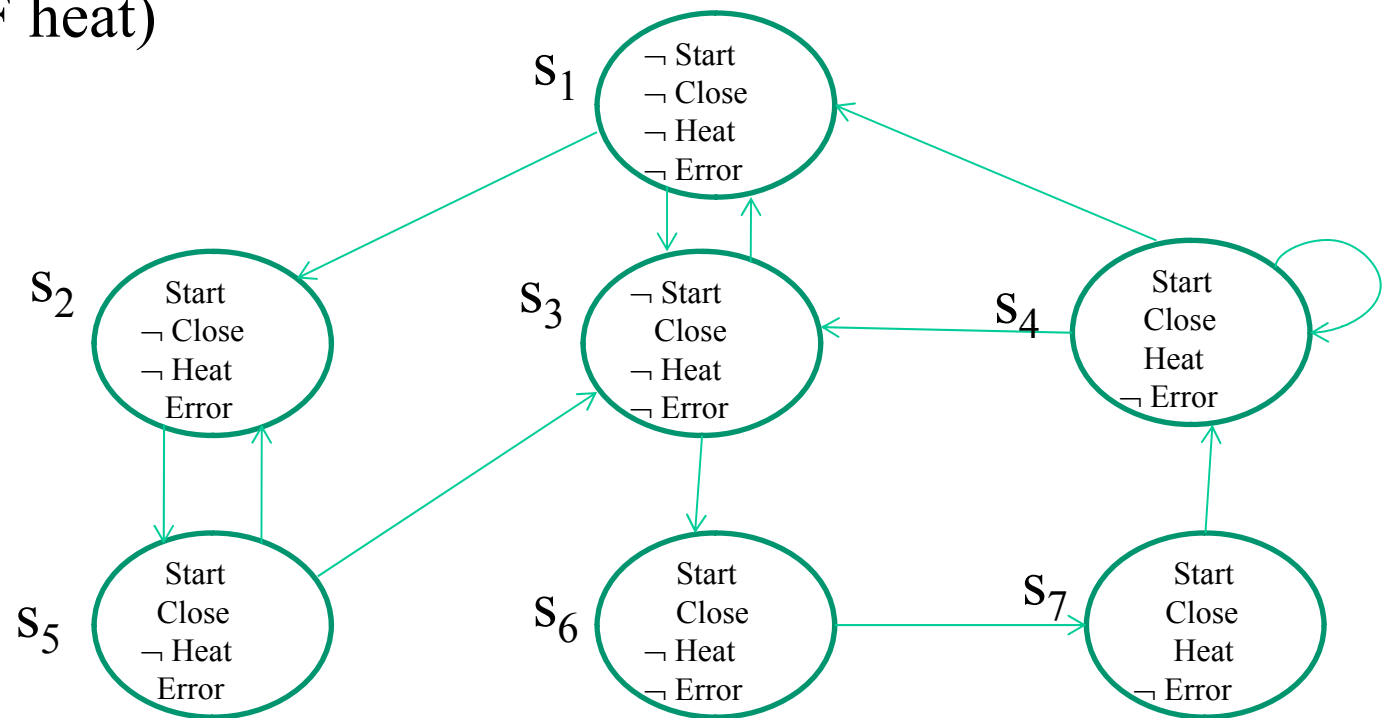


$AG(start \rightarrow AF\ heat)$ $S8 = \{ s_7, s_6 \}$

$S9 = \{ s_6 \}$

Model Checking for the conditions

$AG(start \rightarrow AF\ heat)$



$AG(start \rightarrow AF\ heat)$ $S9 = \{ s_6 \}$

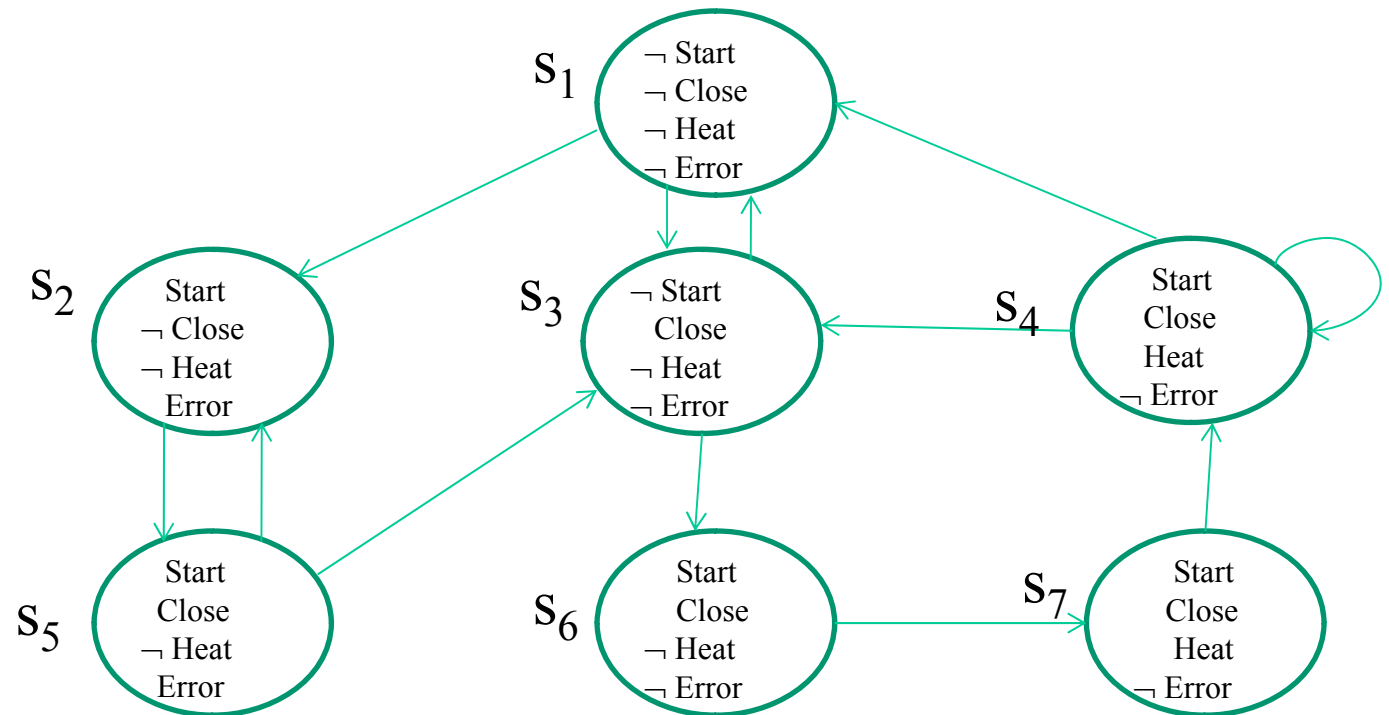
$S10 = \{ \}$

What if Error

- The given specification is not true
 - $\text{AG}(\text{start} \rightarrow \text{AF heat})$
- What to do
 - Revisit the design
 - Look for correct sequence of operation
 - $\text{AG}((\text{start} \wedge (\neg \text{error})) \rightarrow \text{AF}(\text{heat} \wedge (\neg \text{error})))$
 - $\text{AG}((\text{start} \wedge (\text{close})) \rightarrow \text{AF}(\text{heat} \wedge (\text{close})))$

Model Checking for the conditions

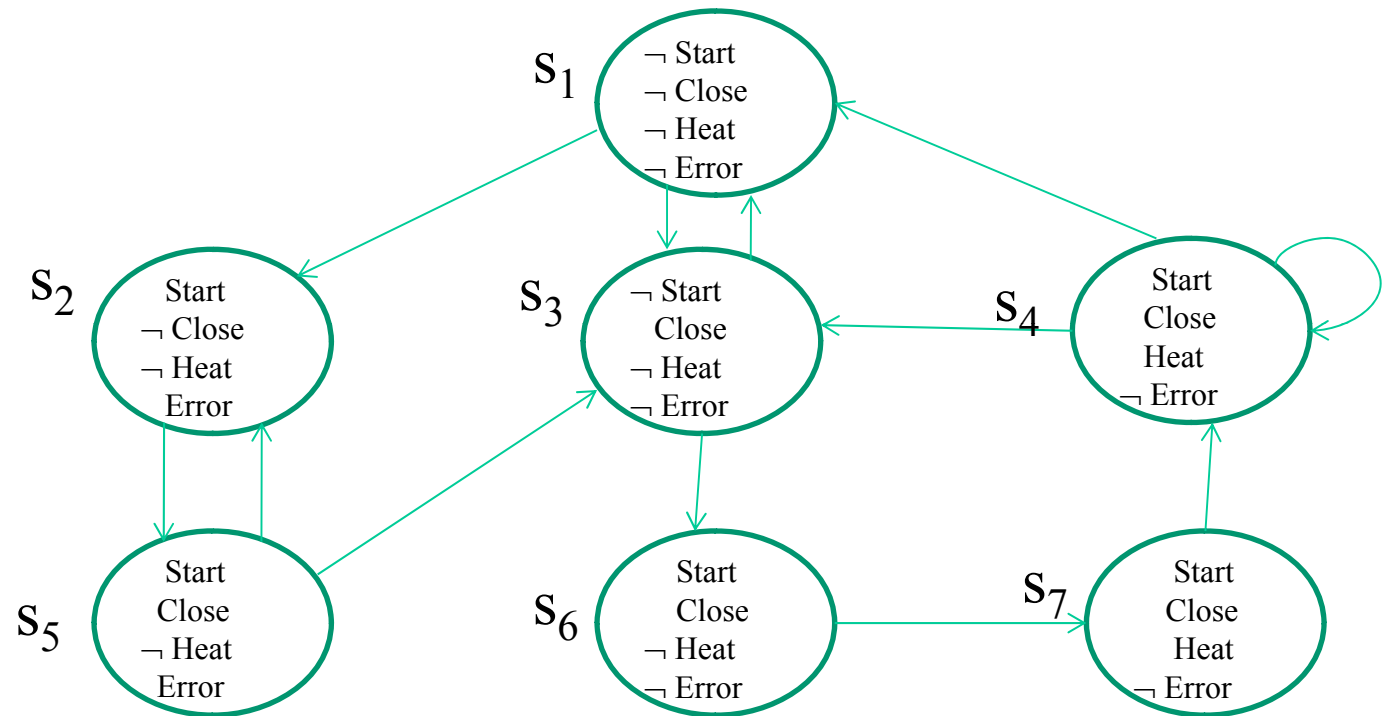
$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$



$heat \wedge (\neg error) \quad S1 = \{s_4, s_7\}$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

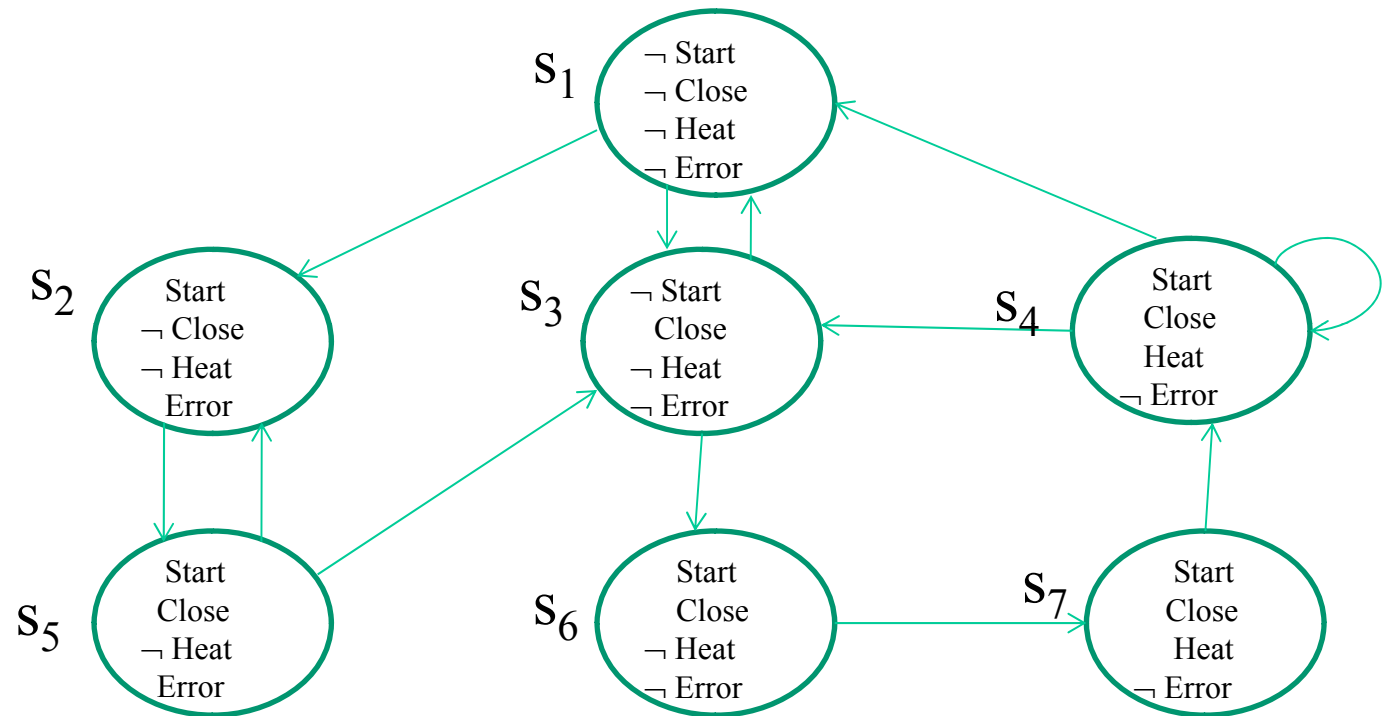


$heat \wedge (\neg error) \quad S1 = \{s_4, s_7\}$

$AF(heat \wedge (\neg error))$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

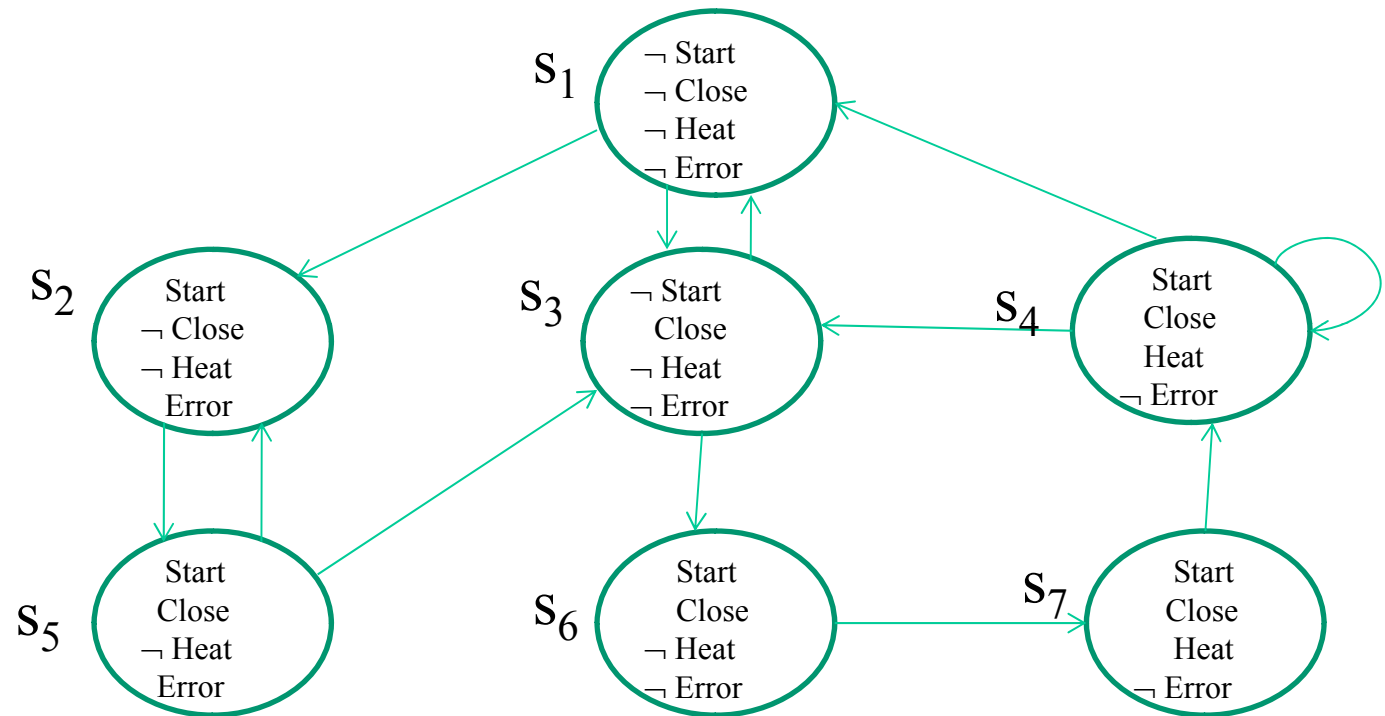


$heat \wedge (\neg error) \quad S1 = \{s_4, s_7\}$

$AF(heat \wedge (\neg error)) \quad S1 = \{s_4, s_7\}$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

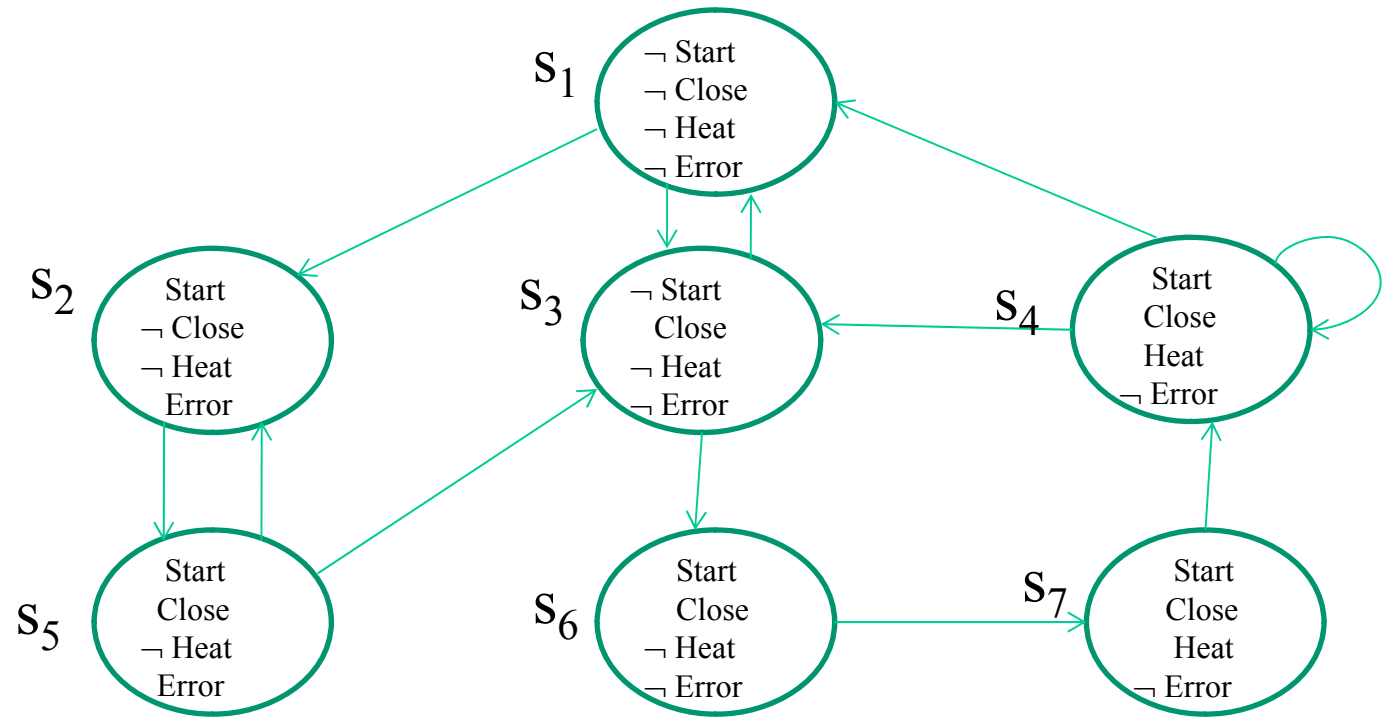


$heat \wedge (\neg error) \quad S1 = \{s_4, s_7\}$

$AF(heat \wedge (\neg error)) \quad S1 = \{s_4, s_7\} \quad S2 = \{s_4, s_7, s_6\}$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

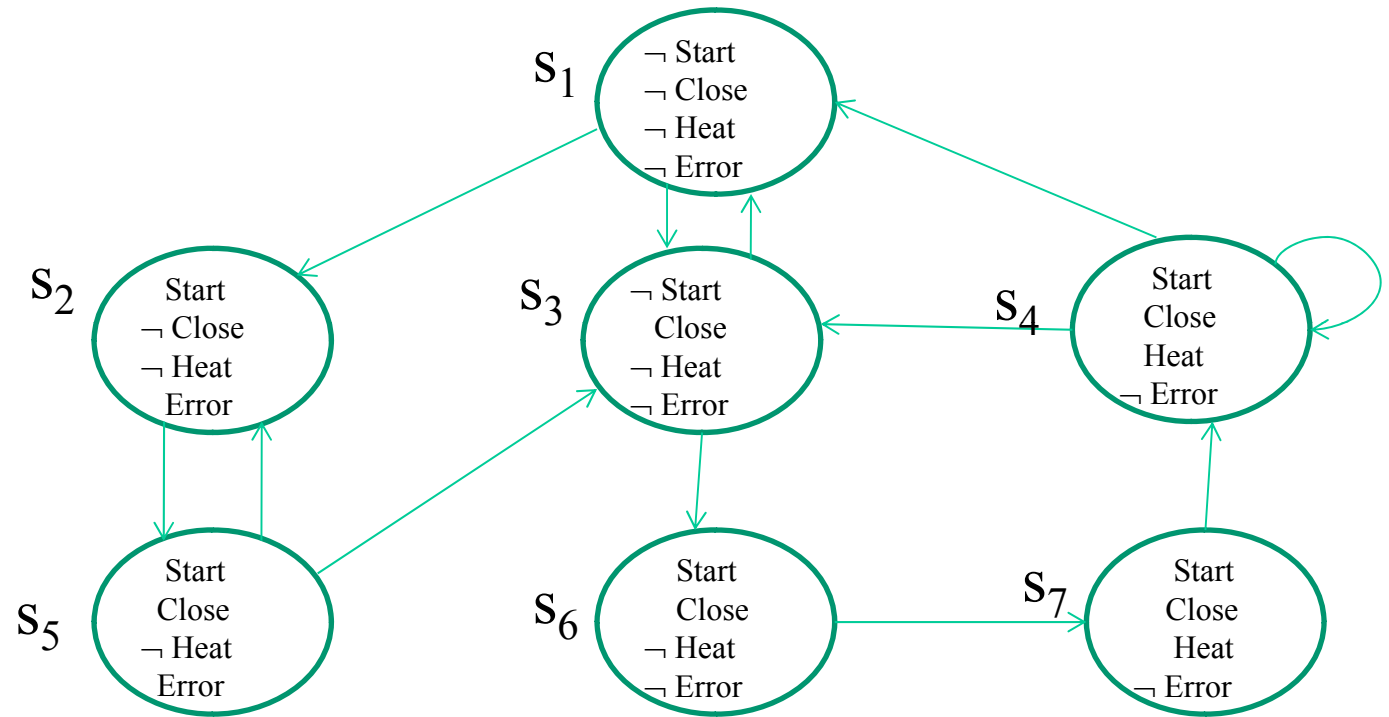


$heat \wedge (\neg error) \quad S1 = \{s_4, s_7\}$

$AF(heat \wedge (\neg error)) \quad S1 = \{s_4, s_7\} \quad S2 = \{s_4, s_7, s_6\} \quad S3 = \{s_4, s_7, s_6\}$

Model Checking for the conditions

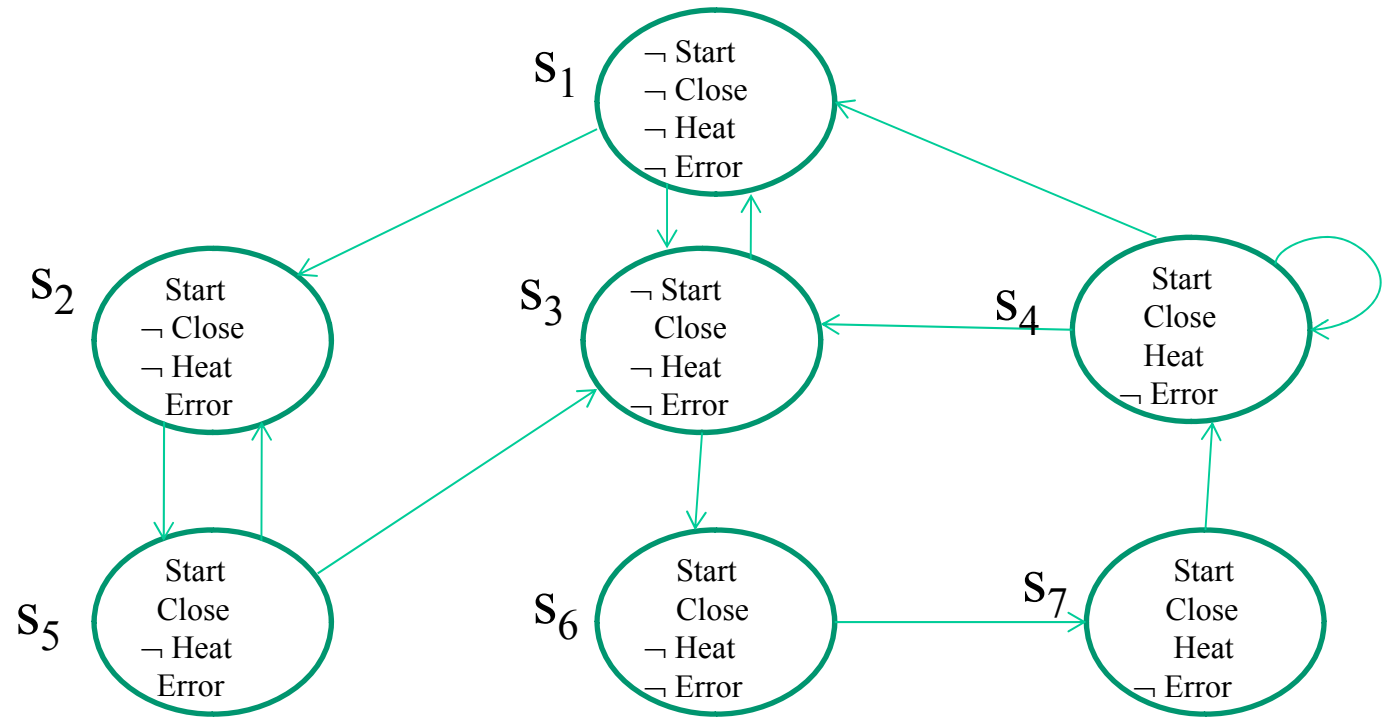
$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$



$(start \wedge (\neg error))$	$S1 = \{s_4, s_7\}$
$AF(heat \wedge (\neg error))$	$S2 = \{s_4, s_7, s_6\}$
$(start \wedge (\neg error))$	$S3 = \{s_4, s_7, s_6\}$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$



$(heat \wedge (\neg error)) \quad S1 = \{s_4, s_7\}$

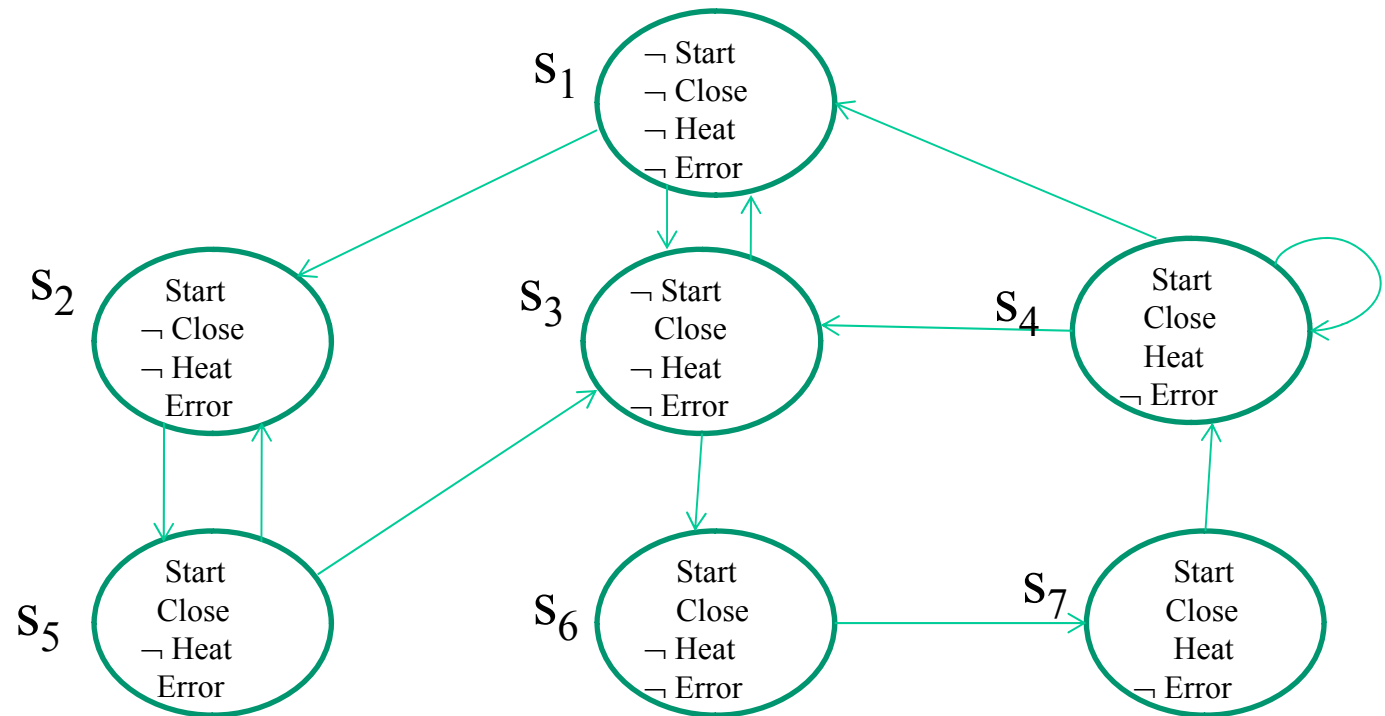
$(start \wedge (\neg error)) \quad S3 = \{s_4, s_7, s_6\}$

$AF(heat \wedge (\neg error)) \quad S2 = \{s_4, s_7, s_6\}$

$(start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)) \quad S4 = \{s_4, s_7, s_6, s_1, s_3, s_2, s_5\}$

Model Checking for the conditions

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

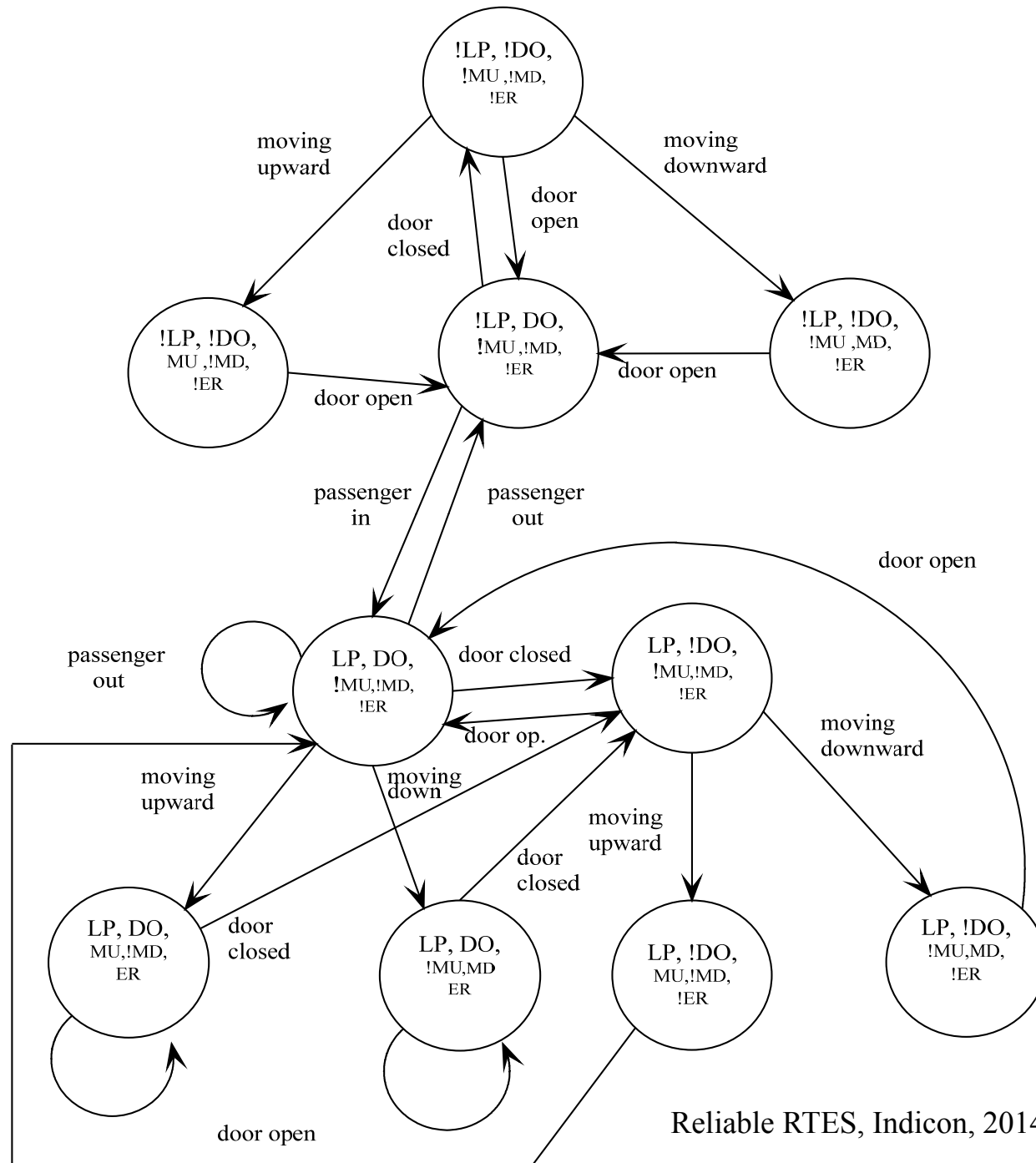


$S5 = \{s_4, s_7, s_6, s_1, s_3, s_2, s_5\}$

$AG((start \wedge (\neg error)) \rightarrow AF(heat \wedge (\neg error)))$

True in all states

Example 2: Lift Control



Reliable RTES, Indicon, 2014

Example 2: Lift Control

MU: elevator is moving in the upward direction.

MD: elevator is moving in the downward direction.

DO: door is open.

LP: elevator is loaded with passengers,

ER: some error occurred.

Model Checking for the conditions

- Specification:
 - **If the lift is moving then in the next state it will stop at somewhere and the door will open.**

$$(MU \wedge MD \wedge \sim ER) \rightarrow AX(DO \wedge \sim MU \wedge \sim MD \wedge \sim ER).$$

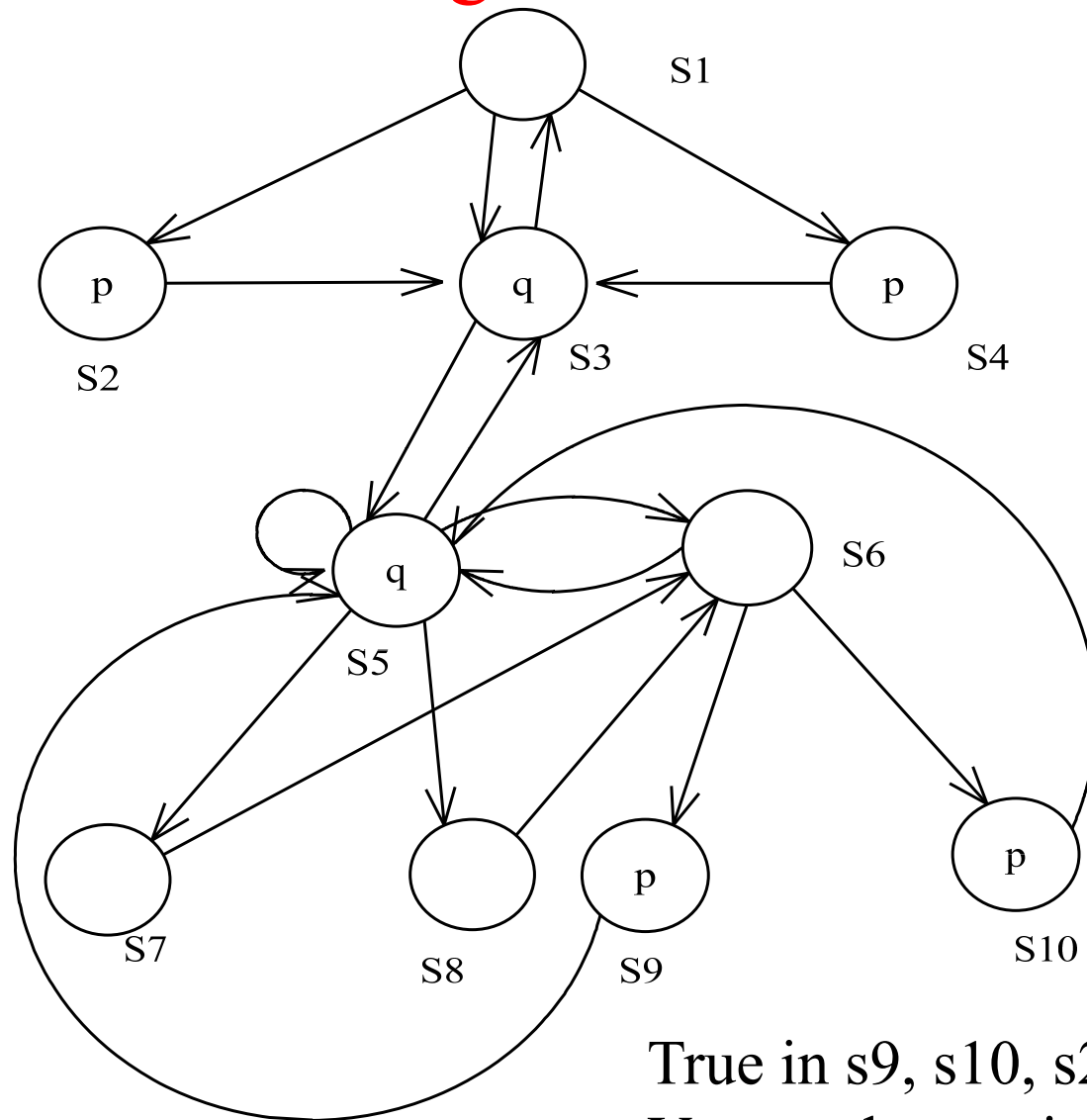
Model Checking for the conditions

$$(MU \wedge MD \wedge \neg ER) \rightarrow AX(DO \wedge \neg MU \wedge \neg MD \wedge \neg ER).$$

Let $p = (MU \wedge MD \wedge \neg ER)$, so $SAT(p) = \{S2, S4, S9, S10\}$.

$q = (DO \wedge \neg MU \wedge \neg MD \wedge \neg ER)$ So $SAT(q)$ is $\{S3, S5\}$.

Model Checking for the conditions



True in s9, s10, s2, s4
Vacuously true in all other states₈₃

Model Checking for the conditions

- Specification:
 - Always it is possible to reach the state where the lift is static, doors are closed and waiting for the passenger.

$p = EF(\sim DO \ \sim LP \ \sim MU \ \sim MD \ \sim ER)$

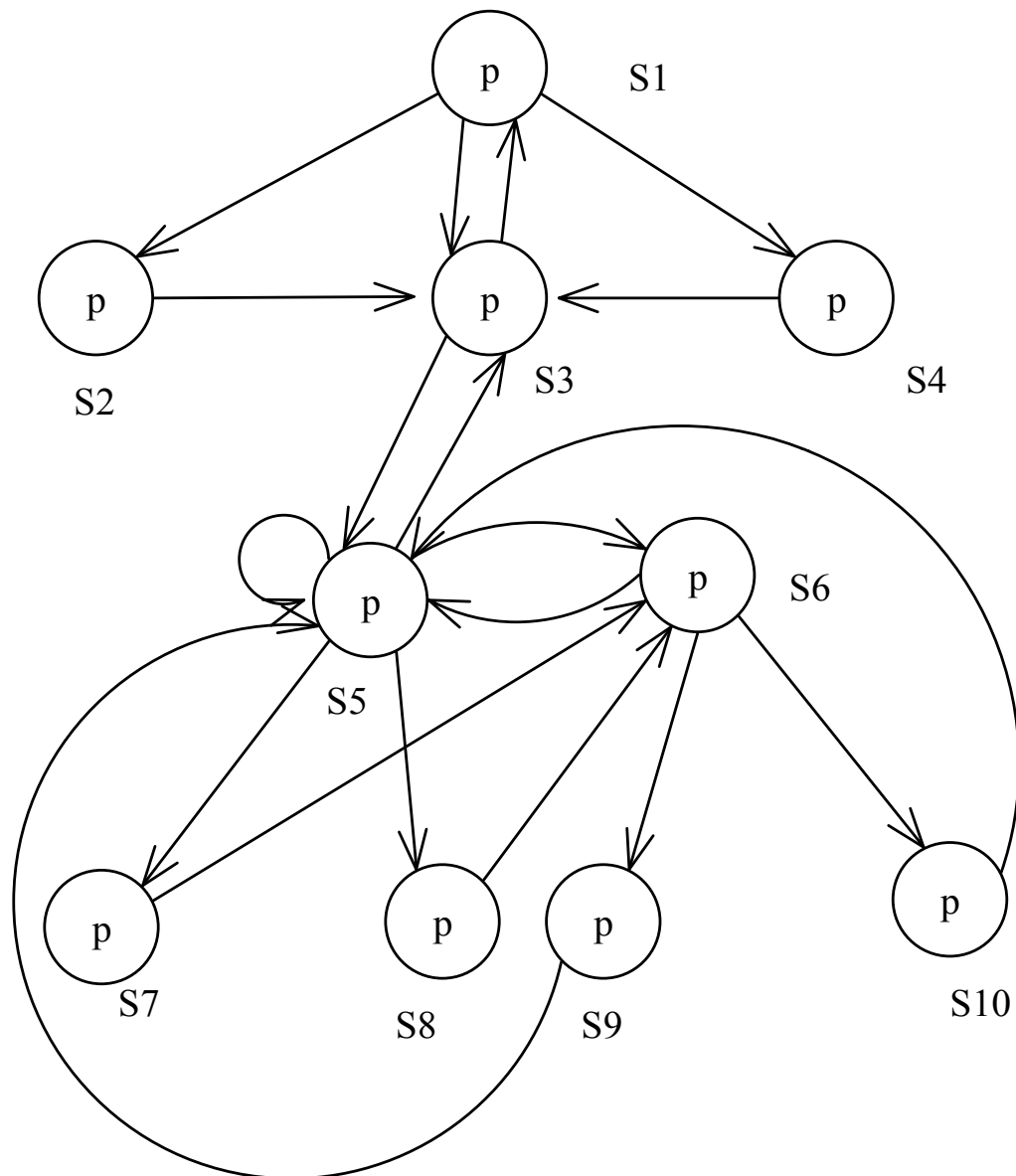
Model Checking for the conditions

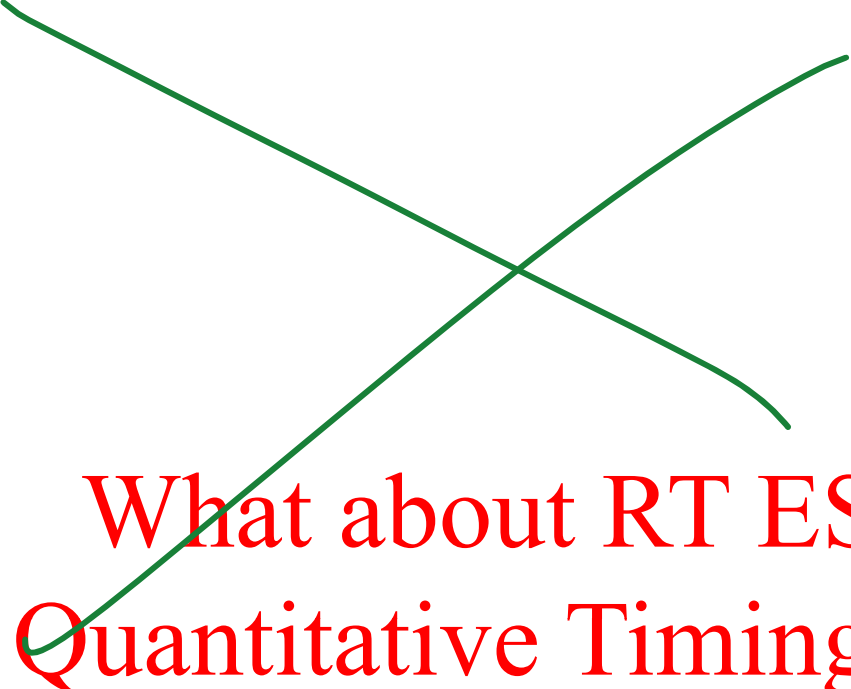
- Here, $Y = \text{SAT}((\sim\text{DO} \ \sim\text{LP} \ \sim\text{MU} \ \sim\text{MD} \ \sim\text{ER}) = \{S1\}$
- Loop 1
 - $X = Y = \{S1\}$
 - $Y = Y \cup \{S3\} = \{S1, S3\}$
- Loop 2
 - $X = Y = \{S1, S3\}$
 - $Y = Y \cup \{S1, S3, S2, S4, S5\} = \{S1, S2, S3, S4, S5\}$

Model Checking for the conditions

- Loop 3
 - $X = Y = \{S1, S2, S3, S4, S5\}$
 - $Y = Y \cup \{S1, S2, S3, S4, S5, S6, S9, S10\} = \{S1, S2, S3, S4, S5, S6, S9, S10\}$
- Loop 4 -
 - $X = Y = \{S1, S2, S3, S4, S5, S6, S9, S10\}$
 - $Y = Y \cup \{S1, S2, S3, S4, S5, S6, S9, S10, S7, S8\} = \{S1, S2, S3, S4, S5, S6, S7, S8, S9, S10\}$
- So we have $EF(\sim DO \sim LP \sim MU \sim MD \sim ER) = \{S1, S2, S3, S4, S5, S6, S7, S8, S9, S10\}$. Let $p = EF(\sim DO \sim LP \sim MU \sim MD \sim ER)$. So, p is satisfied in every state of the model

Model Checking for the conditions





What about RT ES ?
Quantitative Timings!!!
RTCTL and PRTCTL logic

Characteristics of RTCTL logic

What can RTCTL be applied to?

RTCTL logic may be applied to specify time dependencies not only qualitatively (like in CTL), but also quantitatively:

- To give a maximal time distance between two propositions on every /some path of states.
- Globally in future p holds, within k time units $AF^{\leq k} p$

Syntax and Semantics of RTCTL logic

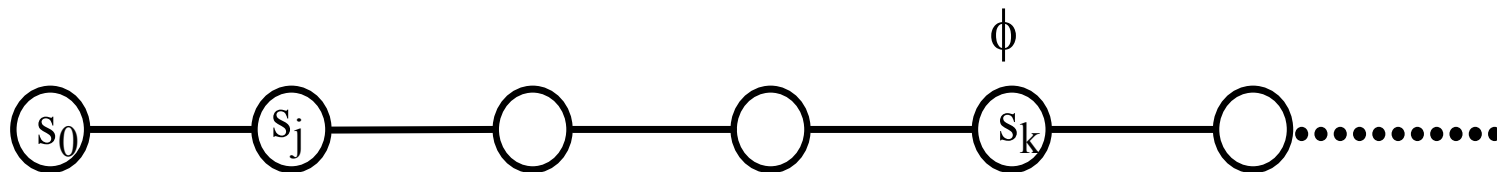
- Time structure (Kripke structure) of RTCTL
- Time bounds for temporal operators

RTTCL Temporal Operator

Future : $F \preceq^l \varphi$

$$(M, S_j) \models F \preceq^l \varphi \Leftrightarrow \exists k, k \geq j, k-j \leq l, (M, S_k) \models \varphi$$

state S_j satisfies as future state S_k satisfies φ and S_k lies within l (in this example 3) time units of S_j .



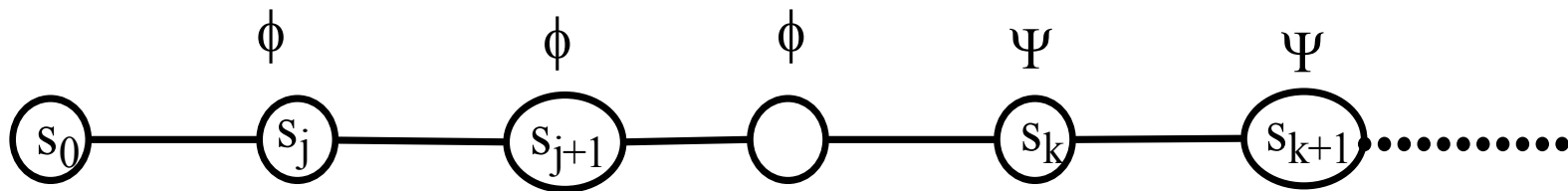
RTCTL Temporal Operator

Until : $(\Psi \text{ U}^{\leq l} \varphi)$

$(M, S_j) \models \Psi \text{ U} \varphi \Leftrightarrow \exists k, 0 \leq k (M, S_k) \models \varphi \text{ and}$

$\forall k, j < k (M, S_j) \models \Psi \text{ where } k-j \leq l$

S_j satisfies $(\Psi \text{ U}^{\leq 3} \varphi)$ because Ψ is true for all states S_i $j \leq i < k$ and then φ is true for state S_k .



Some Tools

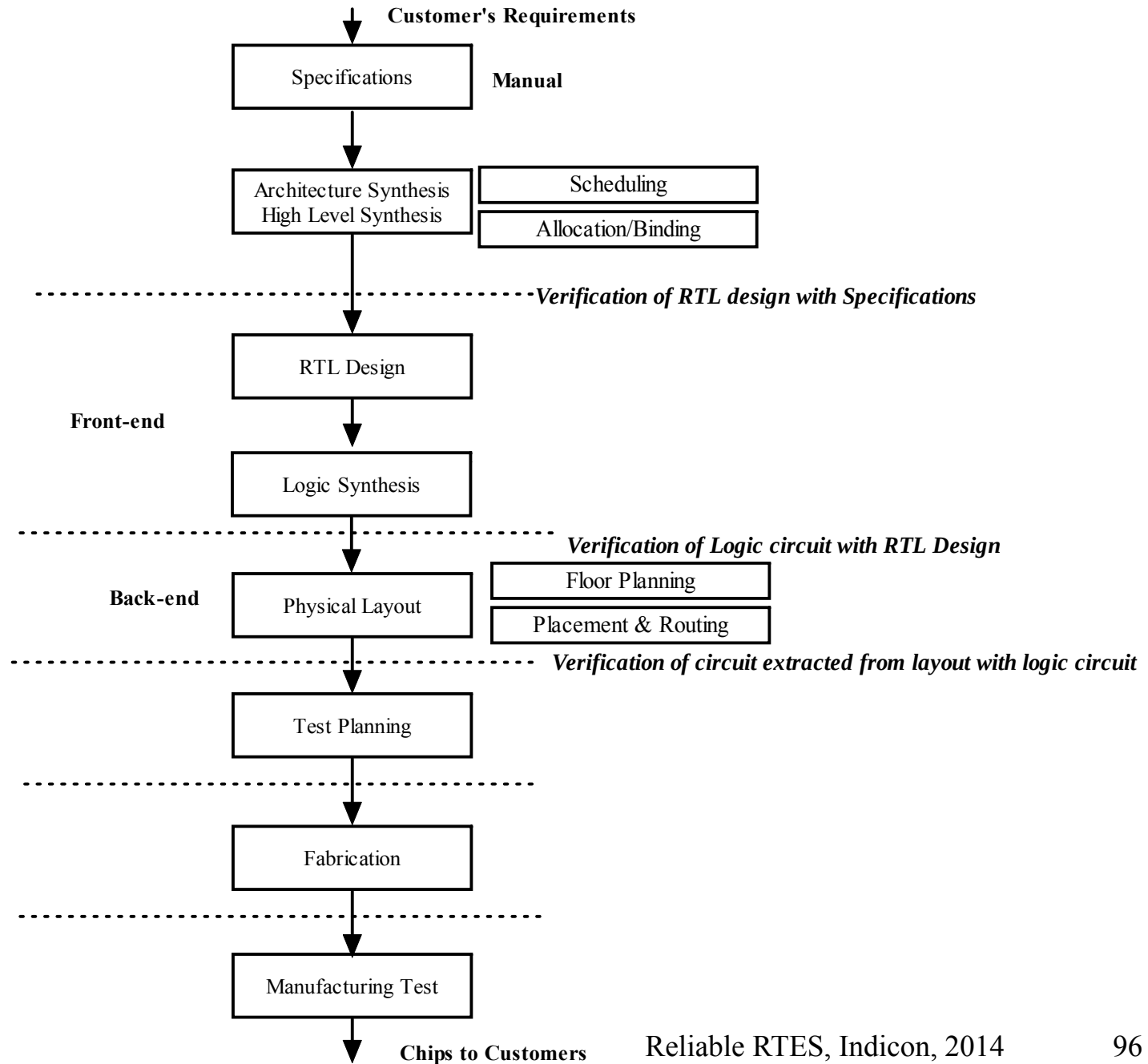
- NuSMV Model checker based on BDDs
- UPPAAL is an integrated tool environment for modeling, validation and verification of real-time systems modeled as networks of timed automata, extended with data types (bounded integers, arrays etc.).
- Ptolemy: A framework for simulating and prototyping heterogeneous systems,

References

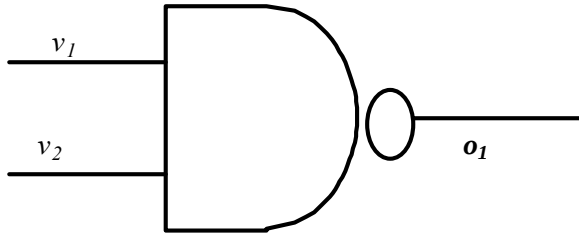
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- E.A. Emerson , ”Temporal and modal logic”, 1995
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- E.A. Emerson et al. “Parametric Quantitative temporal Reasoning”, 1999
- Santosh Biswas and Jatindra K Deka, NPTEL Video lectures and Web course VLSI Design Verification and Test (<http://www.nptel.ac.in/courses/106103116/>)

Testing of RT Embedded Systems

Hardware Design and Test flow

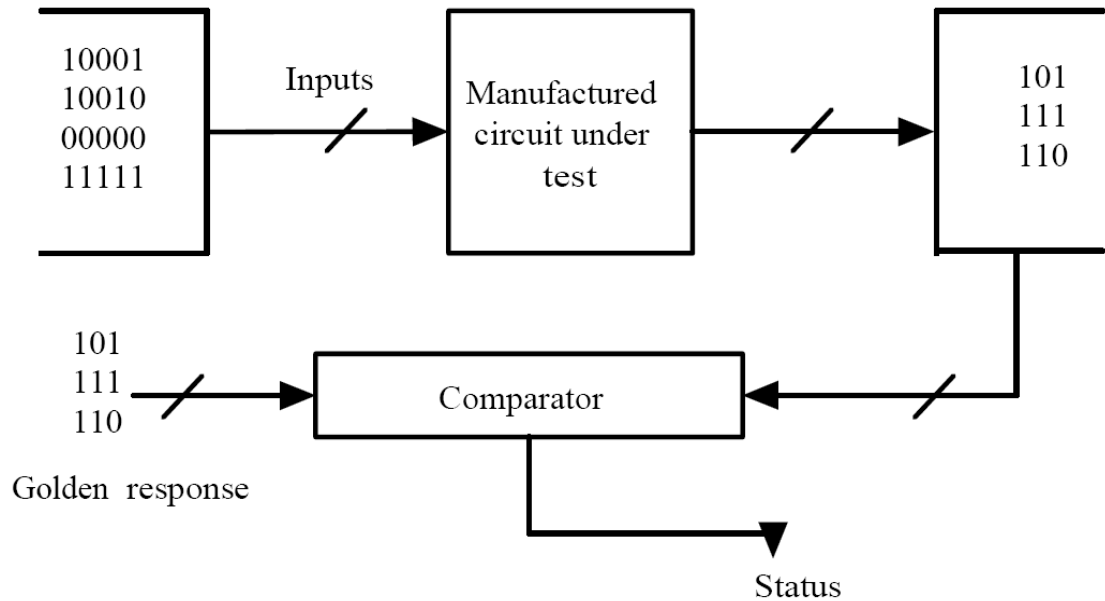


What is testing ?

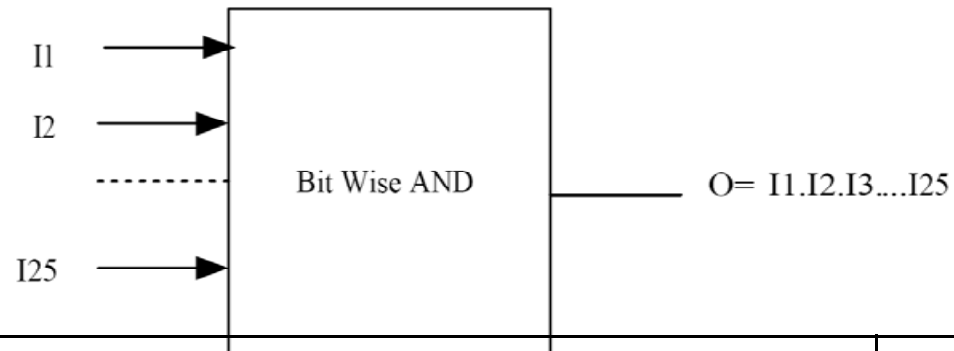


Input		Output
v_1	v_2	o_1
0	0	1
0	1	1
1	0	1
1	1	0

Test patterns
Applied from tester

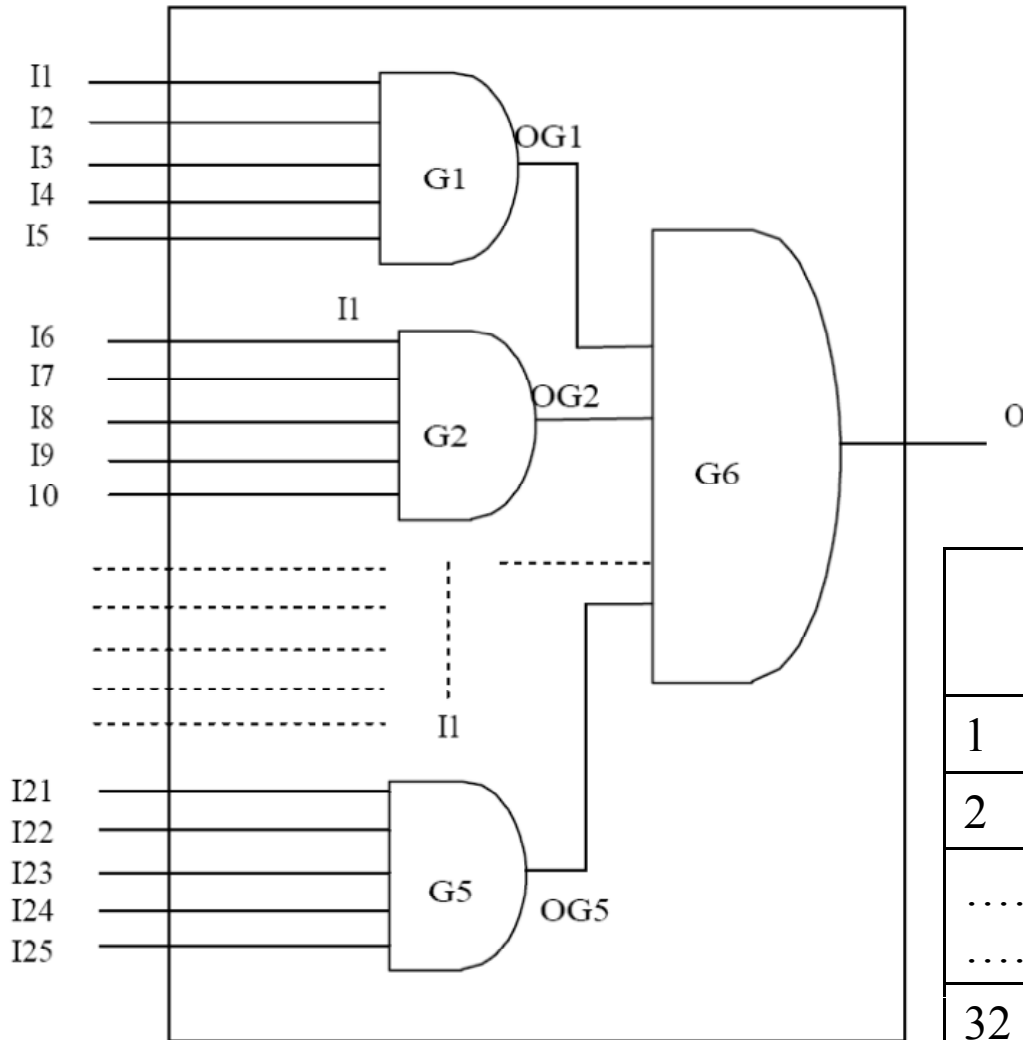


Is Functional Testing Possible



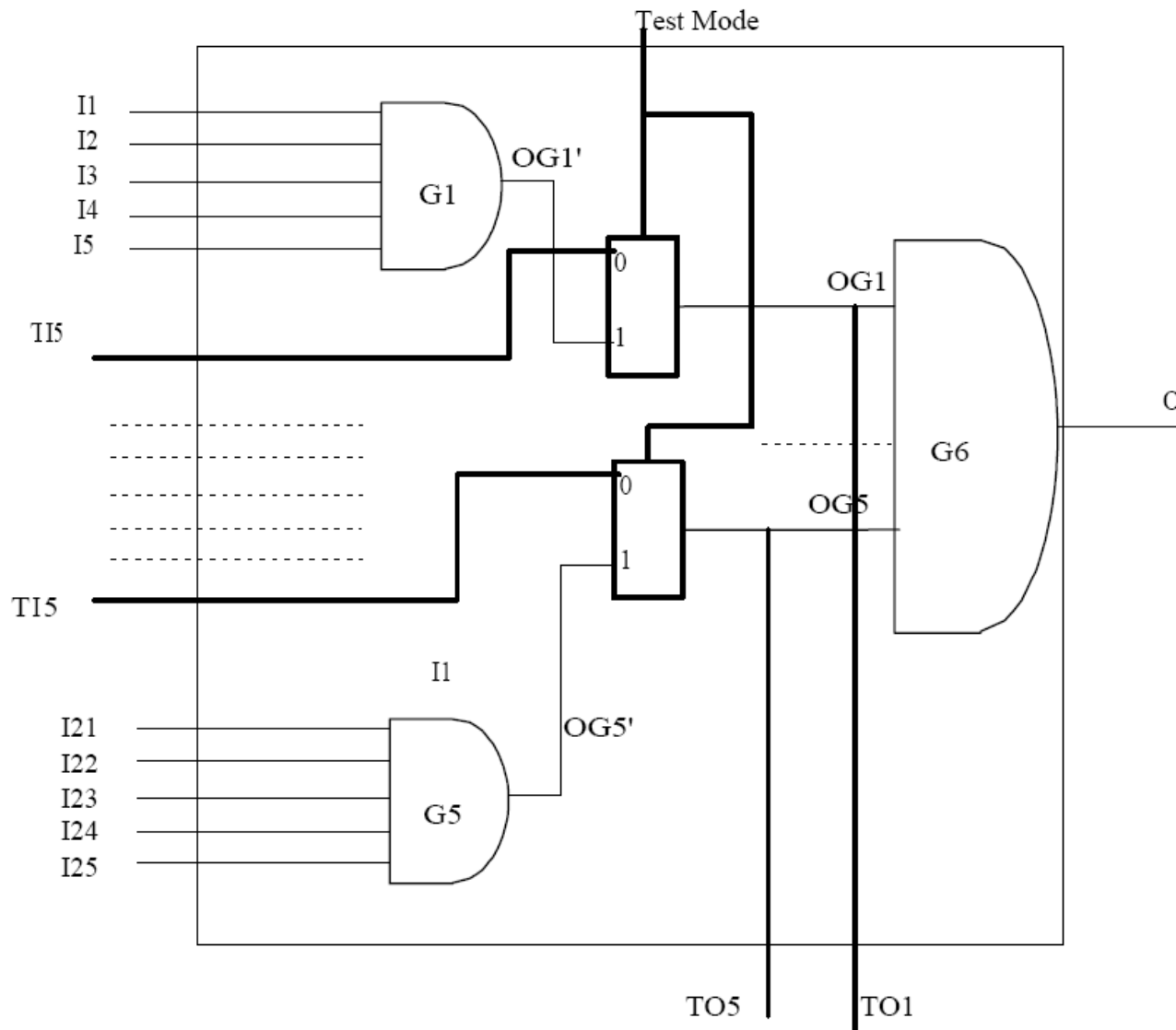
Test Pattern No.	Test Pattern	Output
1	000000000000000000000000000000	0
2	0000000000000000000000000000001	0
..... ...		0
2^{25}	111111111111111111111111111111	1

Is Functional Testing Possible

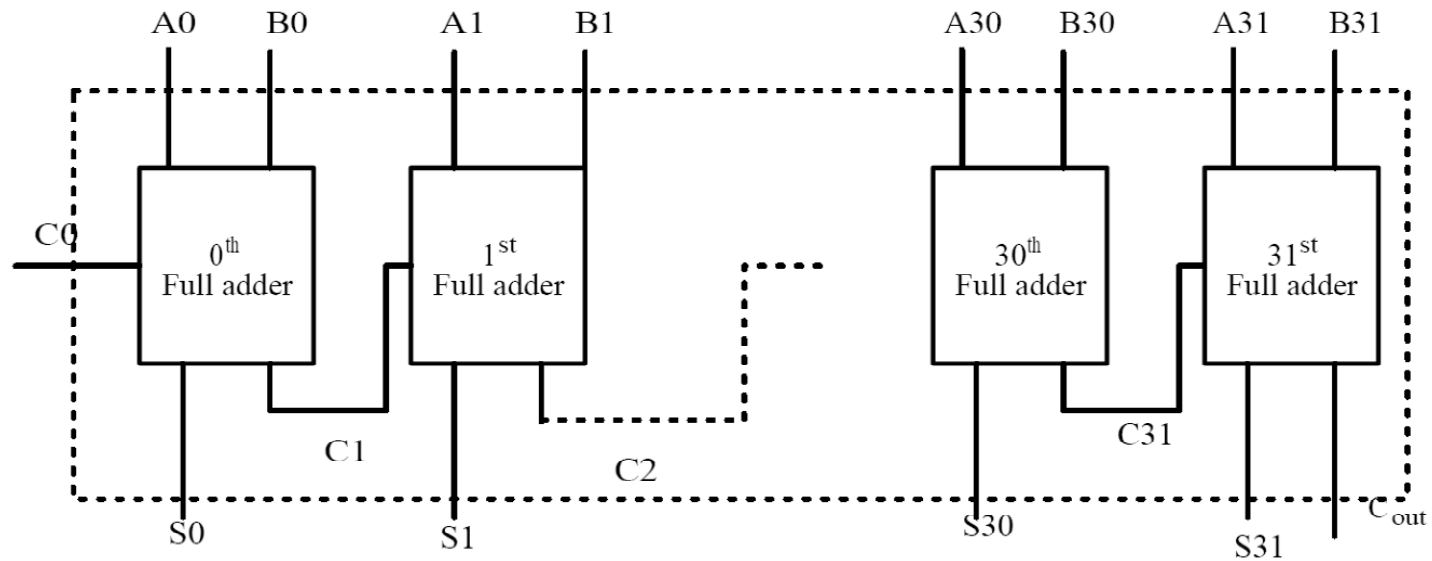


	Test Pattern (I1,I2,I3,I4,I5)	Output (OG1)
1	00000	0
2	00001	0
.....		0
.....		
32	11111	1

Is Functional Testing Possible

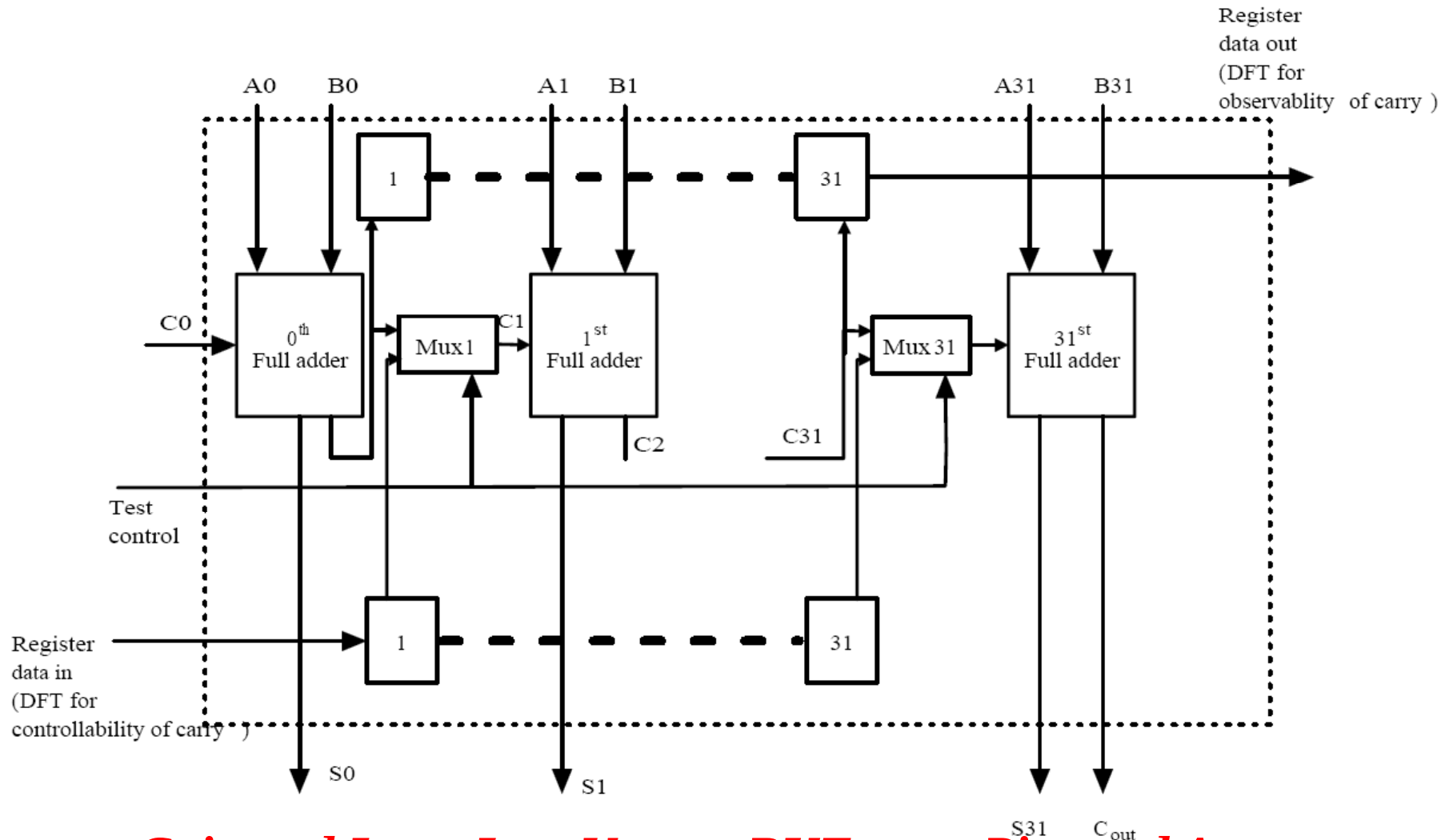


An example of Structural Testing



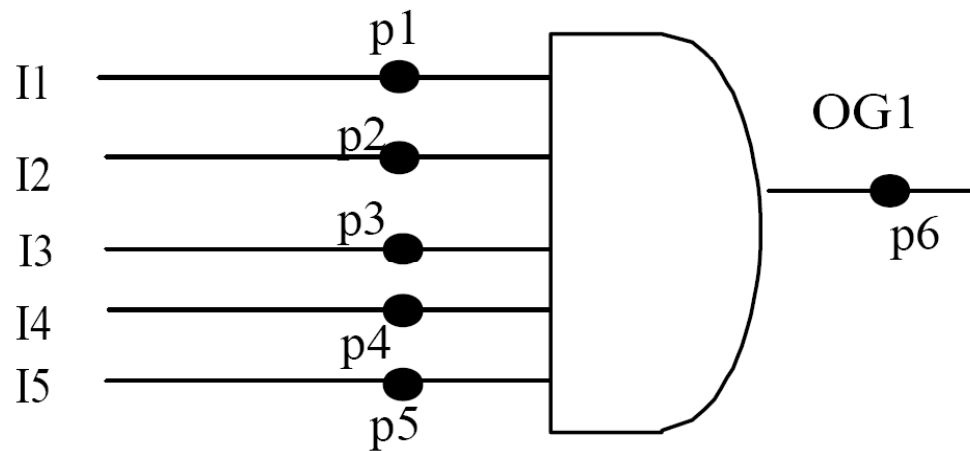
No. of Test Vectors— 2^{64}

An example of Structural Testing



Gain and Loss: Less Vectors BUT more Pins and Area

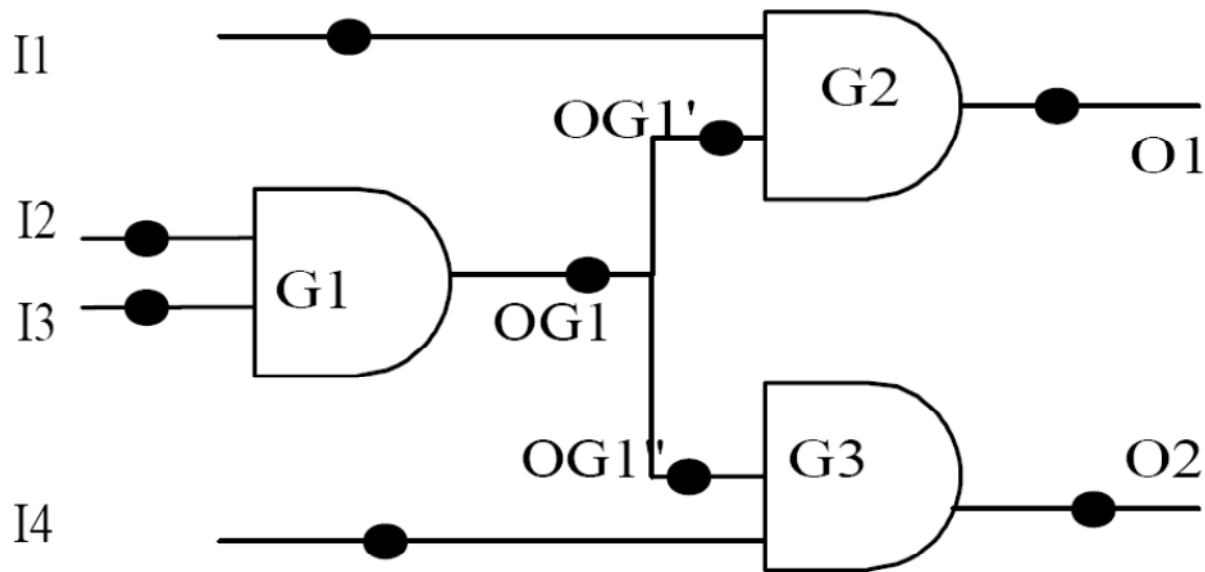
Structural Testing and Fault Model



Each net can have three states: normal, stuck-at-1 and stuck-at-0

Verify No faults

Stuck-at fault locations in a circuit with fanouts

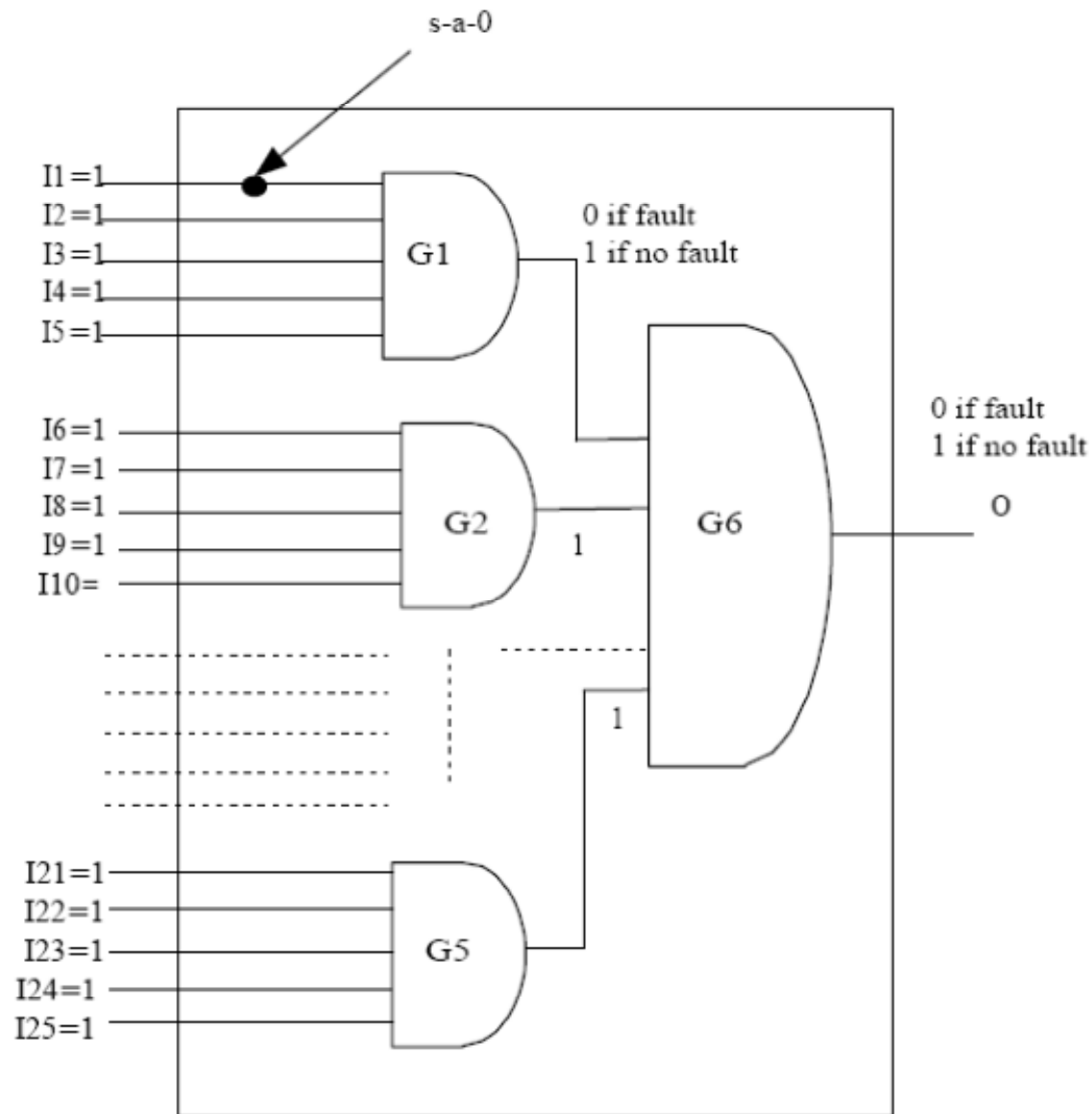


S-a Fault Model

Single stuck-at fault model is characterized by three assumptions:

- Only one net is faulty at a time.
- The faulty net is permanently set to either 0 or 1.
- The branches of a fanout net are independent with respect to locations and affect of a stuck-at fault.

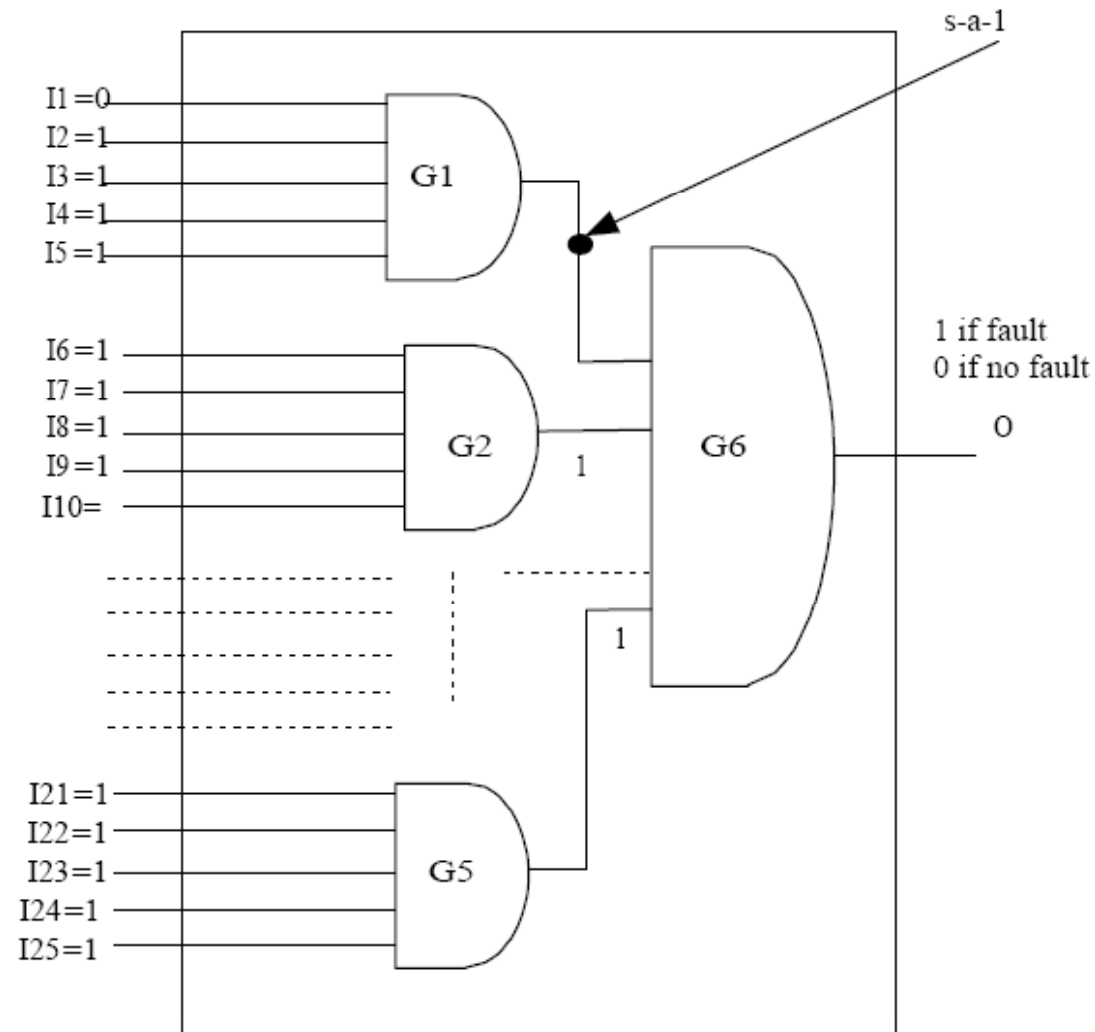
Example: Testing using s-a fault



Comparison of structural and functional testing

Functional testing	Structural Testing
Without fault models.	With fault models.
Manually generated design verification test patterns.	Automatic test pattern generation (ATPG).
Slow and labor intensive.	Efficient and automated.
Fault coverage not known	Fault Coverage is a quantified metric.
More Test Patterns	Less Test Patterns
Can be applied at the operating speed.	Difficult to be applied at the speed the design is expected to work.

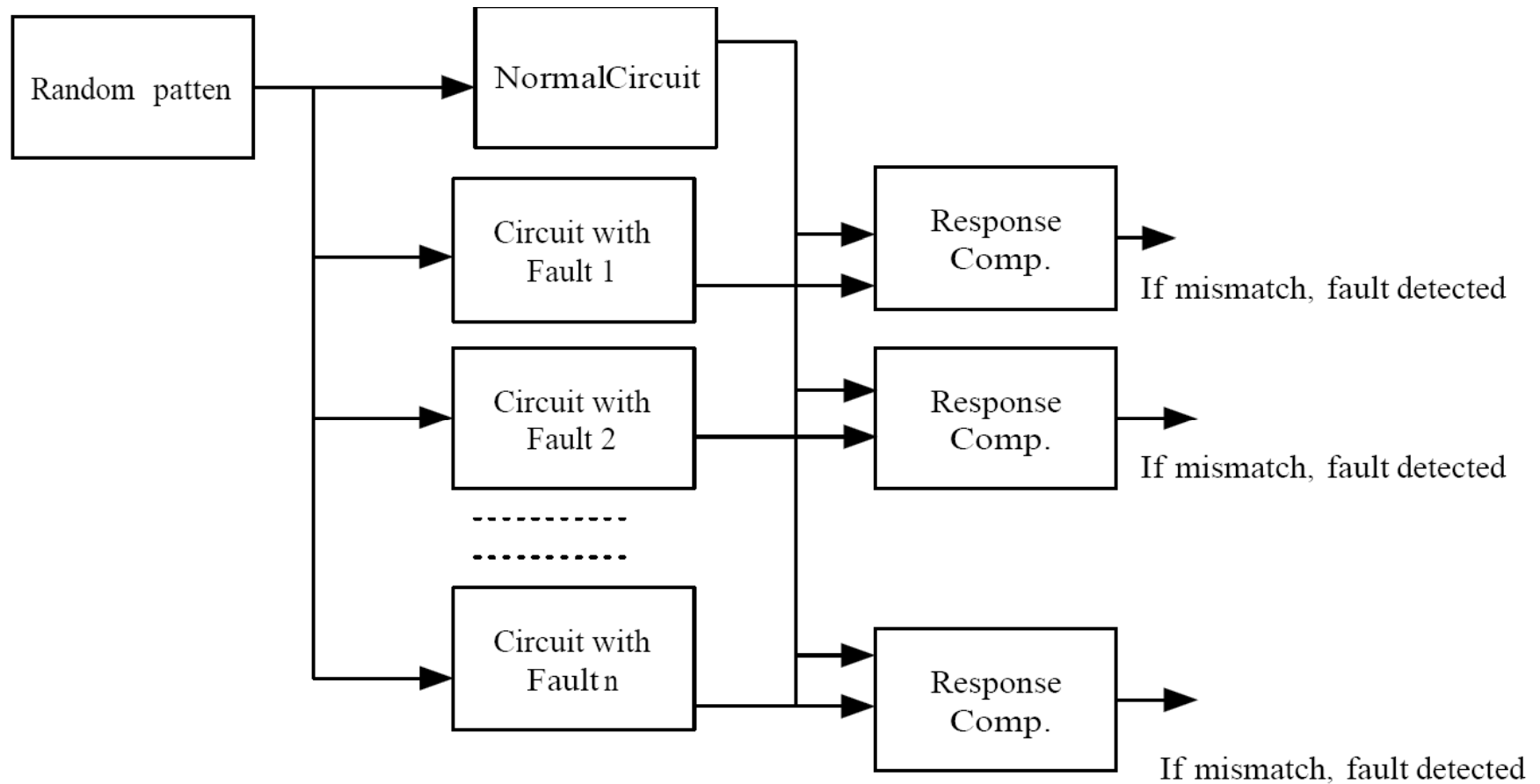
Automatic Test Pattern generation



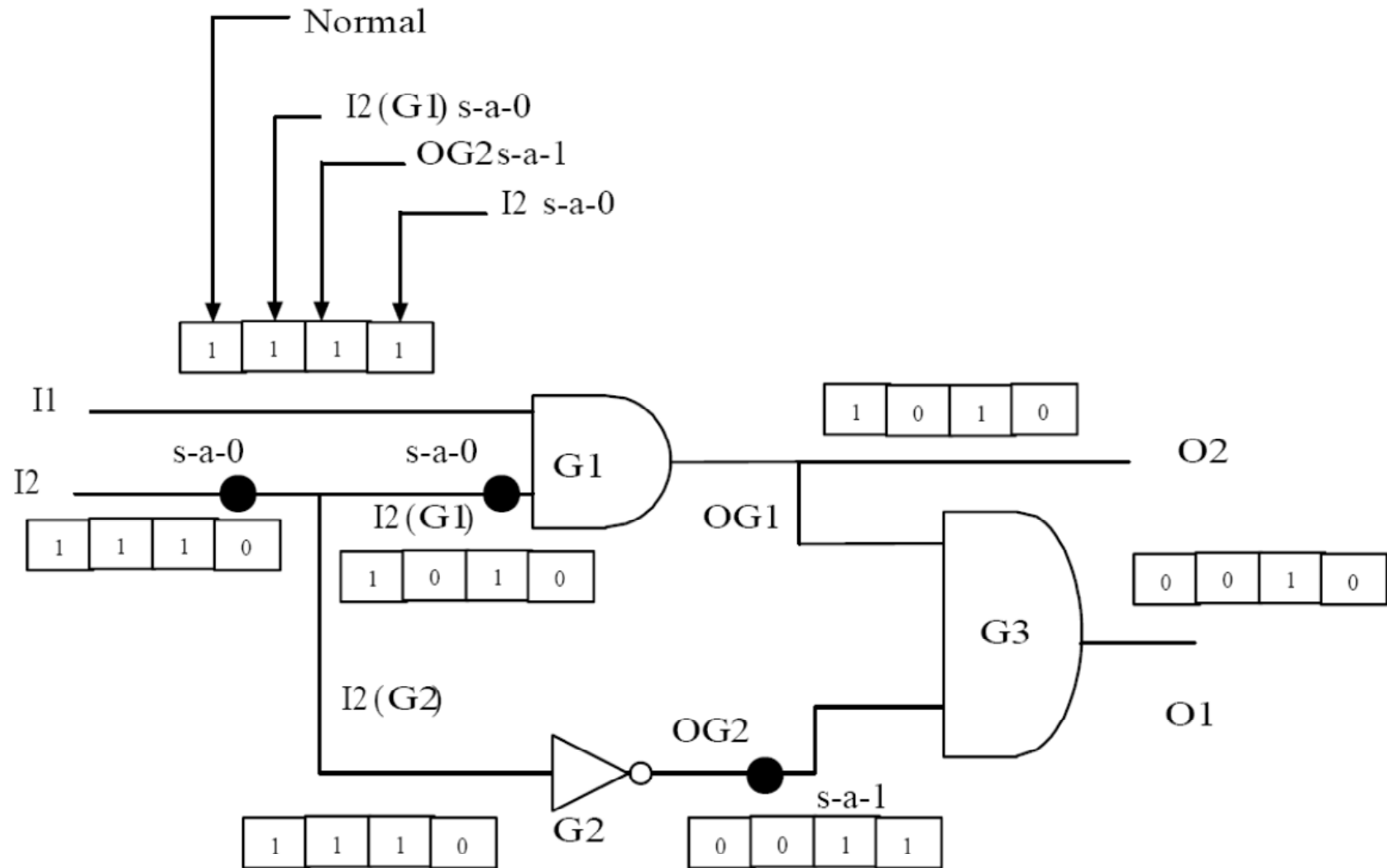
Random ATPG

Pattern No.	Random Pattern I1 I2 I3 I4 I5 I6.....I25	Faults Detected
1	1 0 0 0 1 11111111111111111111	s-a-1 at net “output of G1” s-a-1 at net “output of G6”
2	1 1 1 1 1 11111111111111111111	s-a-0 faults in all the nets of the circuit

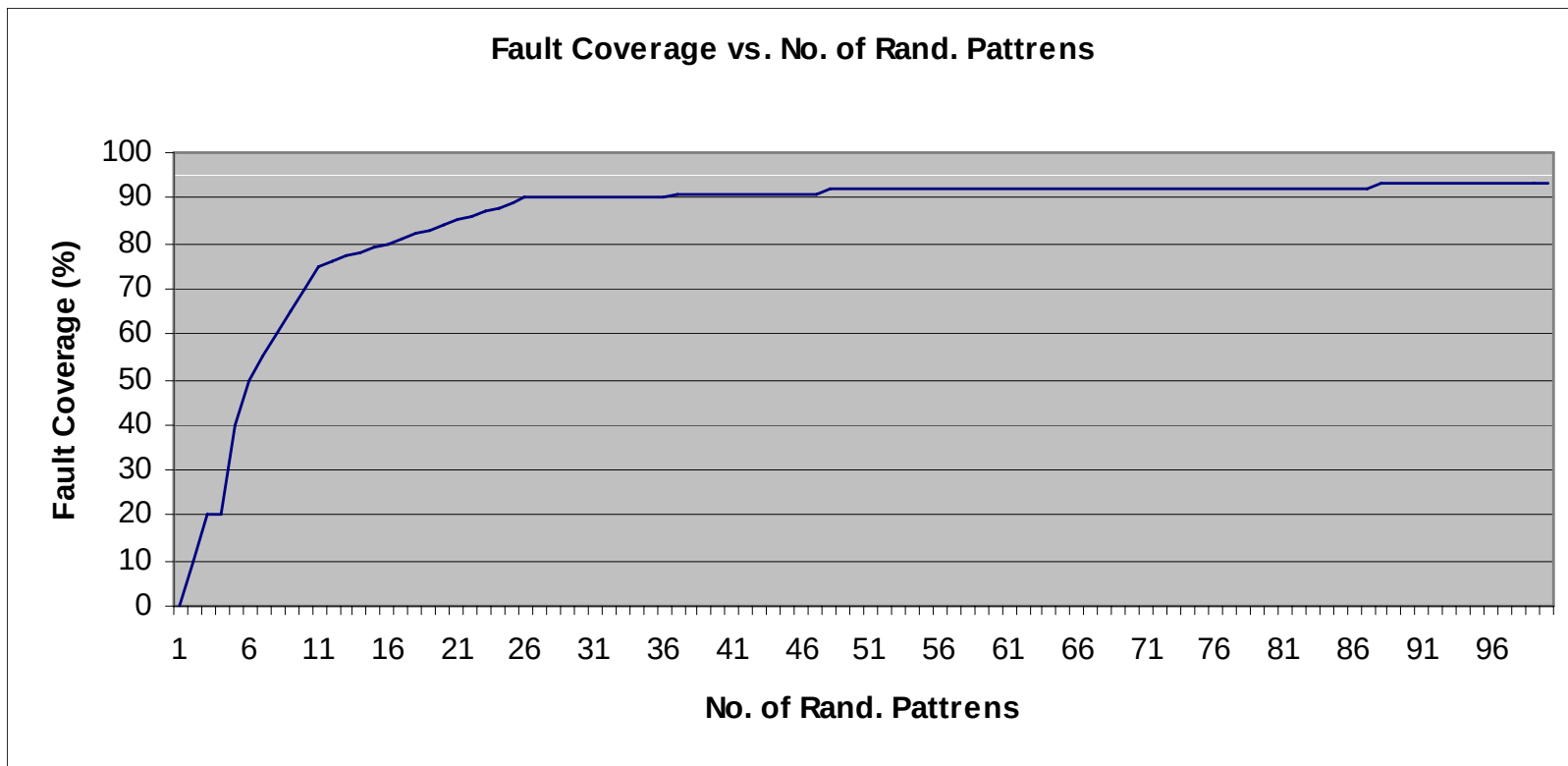
Random ATPG



Random ATPG Example

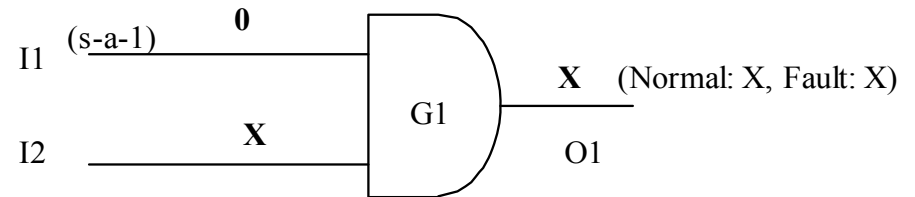
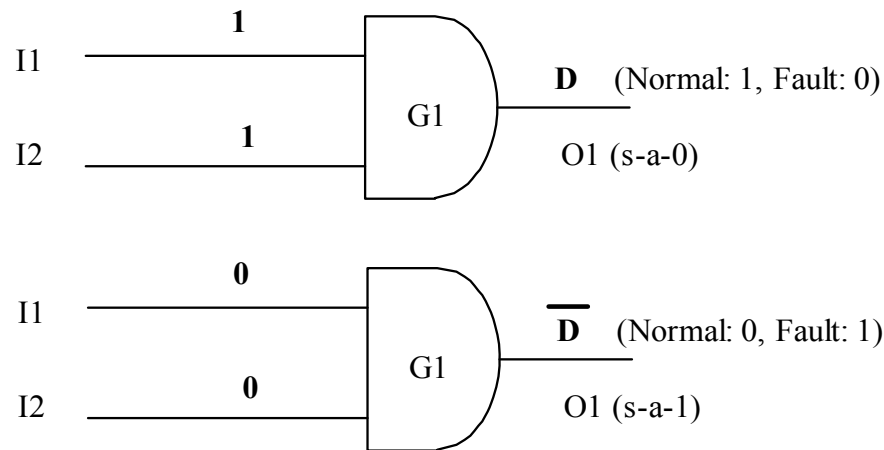


Random ATPG—How Long ?

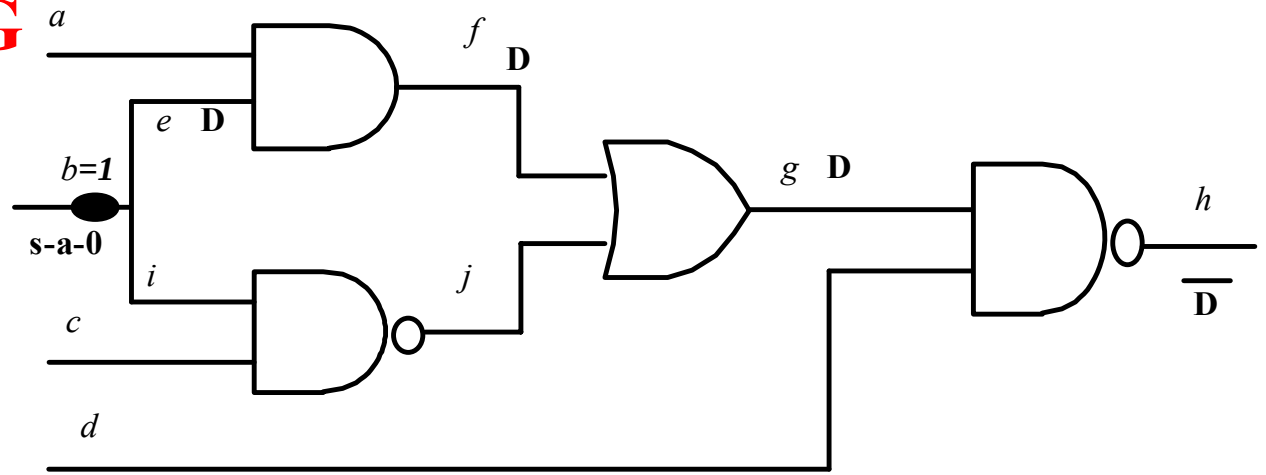


Deterministic ATPG Algebra

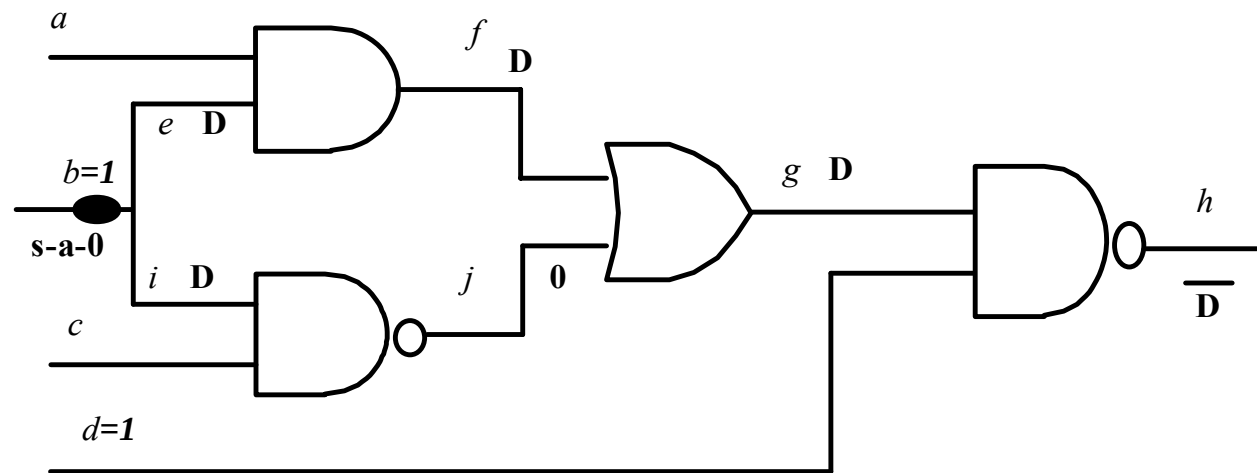
Symbol	Implication	Normal Circuit	Faulty Circuit
0	(0/0)	0	0
1	(1/1)	1	1
X	(X/X)	X	X
D	(1/0)	1	0
$\overline{D}$	(0/1)	0	1



Deterministic ATPG

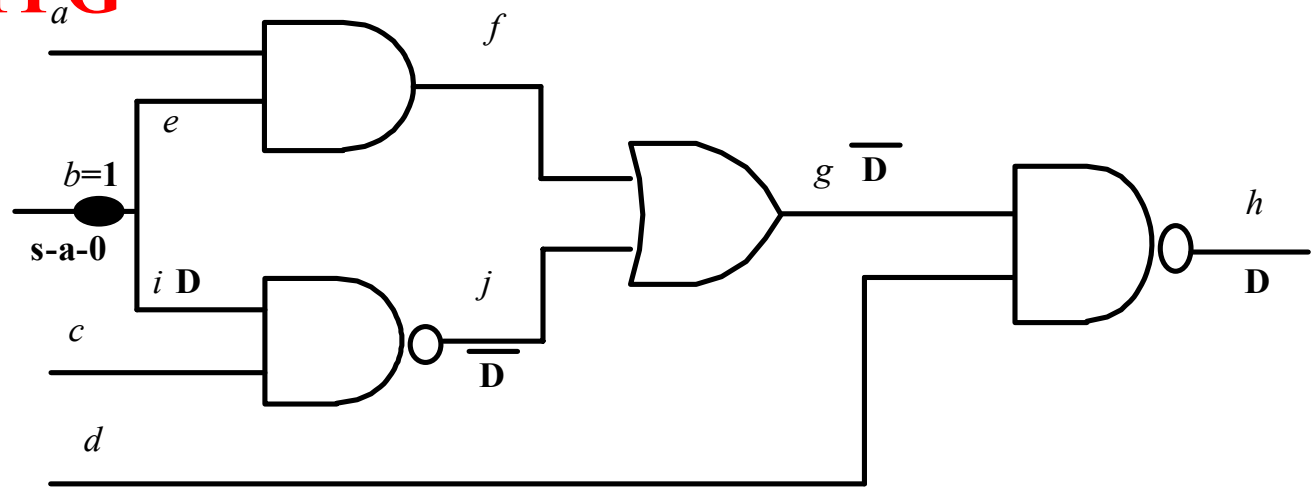


(a) Sensitization by $b=1$
Propagation by path $e-f-g-h$

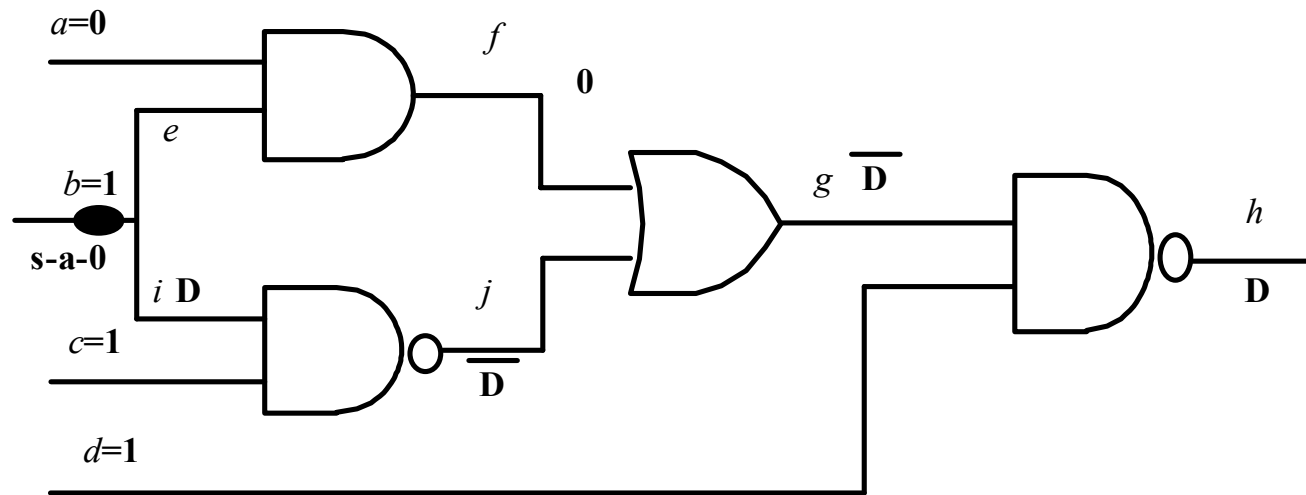


(b) Justification $j=0$
 $j=D$, if $c=1$ and $j=1$ if $c=0$
(conflict at j)

Deterministic ATPG



(a) Sensitization by $b=1$
Propagation by path $i-j-g-h$

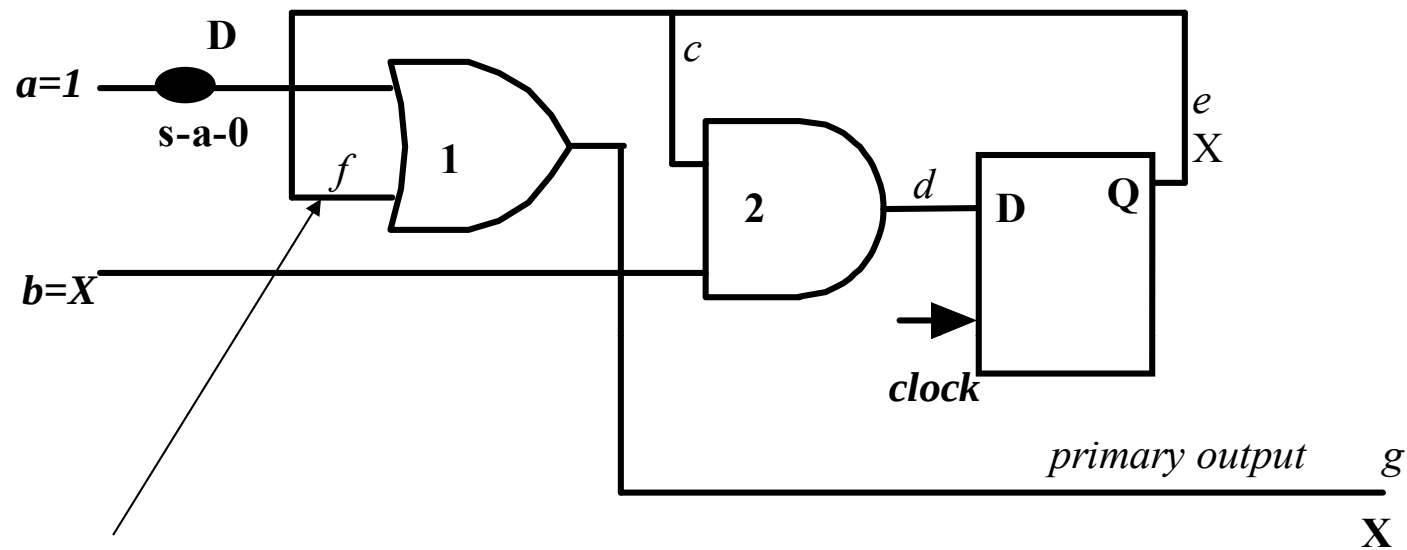


(b) Justification $d=1, c=1, f=0, a=0$
(Successful)

Deterministic ATPG steps

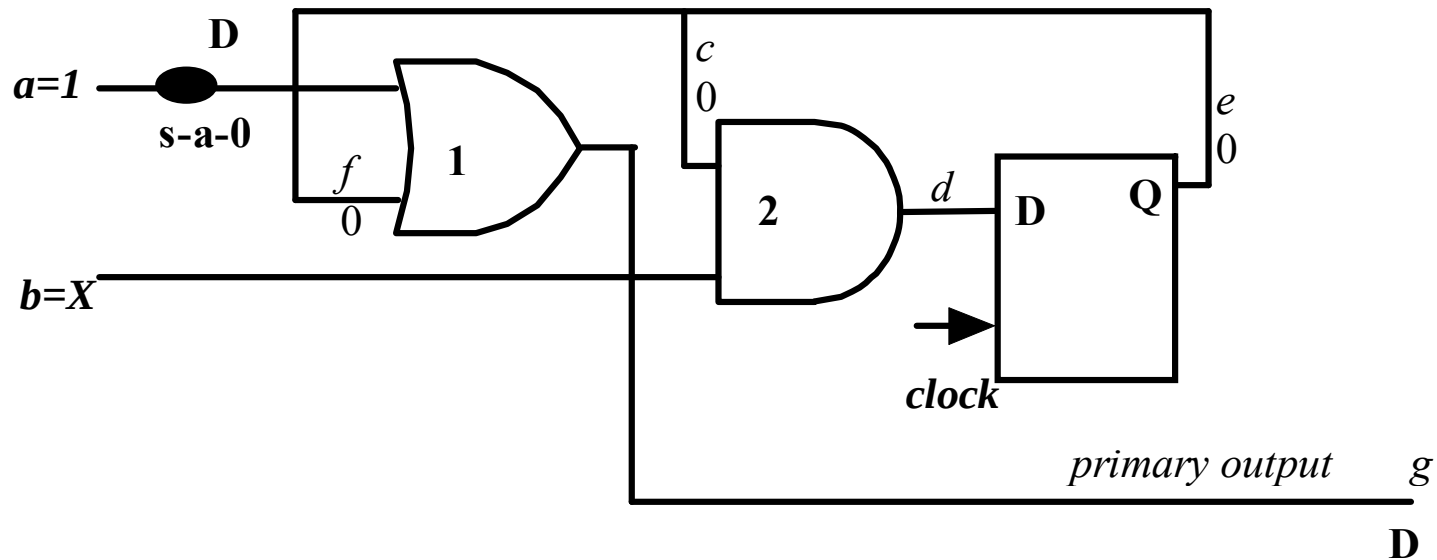
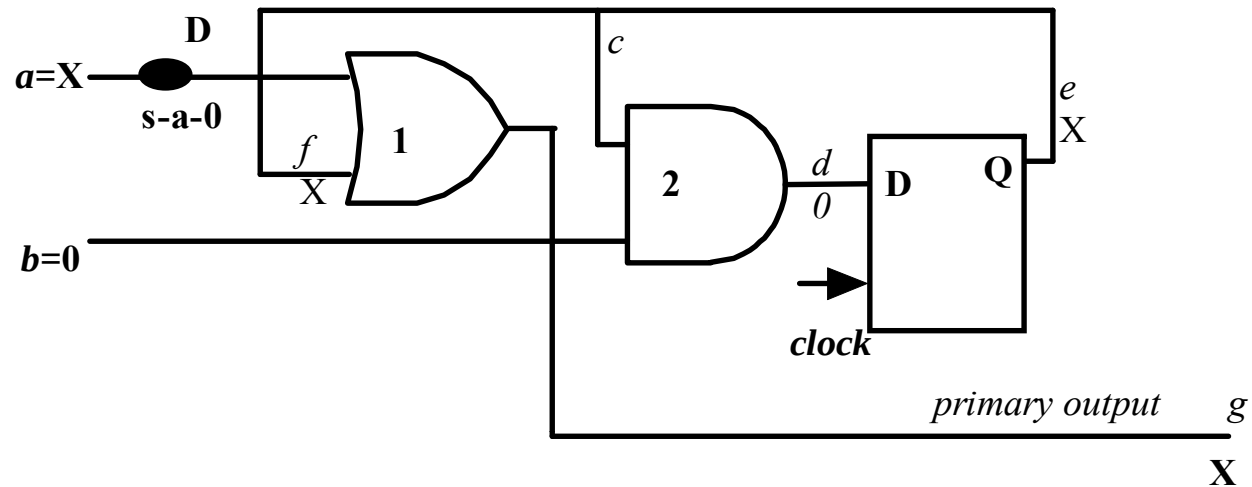
- **Fault sensitization:** In this step a stuck-at fault is activated by setting the signal driving the faulty net to an opposite value from the fault value.
- **Fault propagation:** In this step a path is selected from the fault site to some primary output, where the effect of the fault can be observed for its detection.
- **Line justification:** In this step the signals in (internal) nets or some primary inputs, which were assigned for fault sensitization/propagation, are justified by setting (remaining) primary inputs of the circuit.

How Complicated are Sequential Circuits ?

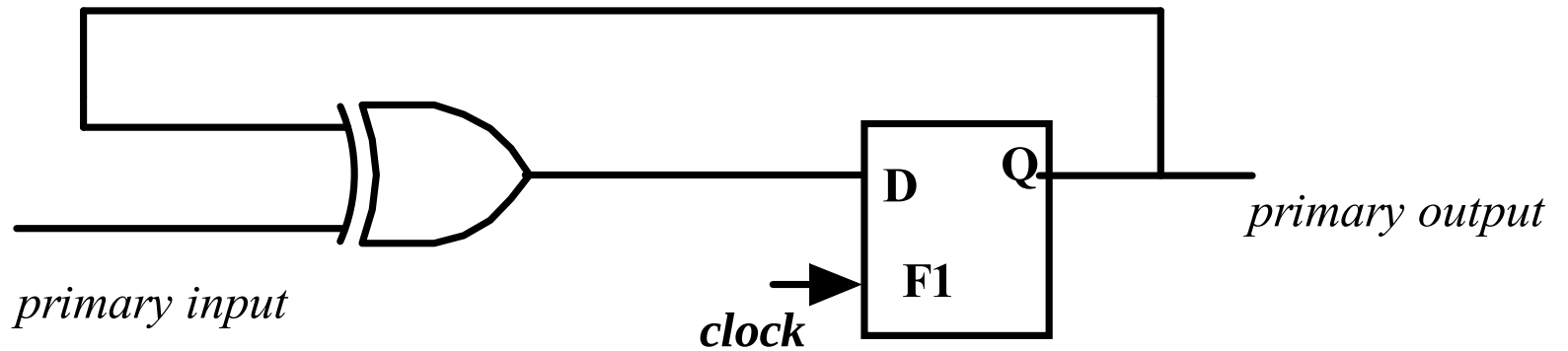


Need 0

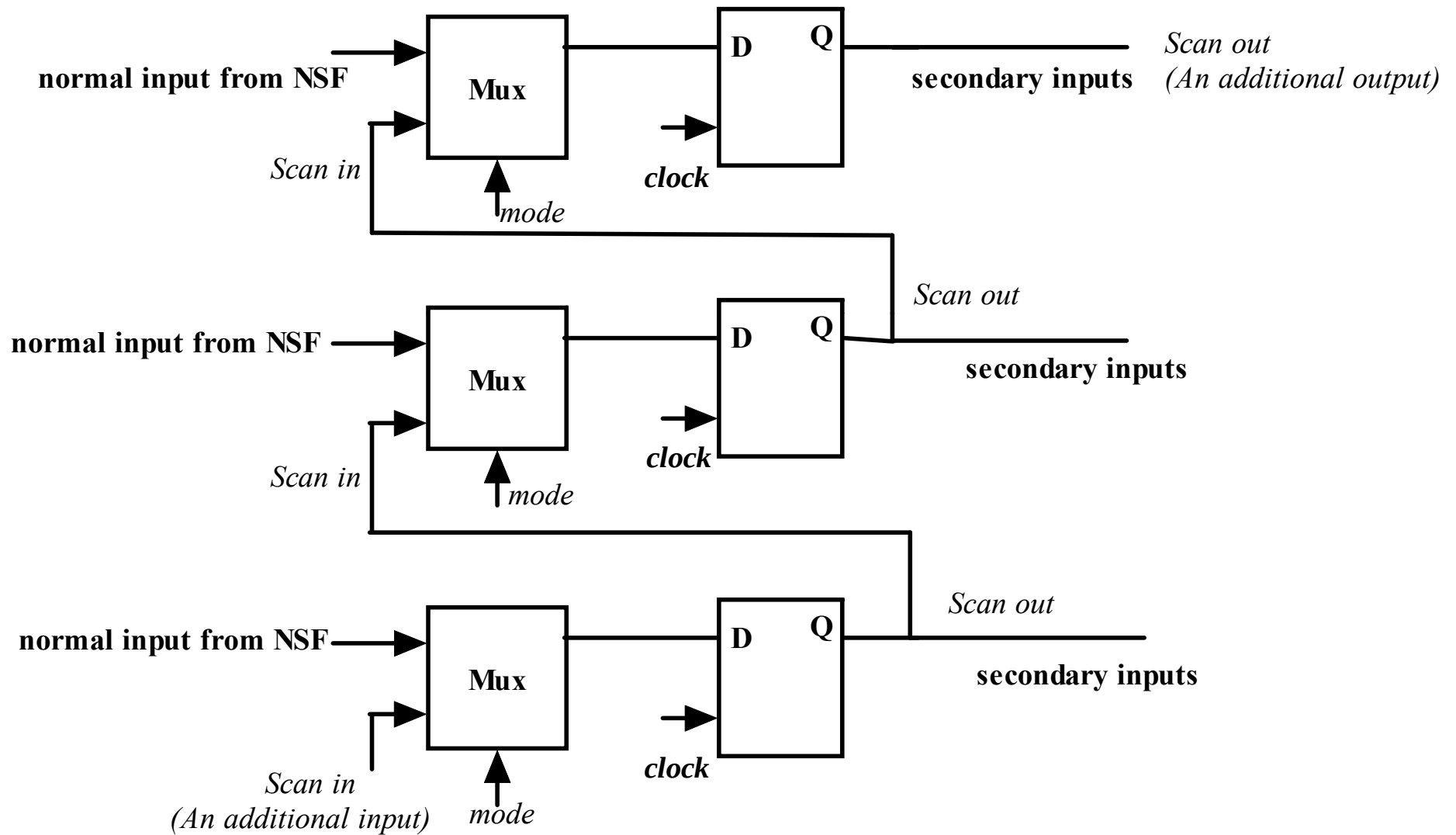
Indirect Control



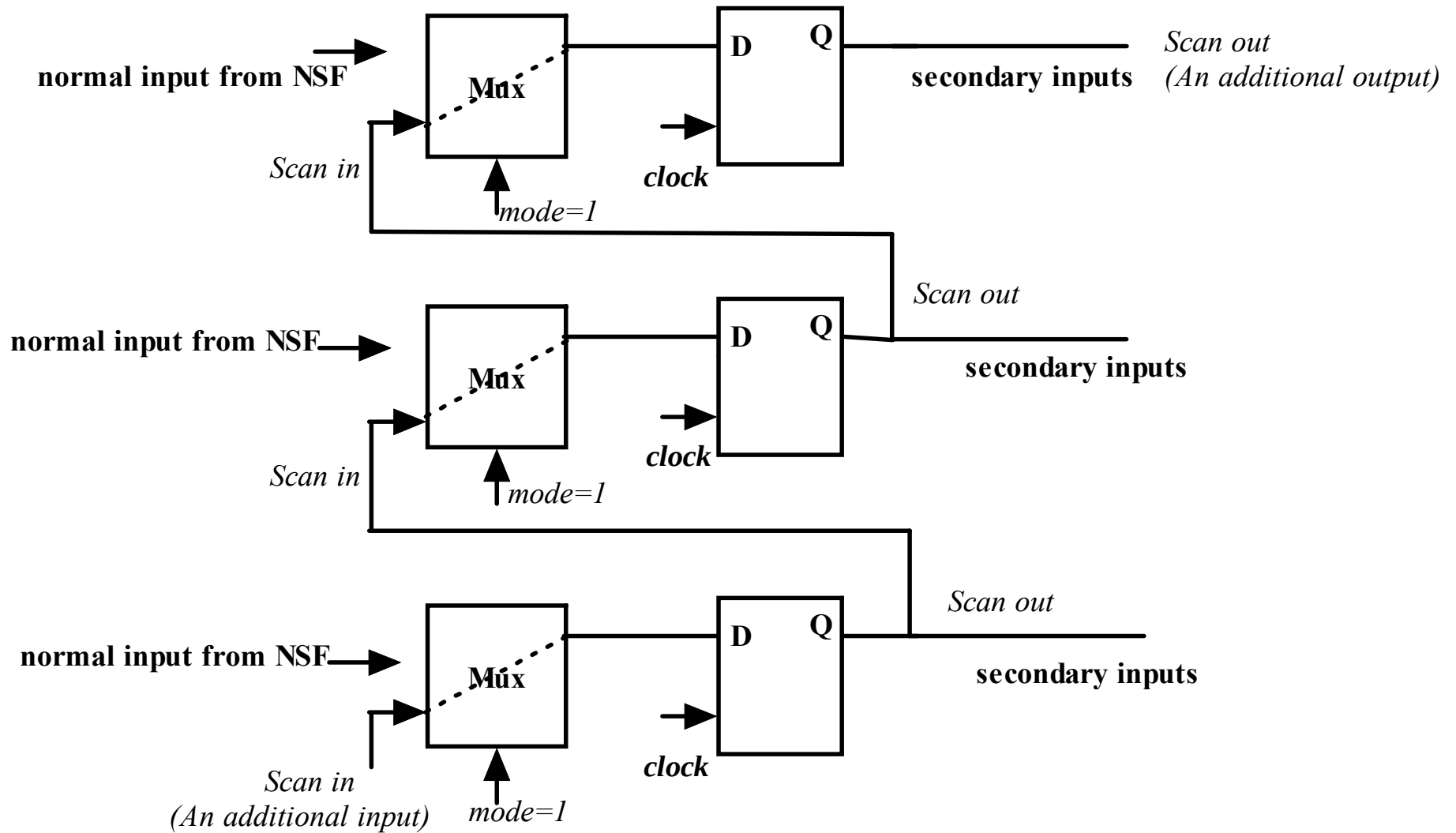
Is it always possible ?



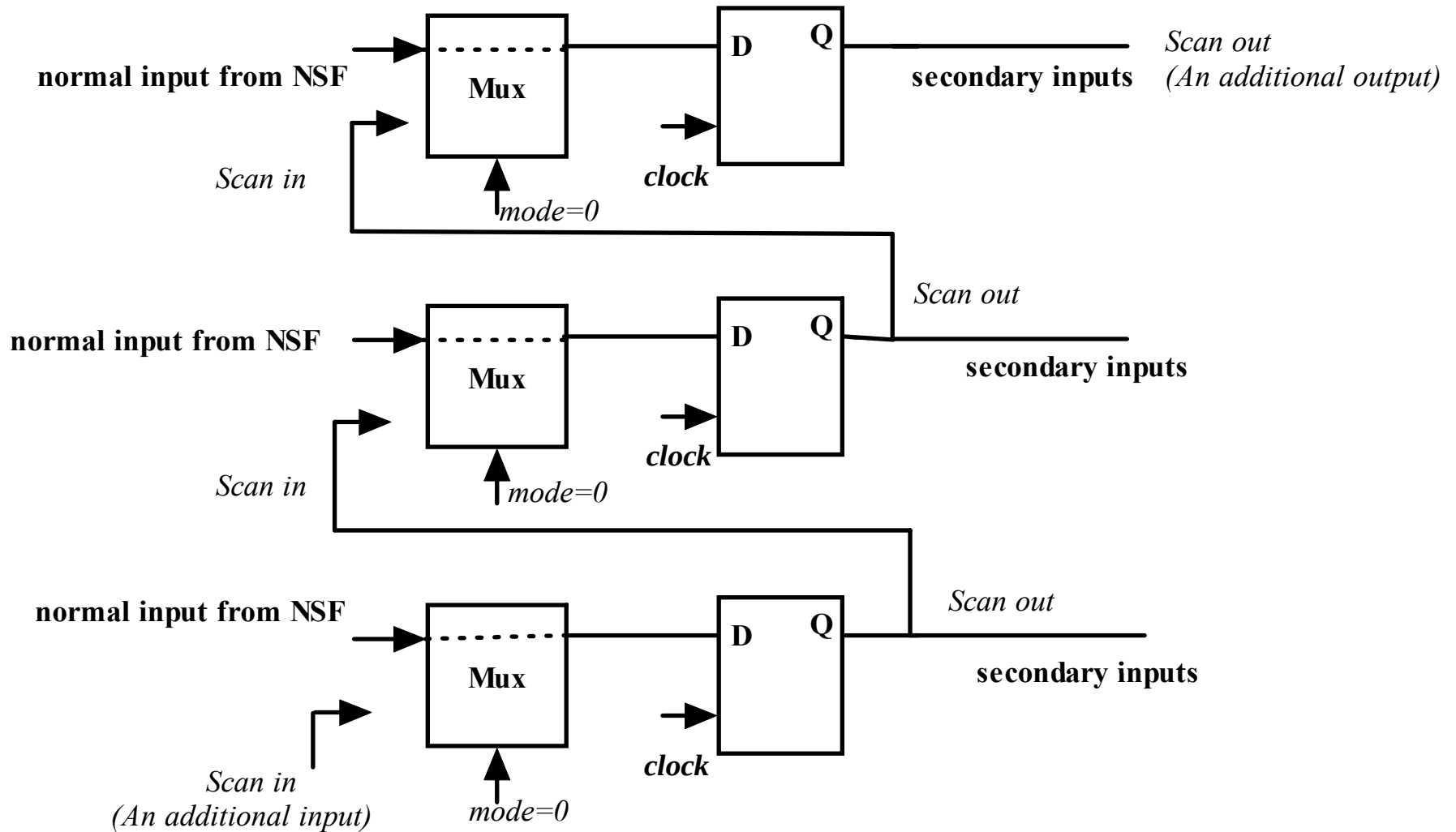
Scan Chains ?



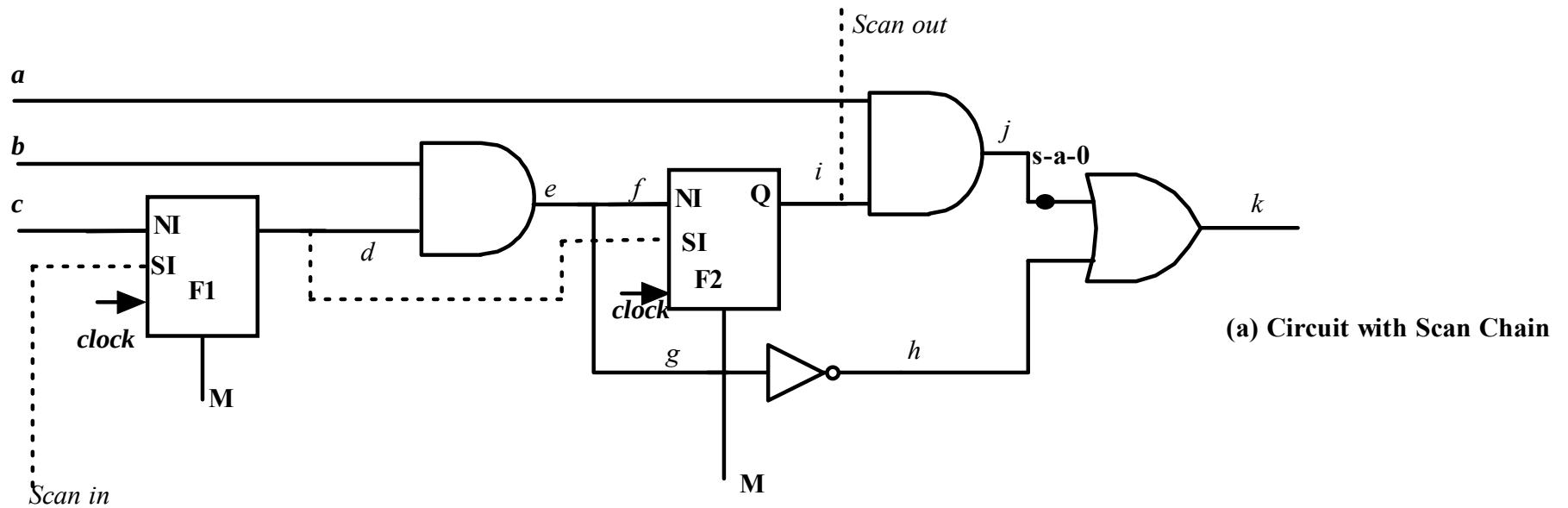
Scan Chains ?

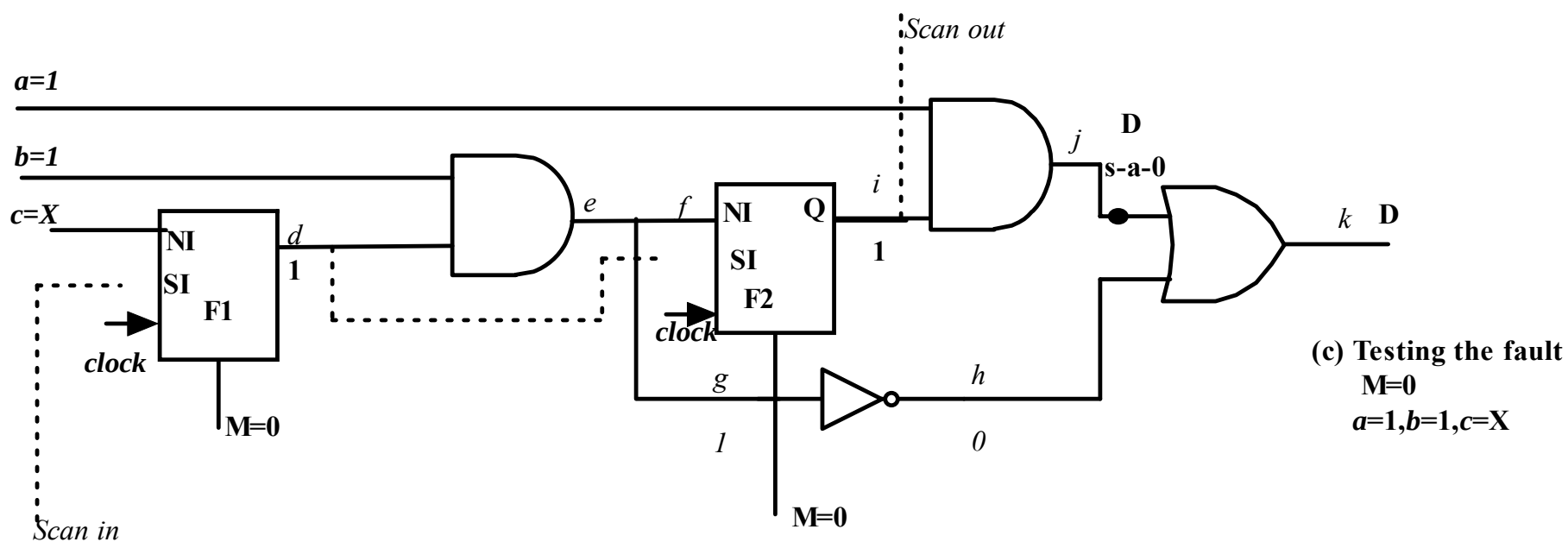
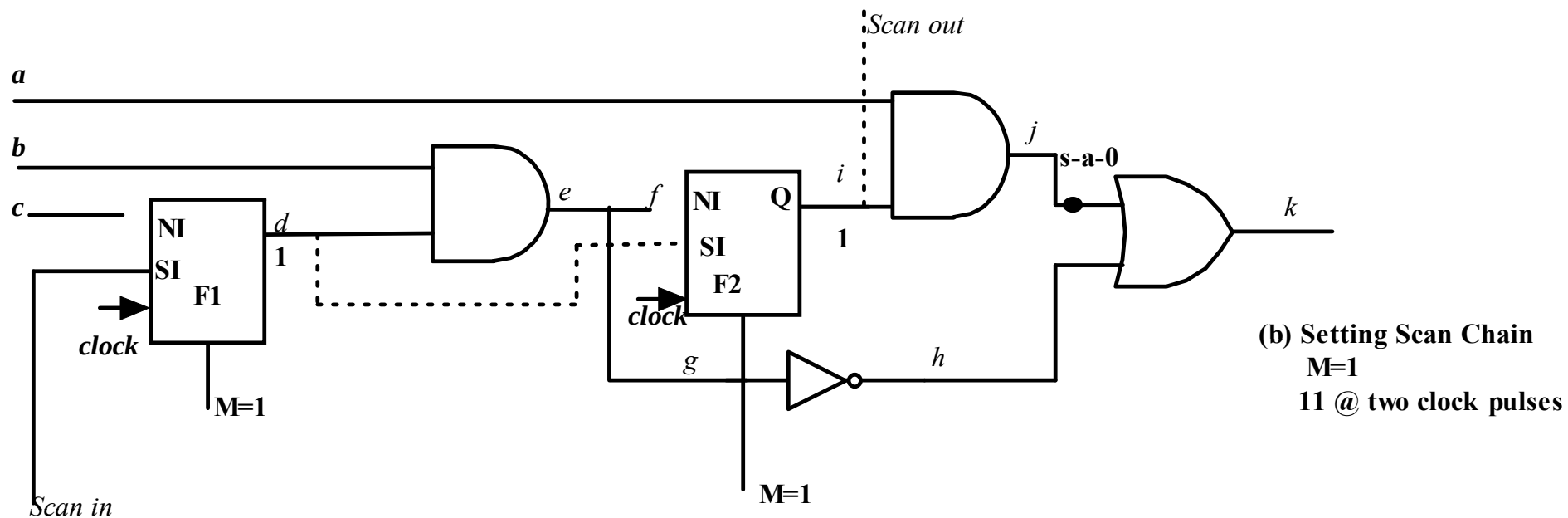


Scan Chains ?



Testing Using Scan Chains - Example





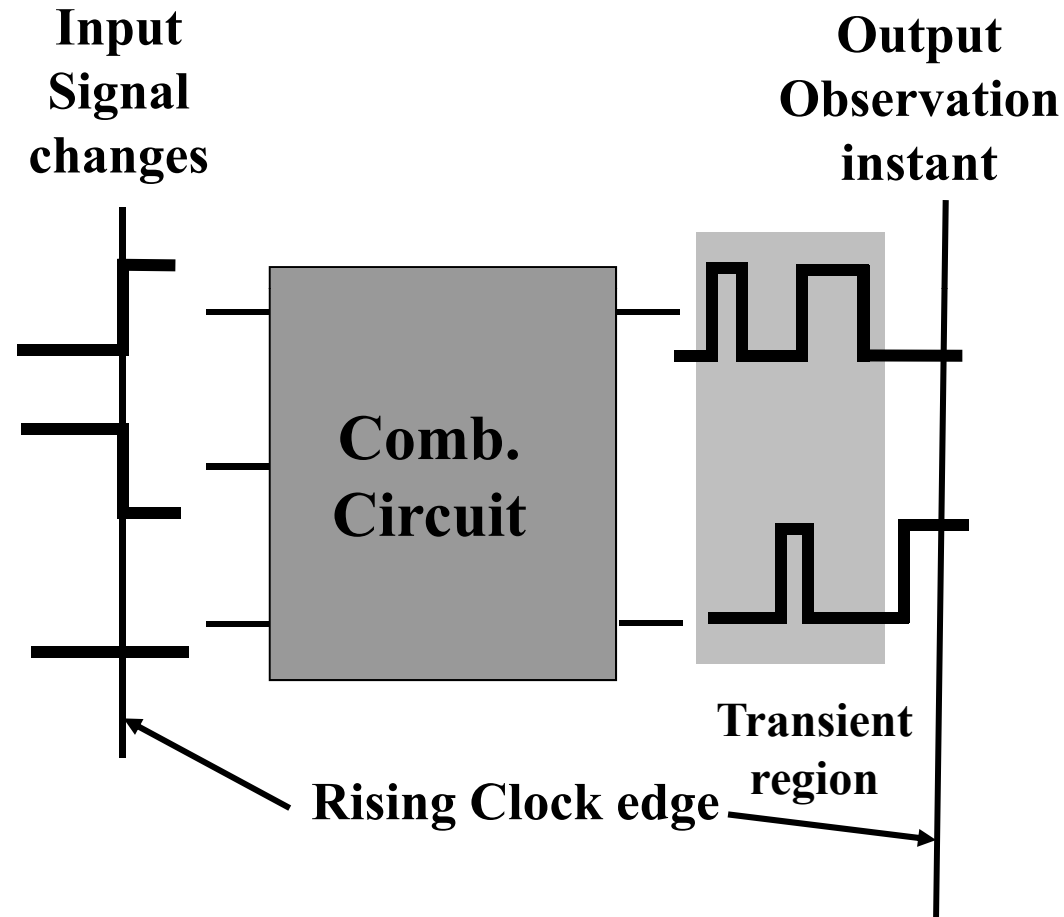
Why Delay Test and New Fault Model ?

- **Limitations of s-a fault model:**
 - Classical fault model and generally do not verify timing correctness.
 - For slower circuits, it was found that s-a fault model covers some delay faults also. This does not hold for high speed circuits
- **Circuits for RT ES:**
 - Need to meet deadlines i.e., correct output and stabilize before clock edge
 - Requires new fault models i.e., path delay fault model.
- **Complicity of Delay test vs. static s-a test:**
 - Single vector for s-a fault while two vectors for delay faults
 - More number of un-testable cases
 - Complex APTG algorithms

What is Delay Fault Model

- Combinational path
 - Start: Primary input or a clocked flip-flop through levels of gates
 - Ends: primary output or a clocked flip-flop.
- Propagation delay = Gate transition delays + transport delays of interconnects on a path.
- Clock period $>$ the propagation delay of all paths
- Delay fault
 - Propagation delay of a combinational path $>$ Clock Period

What is Delay Fault Model

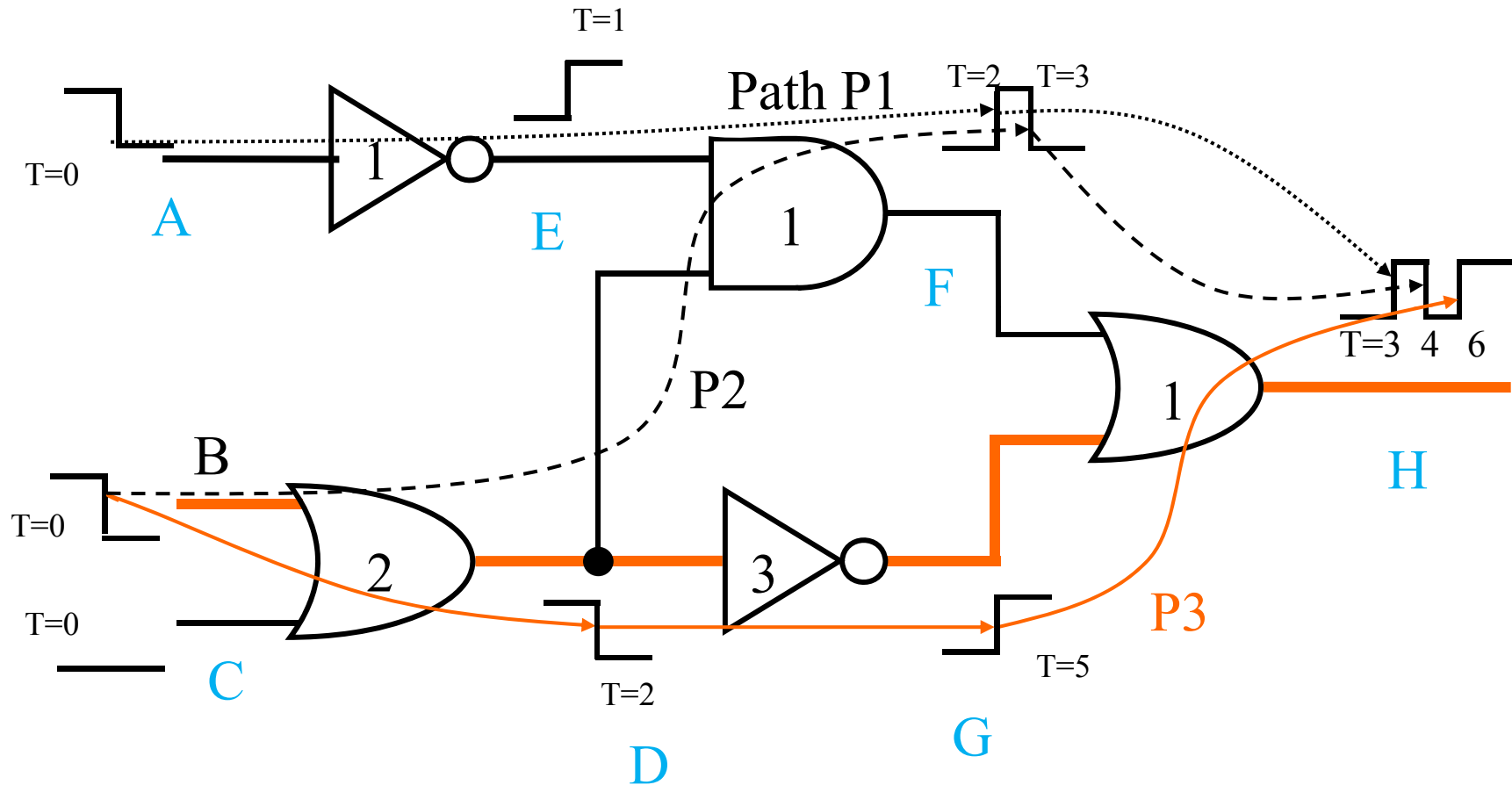


Propagation delay

- Switching or inertial delay is the interval between input change and output change of a gate:
 - Depends on input capacitance, device (transistor) characteristics and output capacitance of gate.
 - Also depends on input rise or fall times and states of other inputs (second-order effects).
 - Approximation: fixed rise and fall delays (or min-max delay range, or single fixed delay) for gate output.
- Propagation or interconnect delay is the time a transition takes to travel between gates:
 - Depends on transmission line effects (distributed R , L , C parameters, length and loading) of routing paths.
 - Approximation: modeled as lumped delays for gate inputs.

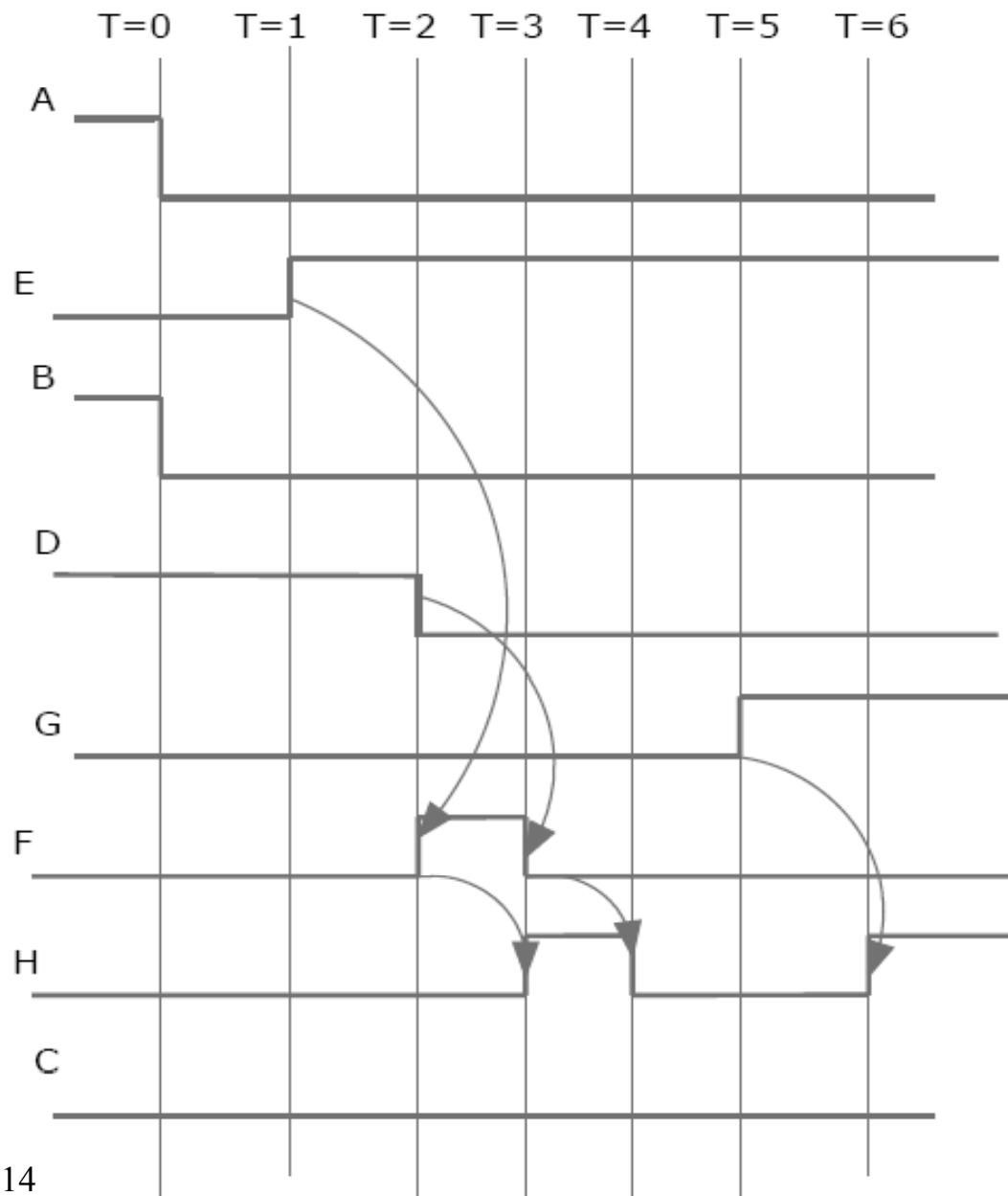
Example

PI transitions assumed to occur without time skew



Time Period is 7

Timing Diagram



Test Pattern pair Conditions

On-path input --source of the output transition.

Off-path input – source that does not have effect on output transition.

- Cond-1: If event is a transition from the controlling value to non-controlling value

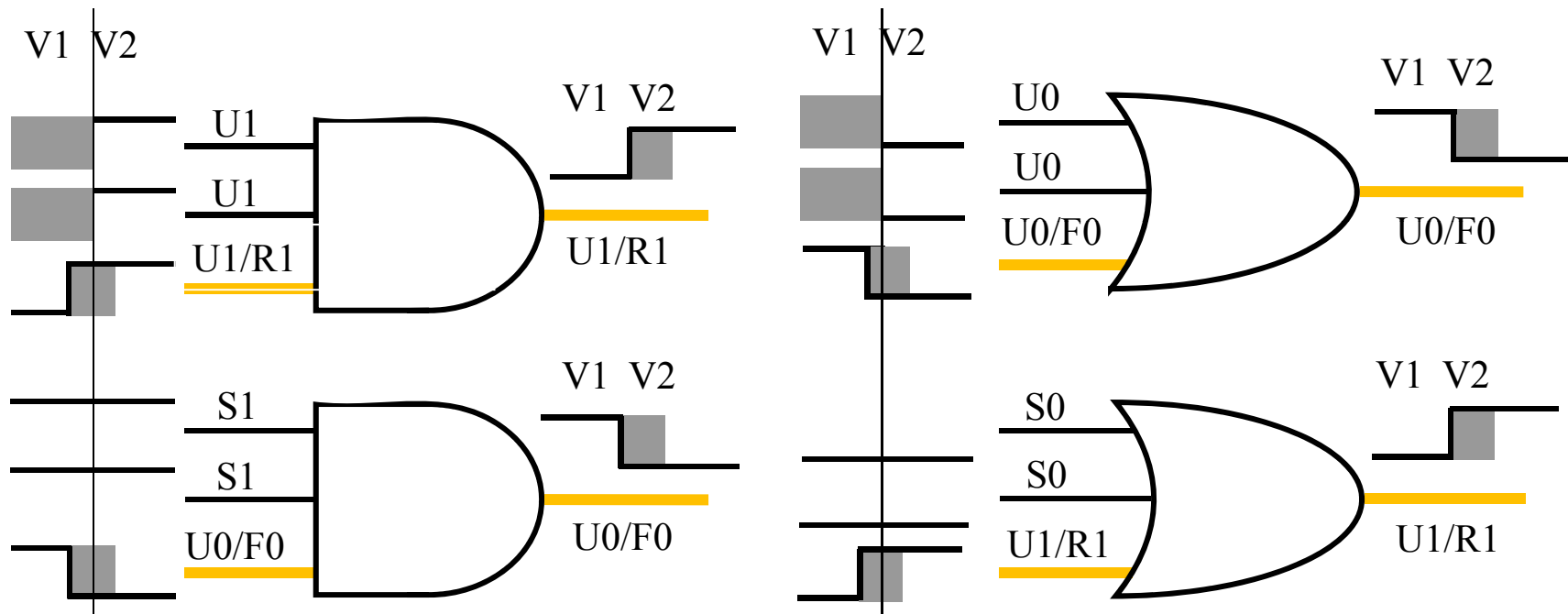
There is no specific requirement for off-path inputs in V1.

All off-path inputs of the gate should have non-controlling value in V2.

- Cond-2: If event is a transition from the non-controlling value to Controlling value

All off-path inputs must have a steady non-controlling value in both V1 and V2.

Test Patten pair Conditions: Examples of Simple Gates



A Five-Valued Algebra

- Signal States: S0, S1, U0, U1, F0, R1, XX.
- On-path signals: F0 and R1.
- Off-path signals: F0=U0 and R1=U1.

AND

		Input 1				
		S0	U0	S1	U1	XX
Input 2	S0	S0	S0	S0	S0	S0
	U0	S0	U0	U0	U0	U0
	S1	S0	U0	S1	U1	XX
	U1	S0	U0	U1	U1	XX
	XX	S0	U0	XX	XX	XX

OR

		Input 1				
		S0	U0	S1	U1	XX
Input 2	S0	S0	U0	S1	U1	XX
	U0	U0	U0	S1	U1	XX
	S1	S1	S1	S1	S1	S1
	U1	U1	U1	S1	U1	U1
	XX	XX	XX	S1	U1	XX

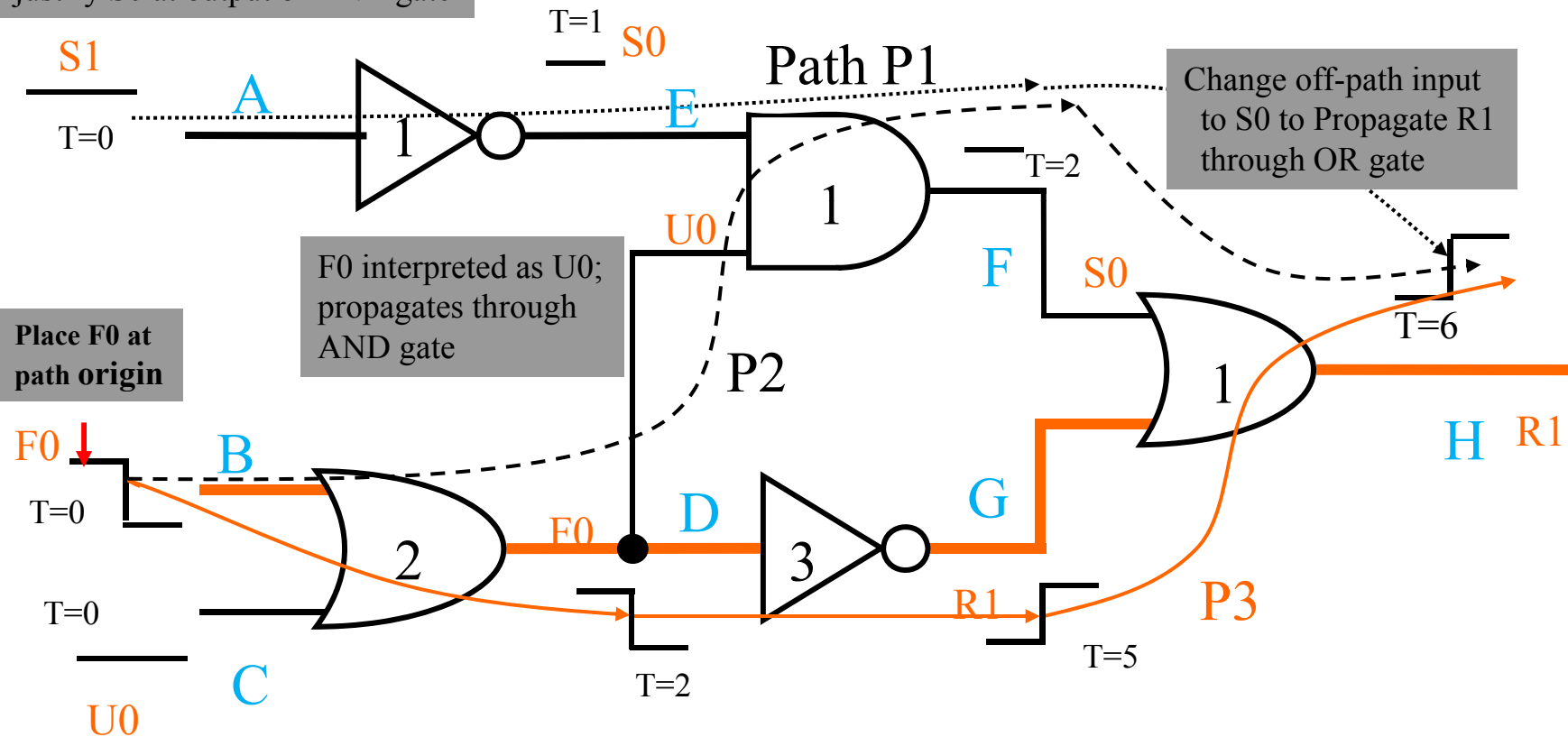
NOT

Input					
S0 U0 S1 U1 XX					
S1	U1	S0	U0	XX	

Example: TP generation

P3—Delay to Fall

Set input of NOT gate to S1 to justify S0 at output of AND gate



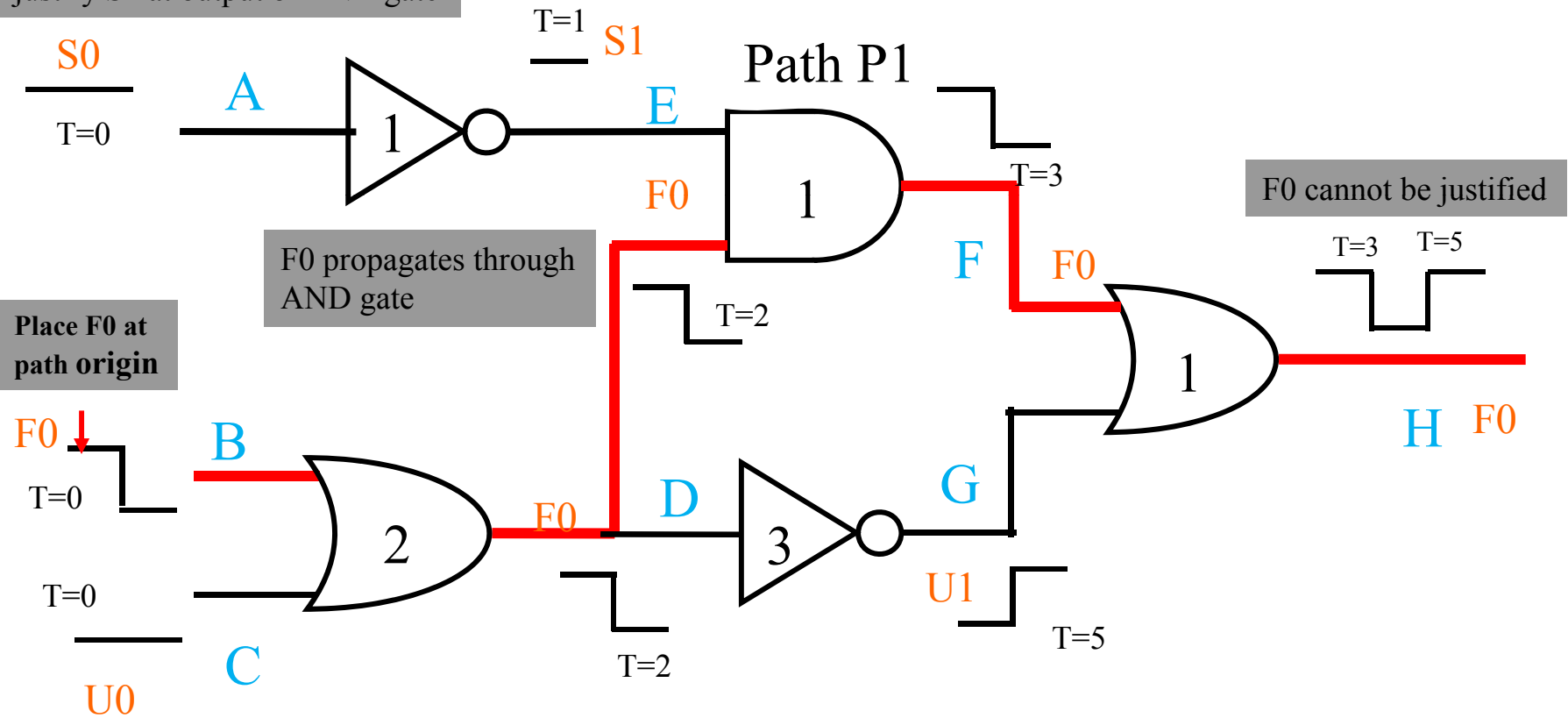
Propagate F0 through OR gate;
also propagates as R1 through
NOT gate

Test Vector pair
11x and 100

Example: TP generation

P2—Delay to Fall

Set input of NOT gate to S0 to justify S1 at output of AND gate



F0 cannot be justified

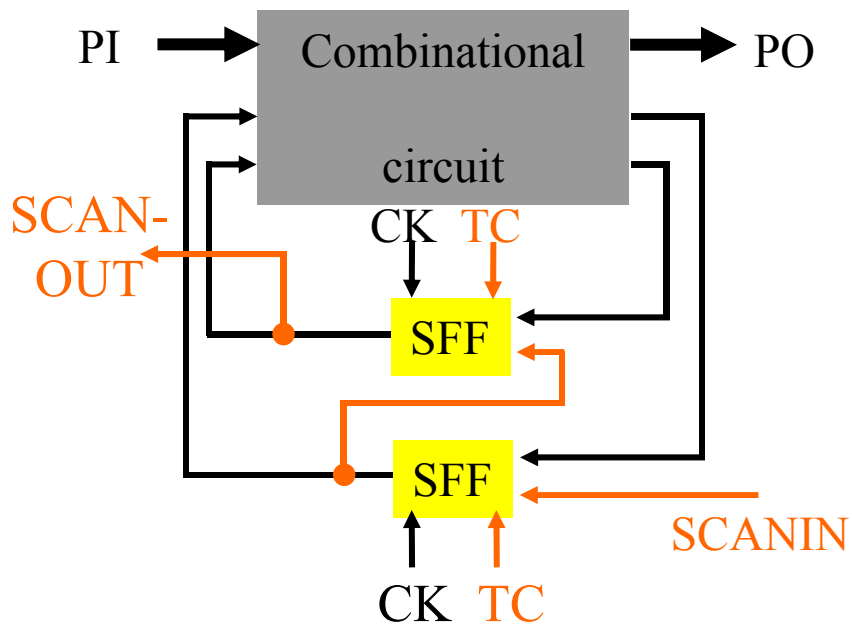
Place F0 at path origin

F0 propagates through AND gate

Propagate F0 through OR gate;

Require: U0
No test Pair

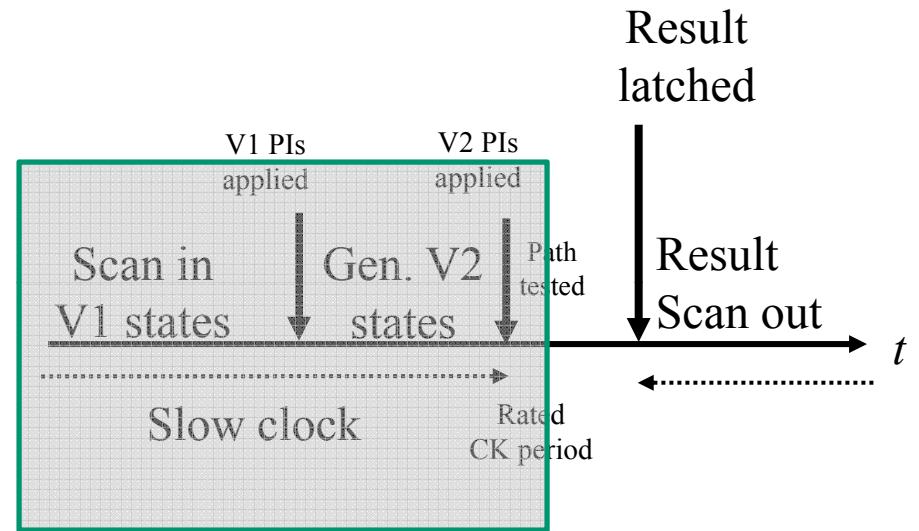
What About Sequential Circuits ?



CK: system clock

TC: test control

SFF: scan flip-flop

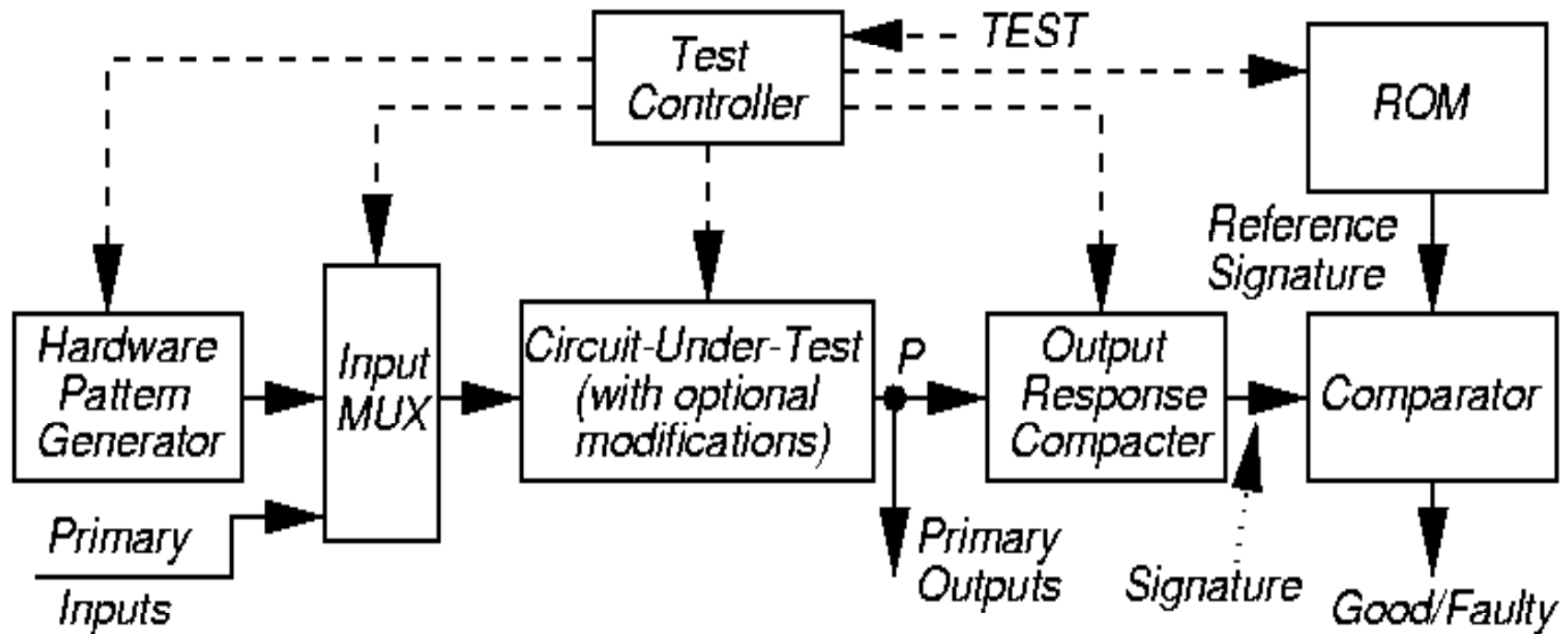


Major Challenge
V1 and V2 Related

At-Speed Functional Test

- Application of test vectors at the rated-clock speed.
- Two Schemes.
- ATE based test:
 - TPs applied and analyzed thorough high speed ATEs
 - High-speed testers are expensive.
- BIST

BIST



Key References

- Bushnell and Agrawal, Essentials of Electronic Testing for Digital, Memory & Mixed-Signal Circuits, Kluwer Academic Publishers, 2000
- M. Abramovici, M.A. Breuer, and A.D. Friedman. Digital Systems Testing and Testable Design. Wiley-IEEE Press, 1994.
- Santosh Biswas and Jatindra K Deka, NPTEL Video lectures and Web course VLSI Design Verification and Test (<http://www.nptel.ac.in/courses/106103116/>)



Thank You